actuators

Micromanipulation

Edited by

Nicola Pio Belfiore

Printed Edition of the Special Issue Published in *Actuators*

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Special Issue Editor

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This is a reprint of articles from the Special Issue published online in the open access journal *Actuators* (ISSN 2076-0825) in 2018 (available at: https://www.mdpi.com/journal/actuators/special_issues/micromanipulation)

For citation purposes, cite each article independently as indicated on the article page online and as indicated below:

LastName, A.A.; LastName, B.B.; LastName, C.C. Article Title. *Journal Name* **Year**, *Article Number*, Page Range.

ISBN 978-3-03897-503-8 (Pbk)
ISBN 978-3-03897-504-5 (PDF)

Cover image courtesy of Alvise Bagolini and Michele Fedrizzi.

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About the Special Issue Editor

Nicola Pio Belfiore, Professor, IEEE Member, teaches ”applied mechanics” and ”functional design” at the University of Roma Tre, Italy. After obtaining his Ph.D. degree, completed at “Sapienza” in cooperation with the University of Maryland of College Park, he won two international awards in the USA and one in Europe. Since 2008, he has been Honorary Professor of Obuda University, Hungary. The author of three textbooks, two patents, and about one hundred scientific papers, he has been the coordinator of several scientific projects, both National and European. In 2013, he was the director of the Second Level Vocational Master in Energy Conversion Efficiency and Renewable Energy. His interests are functional design, MEMS, robotics, tribology, kinematics, and dynamics. In October 2017, he moved from Sapienza to Roma Tre University.

Preface to ”Micromanipulation”

The idea of dedicating a Special Issue to the field of micromanipulation came about from the society's increasing interest in this subject. The demand of microsystems for micromanipulation has grown quite quickly in recent decades, and many fields of applications, nowadays, need new, efficient tools for this peculiar purpose. The problem of actuation is crucial for micromanipulation, and, for this reason, the present book is believed to offer a small contribution to its solution. There is a certain consciousness that this subject must be a highly interdisciplinary one, because the "engineering" offer and the "applications" demand must meet at a certain point. Therefore, the present Issue is composed of papers that are clearly located in the engineering field, although they suggest a variety of implications for many other topics, such as the biomedical field. The purpose of this Special Issue is to stimulate the interest of colleagues, engineers, and students towards a peculiar aspect of engineering with a strong interdisciplinary content: the physical manipulation of micro objects at the micro scale. The development of didactical methods dedicated to this special field has been also covered with two papers. I was very pleased to have received a number of contributions from my colleagues, who agreed to share their results with us, publishing their papers within the present Special Issue. I am very grateful to them. Finally, I would like to thank all the staff of MDPI for supporting and trusting me: my service for "Actuators" has helped me a lot to develop a growing sensitivity towards the world of research and to the dissemination of scientific results. Special thanks is due to Darci Zhou, who has supported me to develop the present Special Issue since February 2018, for her help and enthusiasm.

Nicola Pio Belfiore
Special Issue Editor

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Editorial

Micromanipulation: A Challenge for Actuation

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Received: 12 November 2018; Accepted: 29 November 2018; Published: 3 December 2018

Abstract: Manipulating micro objects has become an important task in several applications. Actuation is a crucial aspect of micromanipulation because there are physical restrictions which affect actuators' performances at the micro or nano scale. One way of getting rid of these limitations is the use of an appropriate mechanical structure which enhances the elasticity of the material or provides mechanical advantage. This Special Issue of Actuators, which is dedicated to micromanipulation, offers a contribution to the development of some promising methods to actuate a microsystem for micromanipulation.

Keywords: micromanipulation; nanomanipulation

1. Introduction

During the last decades, microsystems have been developing very fast thanks to progress in science and technology, giving rise to two main crucial questions, namely,

- how could these micro devices be fabricated and operated?
- how could they improve certain aspects of our lives?

The first question leads to classical issues of engineering, such as design, fabrication and control, which study the most convenient way to create the tools for micromanipulation.

The second question is clearly related to the applications of these new developed tools and to their exploitation, as an opportunity to solve old and new problems for the improvement of certain aspects of our lives.

1.1. Downsizing Effects

Once involved in the development of a microsystem, designers immediately come across the problem of handling the scaling effects. They soon become capable of monitoring how surface and volume properties change their impact on a system; once the latter is downsized by one or more orders of magnitude from the human-size typical dimensions: of course, surfaces or volumes will reduce their quantities by the square or the cube, respectively, of the reduced lengths.

Scaling effects were described by Galileo Galilei in 1638 [1]. He pointed out that "a large animal does not possess simply a bone on a larger scale, but its thickness must increase more quickly than the length of the relevant bone", because resistance and weight scale their quantities differently during downsizing. Similarly, "a giant would never have the same limb ratios of a man with a normal size, but he must have thicker limbs suitable to support its mass".

After about a century, Jonathan Swift described the voyage of Lemuel Gulliver [2] to Lilliput. In the book, the imaginary Surgeon, and then a Captain, reports, literally, that the Lilliputian "mathematicians, having taken the height of my body by the help of a quadrant, and finding it to exceed theirs in the proportion of twelve to one, they concluded from the similarity of their bodies, that mine must contain at least 1724 of theirs, and consequently would require as much food as was necessary to support that number of Lilliputian". Surprisingly, the mentioned human-to-lilliputian food ratio (1724)

only approximates the square of 12 (1728) and, even more curiously, the year of publication of the book (1726).

1.2. Design, Fabrication, and Control

Design, fabrication and control of microsystems for micromanipulation have to face several difficulties because, at the micro or nano scale levels, several paradigms of macro-scale engineering are no longer valid [3]. These and many other consequences and implications have been early underlined by Richard P. Feynman, during his seminal talk given on 26 December 1959 at the annual meeting of the American Physical Society (APS) at the California Institute of Technology [4]. In this speech, Feynman surprisingly mentioned many applications such as computer miniaturization, microsurgery, micro-machining and actuation.

Nowadays, an increasing number of applications are demanding high-precision tools which introduce more and more constraints to their design, following the classical client-to-designer feedback.

The following issues have been particularly discussed in the literature.

- Micromanipulation mechanics [5–7]:

  theoretical modeling,
  numerical simulation, and
  experimental testing.

- Microsystems architecture, components and manufacturing [8–12]:

  design,
  fabrication,
  fabrication constrained design rules,
  packaging and
  biocompatibility.

- Actuation and Sensing [13–16]:

  electrostatic,
  electrothermal,
  electromagnetic,
  piezoelectric.

- Micro-electro-mechanical system (MEMS) integration:

  lab-on-chip systems [17–19],
  MEMS integration with Application Specific Integrated Circuit (ASIC) [20,21].

- Control [22–24]:

  automatic regulation and control of microsystems,
  operational aspects of micromanipulation,
  measurements.

However, while Nanotechnology has found a certain variety of good (although still perfectible) solutions to many problems in MEMS developing [25], what makes micromanipulation particularly difficult is its actuation. In fact, the available sources are often not able to exert an adequate force or torque. Furthermore, the highest performances are achieved only within a rather restricted range for end-effector displacements, and so the mechanical structure of the microsystem must be optimized. In fact, actuators are devices which transform energy (e.g., thermal or electrical, depending on the available source) in mechanical energy. Using the classical sources, such as electrostatic, thermal (shape memory, electrothermal), electromagnetic, or piezoelectric [26], it turns out that it is

rather difficult to achieve both a large force together with a large displacement at the output link and, therefore, a micro mechanism can be employed to gain the mechanical advantage. Fortunately, recent developments in MEMS Technology allows designers to introduce different sorts of micro mechanisms, such as microgrippers [27,28], with multi-hinge and multi-DoF (Degrees of Freedom) properties, and other multi-axes devices [29]. Such opportunity gave rise to the introduction of a design technique based on the rigid-body replacement method [30], which refers to classical issues of mechanism science such as topology [31,32], kinematic synthesis [33–35], kinetostatic indexing [36], isotropic compliance [37,38] and parametric design [39,40].

1.3. Applications

Nowadays, micro or nano manipulation is very attractive in a large variety of applications, ranging from medicine, surgery or biology, to microelectronics, micro mechanics and aerospace. Therefore, the following items represent a non-exhaustive list of much more examples of applications:

- drug delivery [41,42],
- minimal-invasive surgery [43,44],
- tissue or cell manipulation [30,45–47],
- diagnostics [48–51],
- aerospace [52–54],
- micro-assembly [16,55–57], and
- microelectronics [58,59].

1.4. Forthcoming or Emerging Issues

Other than the above-mentioned classical issues, there are some others that are either more recent or, at least, less usual than the former. In fact, once new concepts become well-established and real devices, the interest in their optimization and use increases. Furthermore, some other related issues appear, and so the following topics could soon become new topics for MEMS and microsystem applications.

- Computational intelligence: optimization and control of microsystems [60,61].
- Development of ambient intelligence [62] based on sensors, actuators and standardized internet communication.
- Use of nonlinearity benefits [63,64].
- Configuration management and reconfigurable manufacturing systems for the development of microsystems during lifecycle [65].
- Ethics: ethical issues in the activities of criteria-based decision making autonomous micro-manipulators in the medical, biological, aerospace and industrial fields [66–70].
- Education: new trends in microsystems teaching–learning methods, tinkering, open access, wiki tools [71,72].

2. On the Variety of Demand vs. the Supplying Capability

A selection of 1846 papers concerning "micromanipulation", distributed over seven different meta categories, namely, Biology, Computer Science, Engineering, Medicine, Multidisciplinary issues, Physics/Chemistry, Science have been analyzed. These categories have been named after the classification suggested by one of the most acknowledged database for Science and Technology, that is the Web of Science [73]. A series of queries has been launched on the database and some statistical data have been extracted. At the first sight, among the above mentioned categories, there is one which can be assumed to represent the "supplying capacity" at the actual state of the art, as introduced in Section 1.2, namely,

- Engineering,

The other six categories represent the variety of the demand of micromanipulation technology and its temporary success in applications, as mentioned in Section 1.3, that are

- Biology,
- Physics and Chemistry,
- Medicine,
- Computer Science,
- Multidisciplinary issues, and
- Science.

Statistical analysis has revealed that the struggling for technological readiness, that could be reasonably represented by the filed of engineering, collects almost a half of all contributions, about 45% of the papers. On the other hand, the applications, which could be thought as the "recipient" of the technological progress, gather the other half of the full bunch of papers. More in detail, the Biomedical Sciences, including Biology (18%) and Medicine (11%), form 29% of contributions. General categories such as Science (3%) and Physics/Chemistry (13%) form about 16% of the considered papers. Finally, Computer Science includes 6% of the selected papers, while the last 4% concerns multidisciplinary issues. Figure 1 illustrates the distribution of the selected papers according to the above-mentioned categories.

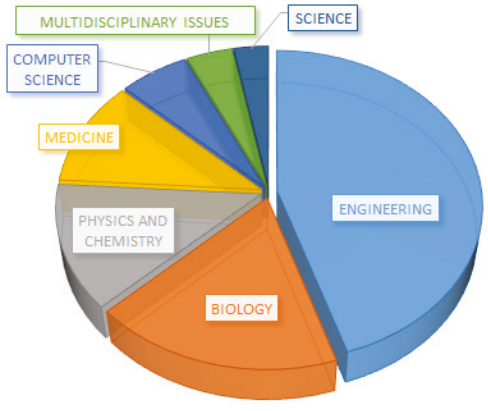

Figure 1. Statistical distribution of 1864 papers concerning "micromanipulation" along seven different groups, gathered over 83 Web of Science Categories.

The adopted method of paper selection has been based on classical database query tools, as implemented in WoS [73] and, therefore, the search keys, which consist of selected words, may be subject to double meaning errors. This introduces an error that can be roughly estimated by manually checking some elements extracted from a randomly sampled group. On the basis of a rough estimation, the mistake is expected not to exceed 3% of the values.

The results of the present investigation show that research into micromanipulation is still more extended in the field of engineering than in any of the other applicative fields. However, the full group of applications represents half of the analyzed papers, which shows that engineering vs. applications are nowadays in balance.

These results probably reveal the actual struggle of investigators to improve the technical characteristics from a rather low technological readiness level (TRL) [74], to higher TRL values that are more suitable for immediate and commercial applications. It is rather difficult to predict how much

time it takes to make micromanipulation technology ready for more applications in hospitals, labs, cars, aircraft, appliances, and so on, while, on the other hand, it is easy to infer that the number of applications will increase as much as new confidence is acquired in the construction and control of the tools.

Figure 2. SEM image obtained by Alvise Bagolini and Michele Fedrizzi at Micro Nano Fabrication and Characterization Facility, Fondazione Bruno Kessler, Trento, Italy (PS the housefly has been found dead in Villamontagna, Trento, Italy and no harm has been inflicted to the poor insect).

3. Tools for the Observation

As known, the human eye normally barely distinguish objects within the size of a few tens of microns, which makes it difficult for us to monitor micro devices without a proper means of magnification. Usually, optical microscopy could give a first-hand tool for observation, with the advantage of operating in air or liquid environments. However, SEM observation is more helpful to characterize the prototypes, because of their higher resolution. For this reason, in MEMS development,

the moment always comes, eventually, to take an SEM picture of a microsystem. Usually, an image obtained by an SEM instrument declares the adopted magnification factor and other technical data and, therefore, there is no real need to use a side comparison object (whose dimensions are universally known). At the macro level, classical objects can be used to compare the observed subject with familiar objects, such as a one cent coin or the portion of a ruler. However, while moving around either the "brobdingnag" or the "lilliput" worlds, it is not always easy to grasp the real sense of the size of an object which has unusual dimensions, such as gigantic (planets, stars, galaxies) or minuscule (cells, atoms) things.

Figure 2 is an example of a comparison exercise at the microscale. An adult housefly, musca domestica, was positioned over the square portion of a silicon wafer where a microgripper had been fabricated [12]. According to the construction process and the adopted parameters, each microgripper lays within a 5×5 mm^2 window, whereas adult houseflies are usually from 6 to 7 mm in size. That is why the whole insect was as long as the length of a microgripper window. The fly eyes, whose diameter is approximately one millimeter and a half, are visible at the top of Figure 2, while the microgripper shows up in the lower half of the picture. In the upper half of the figure, the compound eyes of the fly appear in all their complexity. In fact, there are almost 3500 ommatidia facets in the musca domestica' eye [75], while the single ommatidium is about 10 μm [76]. This figure make the microgripper a bit less mysterious object to our understanding.

4. Micro- or Nano-Manipulators vs. Micro- or Nano-Robots

Finally, it is worth noting that the terms micromanipulator and microrobot (the same holds for the pair nanomanipulator and nanorobot) are often used as synonymous, while it would be better not to confuse them that way. In fact, developments in technology have made it possible to build devices with different features; for example, their overall size or their positioning or manipulating accuracy.

Considering the scope of developing miniaturized devices at the bottom scale, it seems to be appropriate to refer to such devices as microrobots, micromechanisms, micromachines or, more generally, microsystems (nanorobots, nanomechanism, and so on). On the other hand, when a device (no matter about its size) is required to be able to manipulate objects with micro (or nano) accuracy, then it seems that the term micromanipulators (or nanomanipulators) becomes more appropriate.

This distinction is quite substantial. In fact, in many occasions, it is necessary to handle micro- or nano-objects, with limited concern about the size of the manipulating object. For example, nano-manipulators used inside the chamber of a Scanning Electron Microscope (SEM) may have extraordinary resolution, $<$0.5 nm, with rather a, relatively, large overall size, of about 6 cm [77]. On the other hand, in some other circumstances, it is necessary to have a small system, namely, a microsystem, which is able to operate within specific environments, with very restrictive size limits. Interest in miniaturizing micromechanisms [78], microrobots [79,80], microgrippers [81,82], and microtribometer [83] has been recently expressed. In fact, in all these examples. the size of the microsystem was fundamental for the performance of its function.

5. Conclusions

The present Editorial has been written with the intent of intriguing readers and colleagues who may want to deepen the topic of actuation at the microscale, specially for the purpose of micromanipulation. The actual state-of-the-art shows that this problem is rather far from being exhausted and, therefore, the present Special Issue represents an invitation to accept the challenge to find proper solutions to this endeavour.

Conflicts of Interest: The author declears no conflict of interest.

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Mechanical Response of Four-Bar Linkage Microgrippers with Bidirectional Electrostatic Actuation

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Received: 2 October 2018; Accepted: 7 November 2018; Published: 11 November 2018

Abstract: This paper presents both an experimental and a numerical study concerning the mechanical response of a silicon microgripper with bidirectional electrostatic actuation to externally applied excitations. The experimental set-up is composed of a probe station equipped with mobile probes that apply contact forces. This part of the investigation aims to test the device's mechanical resistance, its mobility capability and possible internal contacts during the system deformation. The second part of the paper is dedicated to the study of the free undamped vibrations of the microsystem. Finite Element Analysis (FEA) is carried out to evaluate the system vibration modes. The analysis of the modes are useful to predict possible mechanical interference among floating and anchored fingers of the actuating comb drives.

Keywords: MEMS; vibration modes; DRIE; microgripper; comb drive actuators

1. Introduction

The introduction of new classes of flexure hinges [1] and the technological progress in mechanical components of MEMS (micro electro mechanical system) [2,3] gave rise to new devices for the manipulation at the microscale. In fact, about a hundred microgrippers [4,5] were designed and fabricated with different purposes and actuation systems. For example, micromanipulation finds important applications in micro assembly processes. Some devices can be fabricated as monolithical structures, whereas others require an assembly step because of particular geometries or different materials [6]. Microgrippers are also employed in optical fibers assembly [7,8]. Another important field of application is biology: manipulation of single cells is an essential step to understand cells behaviors and interactions [9,10]. For example, microgrippers with force sensors where developed for manipulating biological cells [11,12] or to characterize the mechanical properties of biosamples [13,14]. The sensing [15,16] and the control [17,18] of the gripping forces has also been a fundamental issue in developing microgrippers.

However, in spite of the recent progress in nano and micro-machining, there is still a certain difficulty in building multi-hinge and multi-DoF (Degrees of Freedom) MEMS.

The main problem consists of the fact that mobility is granted by flexure hinges and that the latter are still rather complicated to be designed and fabricated in a small portion of the device. This problem is particularly arduous in the design of microgripper for micro manipulation and, therefore, new microsystems equipped with Conjugate Surface Flexure Hinges (CSFHs) have been developed [19] and fabricated [20]. The idea, which dates back to 2012 [21], is based on the partitioning

of a block into either rigid or flexible sub-parts, with mobility being granted by the presence of the flexible parts (flexure). A CSFH is a particular kind of flexure that is made of a curved beam, which provide compliance, and a portion of a conjugate-profiles, which provides accuracy. The two components are designed in such a way that the center of the elastic weights of the curved beam is coincident with the center of the conjugate profiles.

The CSFH had a certain number of applications in microsystems, for example in micro mechanisms [22,23], micromanipulators [24–26], tribometers [27], grippers [13], biomechanics, etc. Some recent experimental investigations showed also that these microsystems, although with some restrictions, can be operated by means of comb drives [28]. However, there are still some concerns regarding the feasibility and robustness of these devices in both static and dynamic conditions.

Actuation is among the most important functions of a microsystem, and so different kinds of solutions to this problem have been proposed, such as electrothermal [29,30], shape memory alloy [31,32], or piezoelectric actuators [33,34]. Electrostatic actuators, in particular linear [35–39] or rotary [40–42] comb drives, offer also a feasible actuation system for micromanipulation.

In this investigation, to evaluate the mechanical robustness of the microgripper towards its application in an operational environment, the mechanical functionality of a four-bar linkage microgripper with bidirectional electrostatic actuation has been examined both in static and dynamic conditions. Static load was experimentally tested within a probe station. Then, the static response of the microgripper under externally applied forces has been numerically simulated by means of Finite Element Analysis (FEA) and some original design charts have been built to predict possible contacts between CSFH conjugate surfaces. This information is useful in the design steps, to optimize the orientation of the four CSFHs embedded in the four-bar linkage. Finite Element Analysis (FEA) has been also applied to analyze the vibration modes. The eigenmodes and eigenfrequencies have been calculated and the shapes of vibration associated with the first six modes have been analyzed.

2. Design

The MEMS consists of a bulk structure which gains mobility thanks to elasticity [43]. Generally, compliant mechanisms can be categorized into two main classes: with lumped or distributed compliance. The CSFHs are particularly suitable to be used as elements of compliant mechanisms with lumped compliance. This opportunity gives rise to new design methods which make use of a topological approach [44,45], such as for example, the rigid-body replacement method [46,47]. By following this approach, starting from a classic four-bar linkage, the mask represented in Figure 1a has been created in a configuration that maximizes the comb drives' rotations (see the comb drive detailed view presented in Figure 1b) in both closure and opening directions. Then, one CSFH replaces each rotational joint, as illustrated in the detailed view of Figure 1c. Figure 1d shows the gap Figure 1e between the conjugate surfaces in a non contact configuration.

Once each one of the four ordinary revolute pairs has been replaced by a CSFH, a fully compliant four-bar linkage is created (lumped elasticity). Of course, two compliant four-bar mechanisms are needed to assure the grasping operation and, therefore, two four-bar mechanisms are symmetrically positioned to allow the gripping jaws to symmetrically approach the micro object. With reference to Figure 1, the design illustrated by means of the mask Figure 1a has been laid out in such a way that a gripping jaw is attached to the coupler link. For example, the left jaw is pointed out in Figure 1f. The four-bar mechanism provides the jaw tip a wide range of motion from the open to the close extreme configurations.

The device is operated by means of two bidirectional electrostatic actuators. The open position of the jaws is obtained by applying a voltage between the pads (i) and (ii) represented in Figure 2a. With reference to the Figure 2b, the mobile set of fingers (iv) rotates counterclockwise, while the coupler link rotates clockwise because of the given configuration of the four-bar linkage. By applying a voltage between pads (i) and (iii) of Figure 2a, the opposite applies, and the comb drive mobile wing (v) rotates clockwise, inducing a counterclockwise rotation of the coupler.

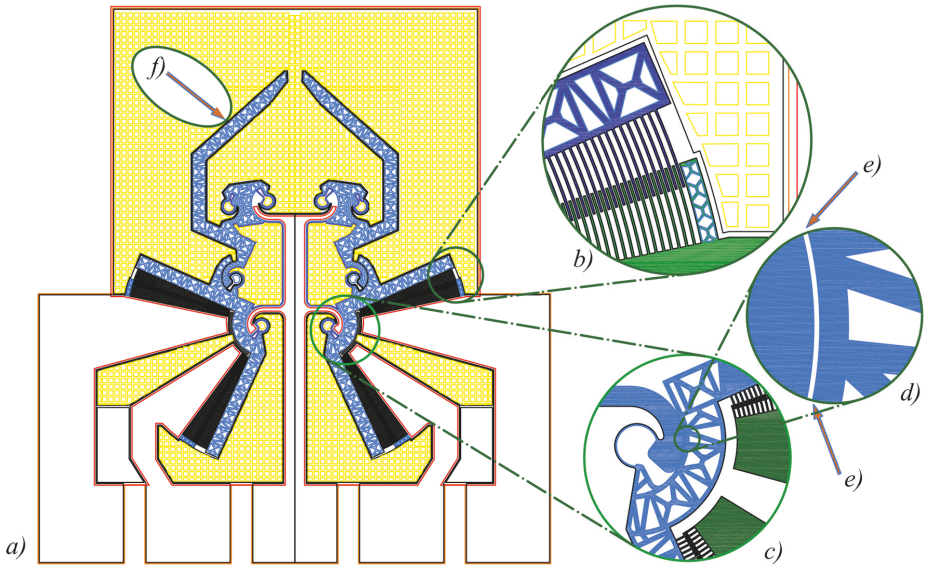

Figure 1. The mask adopted during Deep Reactive-Ion Etching (DRIE) process (**a**); details of the fingers of the comb drive (**b**); the Conjugate Surface Flexure Hinge (CSFH) hinge (**c**); the gap between the conjugate surfaces (**e**) and the left-hand side jaw (**f**).

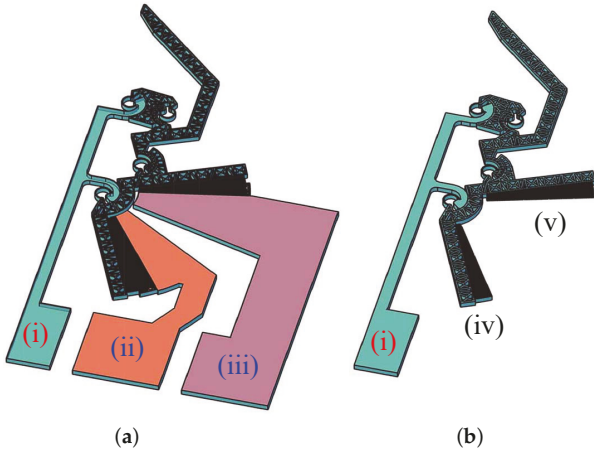

Figure 2. Model of the right jaw of the microgripper with (**a**) and without (**b**) the control pads: (i) device anchor; (ii) opening control pad; (iii) closing control pad; (iv) and (v) mobile fingers engaged with (ii) and (iii), respectively.

3. Fabrication

The fabrication stage has been performed starting from a 6-inch SEMI standard [48] silicon on insulator (SOI) wafer. Boron has been used as dopant for the wafer SOI device layer, with a resultant resistivity equal to 2–4 Ω·cm. Each wafer has a 500-µm handle layer that works as the device support. A 40 µm thick silicon layer works as device layer and is bonded at the top of the wafer. The device (top) layer is the most important one because the suspended moving subparts are therein patterned.

Between the support and the top layer, a 2-μm oxide layer stops etching during silicon patterning. This layer is also necessary to support the top layer before the release of the devices. The support and top layers of the wafer need to be both patterned and so Bosch deep reaction-ion etching (DRIE) process is applied to etch silicon down to the buried oxide layer. The DRIE was performed with an Alcatel AMS200 ICP plasma etcher (Alcatel-Lucent, Paris, France), using a standard two-step bosh recipe with fluorine chemistry. The initial step consists in the deposition of a multilayer mask to provide masking for DRIE etching process on both sides. A 150 nm silicon dioxide is firstly deposited. Then a 200 nm aluminum film is sputtered. A 100 nm film titanium is finally sputtered and standard photolithography is used to pattern this layer stack. The front side is exposed by means of an i-line stepper, while the backside is aligned by means of a broadband mask aligner. Finally, stack etching takes place in IC standard plasma reactors.

After masks patterning, an Alcatel SMS200 (Alcatel-Lucent, Paris, France), etcher is used to etch silicon through DRIE Bosch process. Front and backside etching deepness is equal to 40 μm and 500 μm, respectively.

At the end of this process, the front and back geometry is transferred on the wafer sides. Therefore, the silicon dioxide intermediate layer must be removed by wet etching in hydrofluoric acid based solution. This treatment allows the device to be released and freely move. During wet etching, residual DRIE mask layers are etched as well.

For the sake of electrostatic actuation, some electrical connections are developed by using the physical vapor deposition of an aluminum layer on the front side.

Aluminum hard mask allows the process to greatly reduce the mask layer thickness. This simplifies sub-micron features patterning, eliminating the obstacle of high aspect ratio thick mask etching. Further, aluminum grants an excellent feature size control during pattern transfer from the mask to silicon, as it eliminates mask edge erosion. The three layer structure and the whole fabrication process have been fully described by Bagolini et al. [20]. In the micro-gripper device, an aspect ratio up to 20 is implemented, but the fabrication module is developed for higher aspect ratio which is not part of present work.

4. Contact and Mobility Tests

As mentioned in Section 2, the adopted mechanism is composed of two symmetric four-bar linkages, each one having four CSFH compliant hinges. Since each of the two mobile suspended structures is held in place by only two 20×5 μm^2 cross sectional areas, some concern has arisen about the microsystem capability to bear externally applied loads. Hence, the purpose of the present paper consists in testing the mechanical structure of the developed microgripper. For this reason, the experimental activities have been arranged in such a way to provide information on the resistance of the structure, whereas its operational capability in unloaded conditions have been previously investigated [28,49].

In the present paper, a contact force has been externally applied to the right arm of the microgripper. This external load is much more invasive than the torque which is exerted by the electrostatic actuation on the crank-link, because it transmits to the block of mobile fingers not only a moment, but also a force. Such force has generally a radial component, the latter being much dangerous for the microsystem, because it pushes mobile fingers against the fixed ones.

The microgripper under analysis has been tested by means of an Agilent probestation equipped with binocular microscope and needle probes with tri-axial micrometer positioning control. The kinematic functionality of the device has been verified by monitoring the microgripper response to a force exerted by a probe through the contact area.

A series of mechanical tests have been conducted by using the three-axis micro probe (Suss PH 150 probehead). As reported in Figure 3, the probe was moved in such a way to mimic the electrostatic action exerted by the comb-drive, rotating the floating part toward the corresponding anchor.

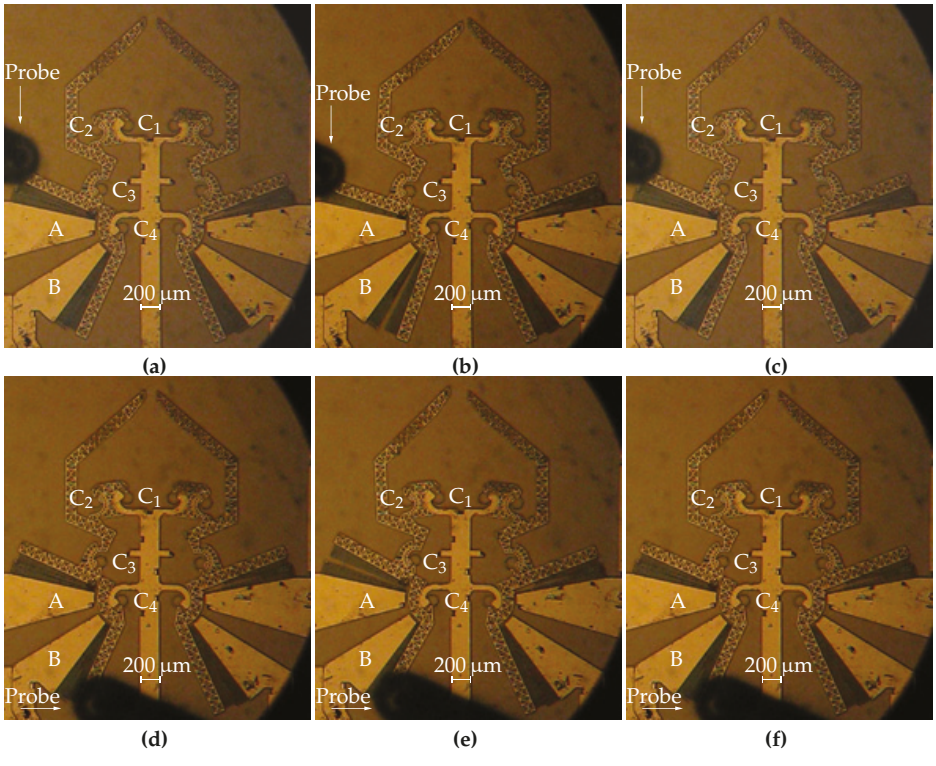

Figure 3. Steps of the mechanical testing procedure: (**a**) Probe positioning near the closing comb-drive (A); (**b**) Close configuration: stroke limit for the closing comb-drive (A); (**c**) Neutral configuration: releasing of the closing comb-drive (A); (**d**) Probe positioning near the opening comb-drive (B); (**e**) Open configuration: stroke limit for the opening comb-drive (B); (**f**) Neutral configuration: releasing of the opening comb-drive (B).

The test is performed considering the following steps:

(a) the probe is positioned in proximity of the closing comb-drive: before contact, the compliant mechanism stands in neutral configuration;

(b) the probe contacts the device and gently pushes the floating part until the maximum rotation is reached. Therefore, the gripper jaw follows a closing trajectory until the device achieve the close configuration (Figure 3b). During this stage, while the rotation angle between floating and anchored parts of the closing comb-drive decreases, the relative rotation between the same parts of the opening comb-drive increases;

(c) the probe is brought back to the initial position; in this phase, the comb-drive is gradually released and the gripper jaw follows an opening trajectory, until the microgripper achieves again its neutral configuration (Figure 3c);

(d) the probe is positioned in proximity of the opening comb-drive: as in the previous case, before contact, the microgripper is in neutral configuration (Figure 3d);

(e) the probe contacts and pushes the floating part to the comb-drive limit position: the gripper jaw follows an opening trajectory until the device achieve the open configuration (Figure 3e). While the rotation between the floating and anchored parts of the opening comb-drive decreases, the relative rotation between the same parts of the closing comb-drive increases (Figure 3e);

(f) the probe is repositioned at the starting point Figure 3d: the comb-drive is released and the gripper jaw follows an closing trajectory, until the neutral configuration is achieved again (Figure 3f).

Figure 4 shows three overlapping images reporting the left arm of the microgripper in close, neutral, and open configurations. The probe force could not be directly measured, due to the particular adopted experimental set. However, the applied direction could be obtained by recording the tip displacements from the neutral position, while the force magnitude was measured by means of FEA. In fact, a magnitude of the applied force equal to 320 μN was calculated as compatible with the observed close and open configurations. Figure 4 presents also a schematic representation of the two forces corresponding to the close and opening maximal configurations.

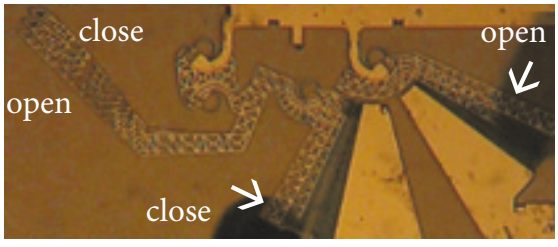

Figure 4. Overlapping images of the microgripper in close, neutral, and open configurations.

Contact Analysis on the Conjugate Surfaces

The CSFH is able to modify substantially the static and dynamic behaviors of the whole microsystem which it belongs to, depending on the occurrence of the contact between the conjugate surfaces. In fact, once the two conjugate surfaces get in touch, a reaction force arises between them. Actually, the purpose of CSFH hinges is to use this reaction force in order to restrict the displacements of the centers of the relative rotations between adjacent links within the size of the gap between the conjugate surface (actually about 5 μm). This makes the microsystem quite stable. However, contacts within a CSFH hinge could be activated too late, namely, after the breaking of the curved beam, which represent the elastic part of the CSFH. Therefore, experimental tests of the system response to externally applied forces are fundamental to validate the FEA model and the structural design of the whole system. The conditions under which this contact occurs are useful to optimize the final layout.

In order to analyze different contact conditions, a force **F**, illustrated in Figure 5, is introduced to represent the action that could be applied to the tip during the gripping task or the positioning phase. The right hand side four-bar linkage is also illustrated together with the CSFH's C_1, C_2, C_3 and C_4. The force magnitude $|\mathbf{F}|$ and orientation φ have been adopted as variable parameters, with the purpose of detecting their critical values in correspondence of which the conjugate profiles of the four embedded CSFH get in touch.

Six different values of the magnitude $|\mathbf{F}|$ have been applied, with the assumption that **F** acts on the device working plane, with $F_z = 0$. The force direction is identified in the plane by means of the angle φ between **F** and the x axis. The full span from 0° and 360° has been investigated for φ. It is clear, from the figure, that **F** induces deformations that are concentrated specially on the curved beams, which form the elastic part of the CSFHs. These flexure elements deflect under the action of the internal loads and therefore the conjugate surfaces may get in contact.

A contact chart has been obtained for each CSFH C_i, as represented in Figure 6, by iterating FEA for discrete number of values of the angle $\varphi = 0, \ldots, 2\pi$ and six values of the force magnitude. Each chart allows designer to immediately understand which are the CSFHs which present contact for the given pair of parameters. For example, contact occurrence in C_i, $i = 1, \ldots, 4$ is highlighted by red colored arches and the red sectors are those corresponding to the directions and magnitudes of **F** which induce contact in C_i hinge.

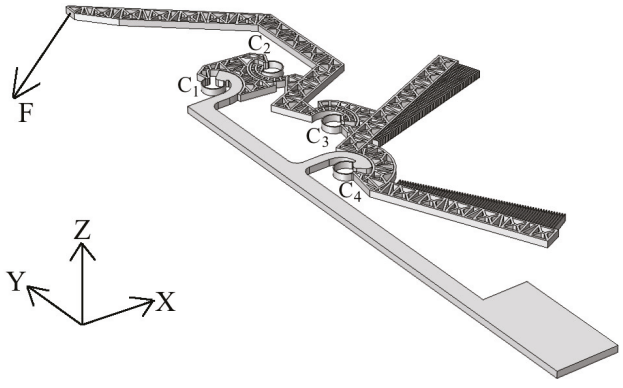

Figure 5. Static load to modify the CSFH contact.

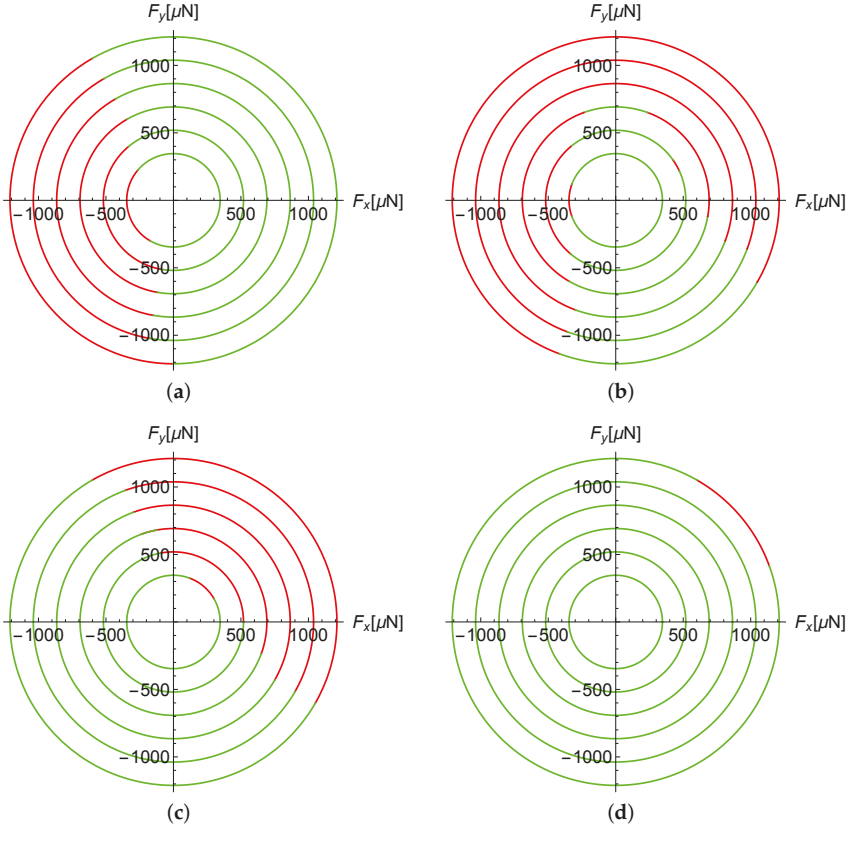

Figure 6. Force vs. contact diagrams: contact charts for the joint C_1 (**a**); C_2 (**b**); C_3 (**c**); and C_4 (**d**), respectively.

It is possible to observe that there are different conjugate surfaces of the CSFHs into contact for different angles. C_2 and C_4 are the CSFHs where the conjugate profiles more or less easily get in

contact, respectively. Figure 7 shows, in particular, the various CSFHs configurations for different angles φ. The magnitude of the adopted force $|F|$ (some hundreds of μN) is comparable to the value of the forces applied to the comb drive mobile finger set (320 μN, as depicted in Figure 4) to fully close or open the configuration.

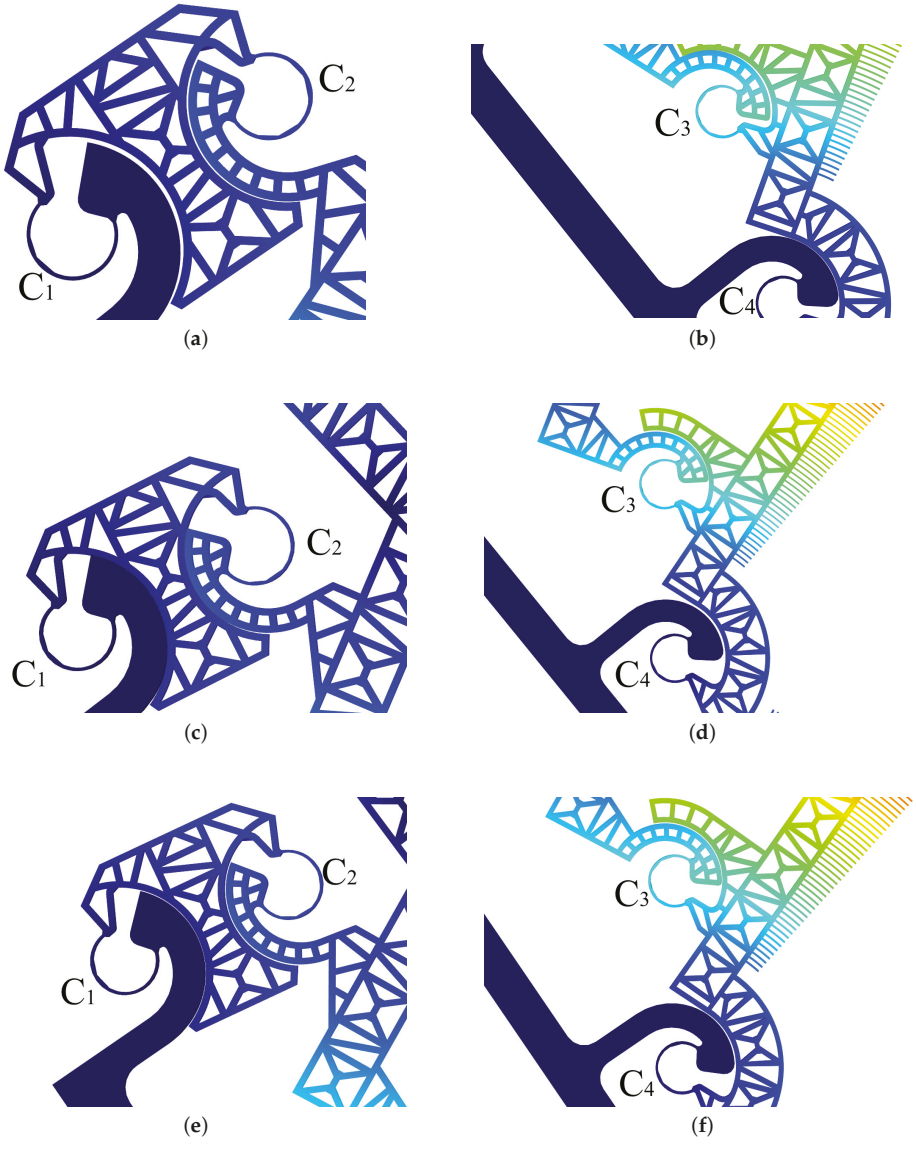

Figure 7. Configurations of CSFHs with $|\mathbf{F}| = 1 \times 10^{-5}$ (N): (**a**) C_1 and C_2 at $\varphi = 50°$; (**b**) C_3 and C_4 at $\varphi = 50°$; (**c**) C_1 and C_2 at $\varphi = 180°$; (**d**) C_3 and C_4 at $\varphi = 180°$; (**e**) C_1 and C_2 at $\varphi = 280°$; (**f**) C_3 and C_4 at $\varphi = 280°$.

5. Vibrational Mode Analysis

The versatility of the CSFH makes them useful in many applications such as micro mechanics, biology, etc. However, in many of these fields the analysis of the vibrations is essential on both negative and positive effects. In fact, the induced vibrations can lead to significant positioning errors [50,51] or undesirable contacts between the fingers. Likewise, jaws vibrations can be exploited to release micro objects [52] or to analyse the mechanical characteristics of the soft biological tissues to study possible pathologies [53].

Direct experimental acquisition of the vibrational modes is not easy for MEMS, although there are some successful cases. For example, in-plane low-frequency vibration was measured by means of circular grating Talbot interferometer [54]. With this method vibration from 1 to 25 Hz has been monitored in a plate system. Scanning laser Doppler vibrometry and experimental modal analysis have been also successfully applied to a pair of micro-cantilevers [55]. A charge-coupled device camera and synchronized pulsating illumination has been also used to measure sub-micrometer in-plane dynamics of MEMS devices with nanoscale precision [56]. Another interesting case of optical detection consists in the acquisition of the electromechanical response of a MEMS-technology based micro-mirror used in scanning pico-projectors [57].

In this paper, the experimental activities have been restricted to the static force analysis only and therefore the vibrational modes have been conveniently obtained via Finite Element Analysis. The modal analysis has been performed with the ANSYS © (Canonsburg, PA, USA) software [58]. The first six, eigenmodes and eigenfrequencies have been reported in Figure 8. In this analysis the device anchor has been considered fixed.

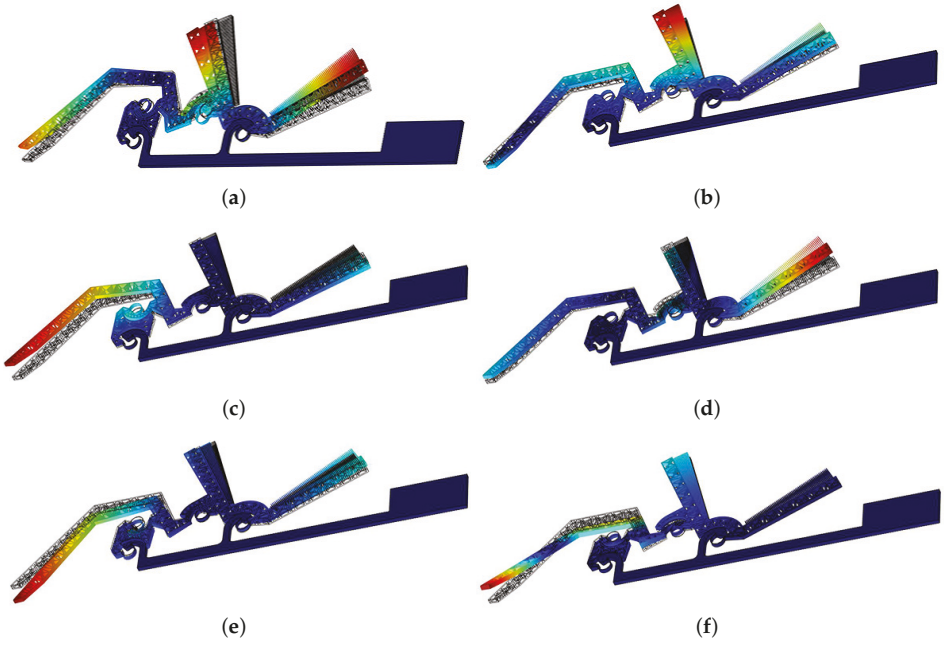

Figure 8. Eigenmodes and eigenfrequencies: first ($f_1 = 1.5147$ kHz) (**a**); second ($f_2 = 1.9433$ kHz) (**b**); third ($f_3 = 2.4749$ kHz) (**c**); fourth ($f_4 = 2.9982$ kHz) (**d**) ; fifth ($f_5 = 6.4103$ kHz) (**e**); and sixth ($f_6 = 6.4618$ kHz) (**f**) eigenmode, respectively.

The relative displacements between the anchored and the rotating fingers are tolerable only if they correspond to a relative rotation where the center is coincident with the center of the conjugate profiles. Therefore, it is very important to understand the nature of the relative motion for all the possible vibration modes. In particular, the first, second and fourth modes present radial displacements for the fingers and therefore finger contact appears to be theoretically possible.

6. Results and Discussions

The results obtained by means of FEA can be displayed, for each vibrational mode, as animated sequences of intermediate deformed configurations. This opportunity has been taken to identify, for each mode, those elements with the minimum displacements during the vibration motion, when a certain mode is excited. Considering also the numerical data, these elements can be easily identified for the first 6 modes and the behavior of the structure can be physically interpreted by introducing nodal oscillation axes.

When the system is excited at the first mode (i), three subparts of the structures behave as pseudo-rigid bodies that correspond to the two rockers and the coupler links. Motion is provided by the four flexure hinges and therefore a relative rotation axis appears for each hinge. During this motion all the subparts move within the mask plane and so the structure is characterized by an in-plane motion. The relative rotation axes are orthogonal to the plane and their intersections with the plane are represented in Figure 9a. Hence, first natural mode consists of a motion that is coincident with the motion for which the system has been designed. As a consequence, this kind of deformation is compatible with the geometry of the fingers because the relative rotation axes are practically coincident with the CSFH rotation axes.

The structure oscillations related to the second mode (ii) reveal that the whole structure behaves approximately as if it was a whole plate which rotates around the axis passing through the centers of the framed CSFH, as depicted in Figure 9b. This axis belongs to the main plane and therefore the oscillation will take place out of the plane. However, the mobile fingers are completely positioned by one side of the axis and so their motion goes along a direction which is orthogonal to the gap between the fingers. This means that finger contact remains quite unlikely.

Considering the third mode (iii), a nodal axis through the centers of two adjacent hinges has been identified and illustrated, as the axis (iii), in Figure 9b. In case the system is excited with the third mode, it will behave, approximately, as it was composed by a flexible plate that oscillates around the axis (iii) with additional deformations due to the presence of the anchored parts. Once again, axis (iii) belongs to the main plane and therefore the fingers will be affected also by an out-plane motion, with limited effect on reciprocal contact likelihood.

Taking into account the fourth mode (iv), the modal analysis shows that the microgripper inflects around two the parallel axes (iv)-a and (iv)-b depicted in Figure 9c. These axes belong to the plane and so the displacements will be out-planar directed. The system is roughly comparable to a flexible plate oscillating around nodal axes (iv)-a and (iv)-b, with three different zones. For example, while the central zone is up, the lateral parts will be down, and vice versa.

In view of preventing comb drives from finger contact, the fifth node of vibration presents the most difficult circumstance under which the microgripper behaves, approximately, as a pseudo-rigid plate which rotates around a point that is positioned within the internal area of the four-bar linkage, as reported in Figure 9d. This is possible because the CSFHs behave as suspension springs. Unfortunately, this motion is rather dangerous for the comb drive fingers, because the mobile finger sets do not rotate about the CSFH centers. This means that the curved fingers follow no more the natural span of the fixed gaps and so they collide with the fixed sets of fingers. This circumstance is depicted in Figure 10 which shows that the mobile finger set has a radial component of the displacement, having lost the original rotation center, with ineluctable mechanical interference.

The vibrational shape related to the sixth mode is rather complicated to be described. In fact, three nodal axes can be identified (Figure 9e). These axes are all in the fabrication plane. For this case,

the microgripper behaves as a flexible plate which inflects around the nodal lines. Since the axes are in the plane, out-plane displacements are the most significant and so no great problem is expected for the fingers.

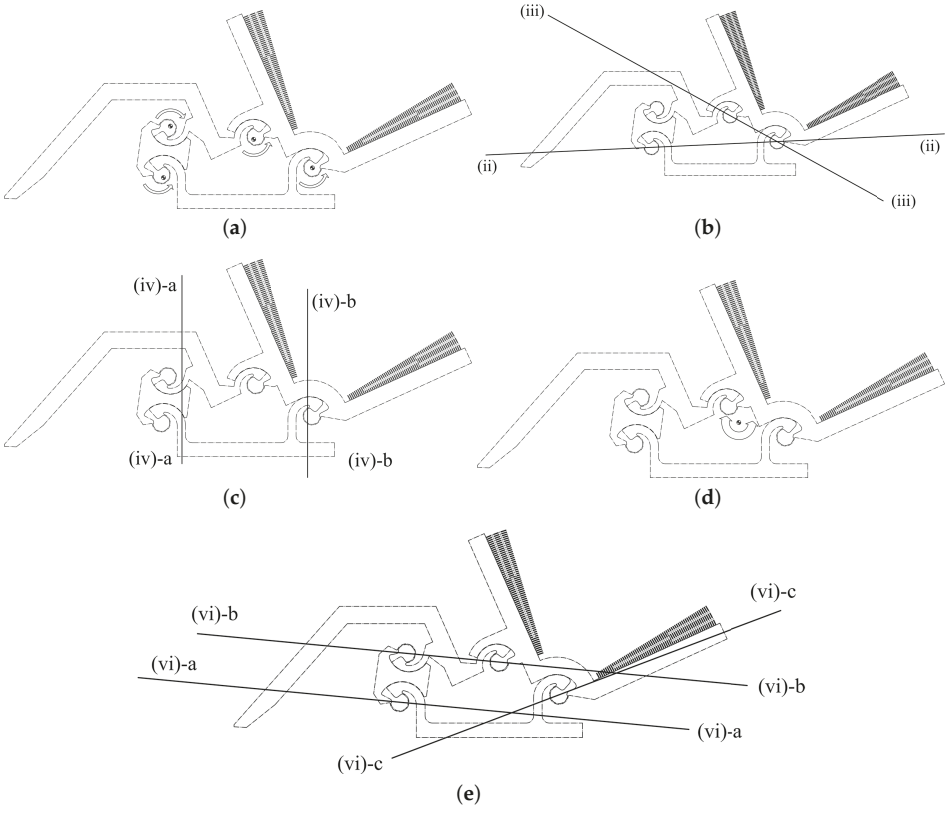

Figure 9. Pseudo-axes of rotation corresponding to the first 6 vibrational modes: (**a**) In-plane rotation axes for mode (i); (**b**) Out-plane rotation axes for modes (ii) and (iii); (**c**) Out-plane rotation axis for mode (iv); (**d**) In-plane rotation axis for mode (v); (**e**) Out-plane rotation axis for mode (vi).

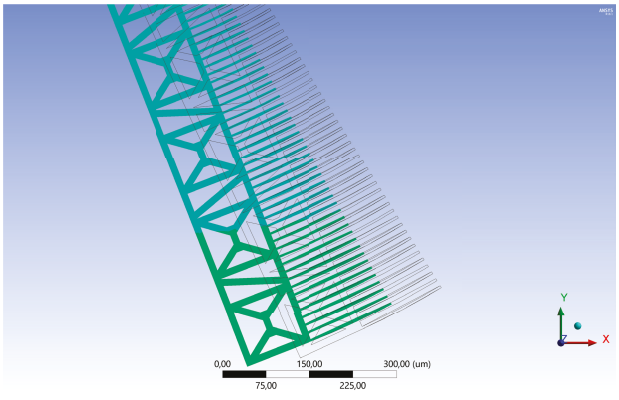

Figure 10. Critical configuration of the comb drive fingers when mode (v) is excited.

Considering that the amplitude of the dumped response to external excitation generally have higher values at the lowest frequencies (see for example Reference [59]), the first six modes only have been considered. Furthermore, given the obtained values of the first six natural frequencies, listed in Table 1, the fifth and the sixth modes will have limited influence with respect to the previous four, because their values is more than twice the fourth frequency. Since fingers contact is expected only at the fifth critical mode, its occurrence likelihood is rather limited.

Table 1. Critical modes for comb drives fingers.

Frequency	kHz	Critical	Plane	Rotation Axes
i	1.51	no	in	4
ii	1.94	no	out	1
iii	2.47	no	out	1
iv	3.00	no	out	2
v	6.41	yes	in	1
vi	6.46	no	out	3

7. Conclusions

The present investigation has shown that the microgripper under analysis it is able to sustain large modifications of the configuration toward both the opening and the closing positions. Considering the size of the embedded flexure hinges, which consist of silicon curved beams with a 5×40 μm^2 cross sectional area, an evident structural robustness has been experimentally observed. In fact, the microgripper has been able to resist to all the actions exerted by the probe. The second part of the investigation has been based on a numerical approach. Firstly, some design charts have been obtained to optimize the likelihood of the contact between the conjugate profiles. Secondly, FEA has been used to detect the main vibrational modes, whose analysis is necessary to prevent fingers contacts. The whole investigation has therefore confirmed that the microgripper under analysis is robust in operational conditions and promising for the applications.

Author Contributions: Conceptualization, all the authors; Methodology, F.B. and M.V.; Software, F.B.; Validation, all the authors; Formal analysis, F.B. and M.V.; Investigation and resources for fabrication, A.B. and P.B.; Data Curation, all the authors; Writing—original draft preparation, Writing—review & Editing & Visualization, M.V. and N.P.B.; Supervision, P.B. and N.P.B.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

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Recursive Least Squares Filtering Algorithms for On-Line Viscoelastic Characterization of Biosamples

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Received: 31 August 2018; Accepted: 15 October 2018; Published: 22 October 2018

Abstract: The mechanical characterization of biological samples is a fundamental issue in biology and related fields, such as tissue and cell mechanics, regenerative medicine and diagnosis of diseases. In this paper, a novel approach for the identification of the stiffness and damping coefficients of biosamples is introduced. According to the proposed method, a MEMS-based microgripper in operational condition is used as a measurement tool. The mechanical model describing the dynamics of the gripper-sample system considers the pseudo-rigid body model for the microgripper, and the Kelvin–Voigt constitutive law of viscoelasticity for the sample. Then, two algorithms based on recursive least square (RLS) methods are implemented for the estimation of the mechanical coefficients, that are the forgetting factor based RLS and the normalised gradient based RLS algorithms. Numerical simulations are performed to verify the effectiveness of the proposed approach. Results confirm the feasibility of the method that enables the ability to perform simultaneously two tasks: sample manipulation and parameters identification.

Keywords: micromanipulation; microgripper; biological samples analysis; visco-elastic characteristic measurement; dynamic parameters estimation

1. Introduction

The mechanical characterization of biomaterials represents a crucial procedure in the fields of tissue engineering, tissue and cell mechanics, disease diagnosis and minimally invasive surgery (MIS) [1–3].

For instance, modeling the mechanical response of brain tissue can be useful in the understanding of traumatic brain injury [4] or blast–induced neurotrauma [5] mechanisms. In tissue engineering, scaffolds should achieve appropriate mechanical properties in order to ensure tissue growth. Therefore, before implantation, it is necessary to verify if these properties meet the specified requirements [6,7]. Many investigations have also been focused on the characterization and on the constitutive modeling of skin [8,9], cornea [10], vocal fold [11], skeletal muscle [12], and blood vessels [1] tissues. Regarding the circulatory system, it is worth mentioning that vascular stiffness is recognized as an important factor in diseases such as pulmonary arterial hypertension, kidney disease, and atherosclerosis [13]. Generally, the variation of tissue stiffness is a hallmark of several disease states, including fibrosis and some types of cancers [14]. Viscosity can also be considered as a biomarker for the metastatic potential of cancer cells [15,16], and the analysis of both elastic and viscous properties could be more effective in detecting and identifying specific diseases [17].

The characterization of the mechanical properties of biomaterials is important not only for in vitro analysis. For example, in MIS operations, tactile information is not available to the surgeon.

Therefore, the capability of on-line detecting and identifying tissues could play a fundamental role in the advancement of MIS procedures [3].

The mechanical properties of tissues have been investigated by using various methods, such as indentation and aspiration, and measuring deformations by means of ultrasound or magnetic resonance imaging techniques [18]. Surgical instruments have also been improved with sensing capabilities [19]. For example, a micro-tactile MEMS-based sensor to be integrated within MIS graspers was presented in Reference [3], whereas a sensor capable of determining the stiffness of biological tissues was modeled and tested in Reference [20].

At the cell level, experimental techniques for investigating mechanical properties include force-application techniques (e.g., micropipette aspiration, atomic force microscope probing, optical trapping), and force-sensing techniques (e.g., traction force microscopy, wrinkling membranes, micropost arrays). The technique selection depends on size of the biological sample, feature to be acquired, detrimental effects on the sample, spatial and force resolution, and accuracy [21,22].

Given the large number of experimental approaches, many mechanical models have been proposed in literature, based on the micro/nanostructural approach and on the continuum approach. The former model targets the sub-cell level, whereas the latter one treats the biological sample as a continuum material. Considering the continuum approach, many constitutive models have been proposed in literature, such as biphasic model, liquid drop models, solid viscoelastic models (Kelvin–Voigt, Maxwell, Zener) [23,24]. Once the mathematical model is available, a parameter estimation problem can be formulated and solved by means of different methods, such as least-squares [25], Kalman filtering [26], or inverse finite element [18,27] methods.

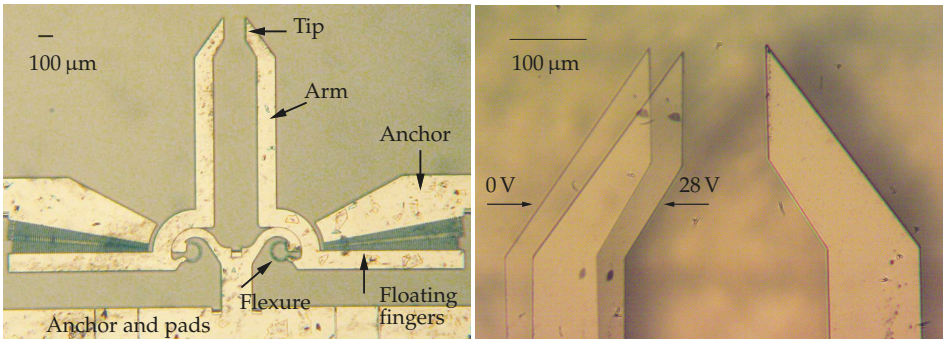
In the last decade, microgrippers became essential tools in the manipulation at the microscale [28–33], and can be adopted to develop novel techniques for the viscoelastic characterization of soft materials. In their previous investigation [34], the Authors proposed a method to evaluate the mechanical properties of a biomaterial sample based on the use of a MEMS microgripper. The mechanical model was developed considering the gripper–sample kinematics and the Kelvin–Voigt constitutive law of viscoelasticity, and a PID controller to determine stiffness and viscous parameters of the sample.

In this work, a parameter estimation problem is formulated starting from the model presented in Reference [34]. Recursive least squares filtering algorithms are implemented to characterize the mechanical properties of soft biomaterials, enabling the *on-line* identification of stiffness and viscosity coefficients. With respect to the other experimental techniques briefly mentioned above, that generally are implemented only for the mechanical testing procedure, the proposed approach enables the ability to perform the mechanical characterization *while* the sample is manipulated for a different purpose in another operation. Furthermore, this method does not require additional elements other than the measurement tool and the specimen as, for example, beads used with optical tweezers. Also, risks related to laser-induced damage and to large deformations of the sample (involved in optical traps and micropipette aspiration, respectively) are avoided.

The paper is organised as follows. The microdevice is introduced in Section 2, whereas the mathematical model is discussed in Section 3. The adopted identification algorithms are described in Section 4, and simulations and numerical results are reported and discussed in Section 5.

2. The Experimental Device

The microsystem considered in this paper consists of two silicon arms actuated by rotary comb drives, as shown in Figure 1. The electrostatic actuators exert the input torques that, through the conjugate surface flexure hinges (CSFH) deflections [35,36], move the arms from the initial position to perform the gripping task [37]. In the neutral configuration, the distance between the jaws is equal to 150 μm. The fabrication method relies on the application of deep reactive-ion etching on silicon-on-insulator wafers [38].



(**a**) (**b**)

Figure 1. Optical microscope images of the device. The whole microgripper (**a**) and two overlapping frames showing the left arm in neutral (0 V) and actuated (28 V) configurations (**b**).

In their previous investigations [34,39], the authors estimated the mechanical characteristic of the sample considering input signals of suitable waveform. According to the proposed method, the sample is gripped with the left arm until the right arm reaches a predefined rotation, that serves as a reference signal for the implemented feedback control scheme. Therefore, the stiffness coefficient can be computed at steady state conditions [39]. In order to estimate also the sample viscosity, a small-amplitude sinusoidal signal was added to the left arm, and the viscous coefficient was obtained as a function of the input torque frequency [34].

In this paper, an estimation algorithm for the elastic and viscosity parameters of the gripped sample under generic control torques is presented. As demonstrated in the next Sections, the algorithm take advantage of the mathematical model structure that, despite its general non linearity, is linear with respect to the unknown parameters.

3. The Mathematical Model

In operative condition, the microdevice is gripping a sample between its jaws, as illustrated in Figure 2. More specifically, Figure 2a shows a schematic drawing of the compliant structure of the gripper. The contact points between jaws and sample are labelled as B and C, whereas the points A and D represent the centers of rotation of the left and right arms, respectively. Starting from the compliant mechanism, a pseudo-rigid body model (PRBM) can be obtained by substituting the constant-curvature CSFHs with revolute joints [35,40–42]. Figure 2b shows the PRBM of the microgripper: the quadrilateral $ABCD$ is a closed chain composed by the links AD (frame), AB and CD (arms), and BC (sample). All the links have a fixed length with the exception of BC, whose compression is at the basis of the measurement procedure. The proposed model takes into account the nonlinearity related to the kinematics of the gripper, but it assumes a linear behavior for stiffness coefficients k_2 and k_4. Moreover, the identification of the cell parameters inherently assumes that its dynamic behavior can be described by the Kelvin–Voight constitutive model of viscoelasticity.

With reference to Figure 2b, the following parameters can be introduced:

- l is the common length of the two links representing the arms, i.e., the distances AB and CD;
- d is the distance between the hinges (AD), i.e., the frame length;
- k_2, k_4 and k are the torsional stiffness of the two arms hinges and the stiffness of the sample, respectively;
- c_2, c_4 and c are the viscous damping coefficients of the two arms and of the sample, respectively;
- I_2 and I_4 are the moments of inertia of the left and right arms around A and D, respectively;
- τ_2 and τ_4 are the input torques generated by the left and right comb drives, respectively.

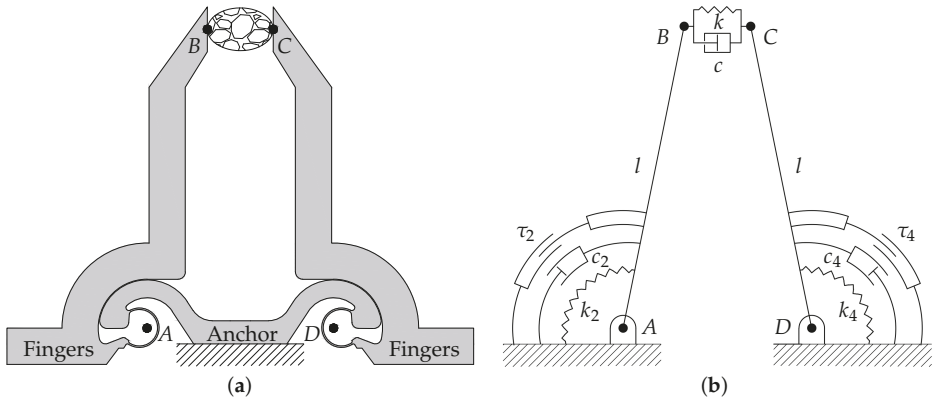


Figure 2. Schematic representations of the microgripper: compliant mechanism (**a**) and corresponding pseudo-rigid body model (**b**).

For the sake of simplicity in the model representation, all the considered variables are referred to the neutral layout: the gripper, in symmetrical configuration, is in contact with the sample but no deformation occurs. By following the notation introduced in Reference [34], and with reference to Figure 3a, the angles $\hat{\theta}_2$, $\hat{\theta}_3$, and $\hat{\theta}_4$ represent the reference orientations of the links AB, BC and DC, respectively, whereas $\hat{u}$ represents the reference length of the link BC. The notation $\tilde{\cdot}$ is used for the parameters in the deformed configuration, as shown in Figure 3b. Therefore, the angles $\theta_i = \tilde{\theta}_i - \hat{\theta}_i$ represent the relative angular displacements of the two links from their neutral configuration, with $i = 2$ for the left link and $i = 4$ for the right one. The orientation of BC follows the same notation, with $i = 3$ and $\hat{\theta}_3 = 0$, whereas the deformation is equal to $u = \tilde{u} - \hat{u}$. The values of the variables in the neutral configuration are given in Table 1.

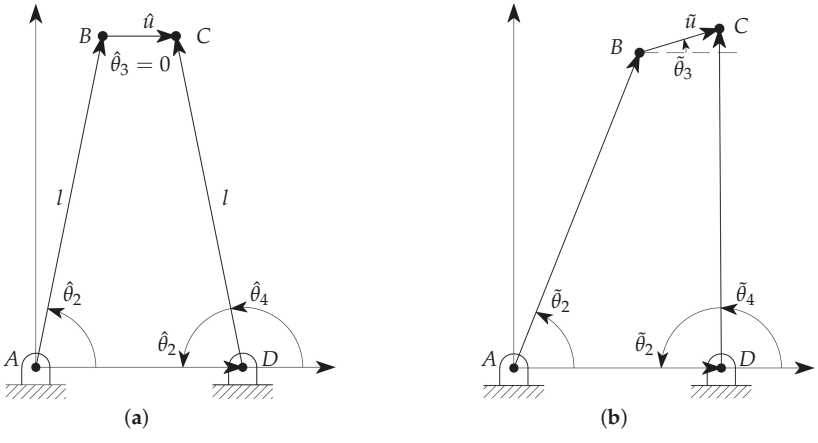


Figure 3. Nomenclature and model parameters in neutral (**a**) and general (**b**) configurations.

Table 1. Constants.

Parameter	Value	Unit
$\hat{\theta}_2$	1.44	rad
$\hat{\theta}_4$	1.70	rad
$\hat{u}$	150×10^{-6}	m

Assuming the inertia of the sample to be negligible, the dynamical model of the links can be written as

$$I_2\ddot{\theta}_2 = -c_2\dot{\theta}_2 - k_2\theta_2 - cl\sin\left(\tilde{\theta}_2 - \theta_3\right)\dot{u} - kl\sin\left(\tilde{\theta}_2 - \theta_3\right)u + \tau_2\,, \tag{1}$$

$$I_4\ddot{\theta}_4 = -c_4\dot{\theta}_4 - k_4\theta_4 + cl\sin\left(\tilde{\theta}_4 - \theta_3\right)\dot{u} + kl\sin\left(\tilde{\theta}_4 - \theta_3\right)u + \tau_4\,, \tag{2}$$

where the subscripts 2 and 4 refer to the left and right joints, respectively. For the device here considered, the values of the parameters appearing in (1) and (2) are reported in Table 2.

Table 2. Numerical values of the parameters.

Parameter	Value	Unit
d	5.47×10^{-4}	m
l	1.50×10^{-3}	m
I_2, I_4	1.25×10^{-14}	kg m^2
k_2, k_4	0.30×10^{-6}	N m^2
c_2, c_4	1.24×10^{-12}	N m^2 s^{-1}

The system has two degrees of freedom and it is fully described by Euqations (1) and (2). The variables θ_3 and u, and their time derivatives, can be computed as functions of the state variables θ_2, θ_4, $\dot{\theta}_2$ and $\dot{\theta}_4$ [34]:

$$\theta_3 = \arctan\frac{-l\sin\tilde{\theta}_2 + l\sin\tilde{\theta}_4}{d - l\cos\tilde{\theta}_2 + l\cos\tilde{\theta}_4}\,, \tag{3}$$

$$u = \sqrt{\left(d - l\cos\tilde{\theta}_2 + l\cos\tilde{\theta}_4\right)^2 + \left(-l\sin\tilde{\theta}_2 + l\sin\tilde{\theta}_4\right)^2} - \hat{u}\,, \tag{4}$$

$$\dot{u} = \dot{\theta}_2 l\sin\left(\tilde{\theta}_2 - \theta_3\right) - \dot{\theta}_4 l\sin\left(\tilde{\theta}_4 - \theta_3\right)\,. \tag{5}$$

4. Mechanical Characteristics Estimation

As briefly described in the Introduction, an approach to determine both the elastic and viscous characteristics of a biosample by using a microgripper was proposed in Reference [34], where particular waveforms were required for the input torques. However, this method is not suitable to perform simultaneous manipulation and measurement tasks.

To overcome this limitation, a measurement scheme adaptable to various operative conditions should be considered. Therefore, an on-line dynamical estimator could be implemented as a more efficient strategy for the mechanical characterization of the sample. Recursive least square methods can be successfully applied when the unknown parameters are the linear coefficients of the system dynamic equations. In such cases, quite usual in literature [43–46], the model can be rearranged as a linear time-varying system with respect to the parameters to be estimated, whereas the other terms, supposed measurable, are functions either of the state or of the output variables.

The model dynamics has to be rearranged as

$$y_i(t) = M_i(t)\omega_i(t)\,, \quad i \in [1, \ldots, m], \tag{6}$$

where m is the number of degrees of freedom of the system ($m = 2$ in the considered case), $\omega_i(t)$ is the vector of unknown parameters, and $y_i(t)$, $M_i(t)$ are known quantities depending on the system dynamics. It is worth noting that all the terms in Equation (6) are time dependent.

Equations (1) and (2) can be rewritten, according to the notation in Equation (6), in linear form with respect to the parameters as

$$y_i(t) = I_i\ddot{\theta}_i + c_i\dot{\theta}_i + k_i\theta_i - \tau_i\,, \tag{7}$$

$$M_i(t) = \mp l\sin(\theta_i + \hat{\theta}_i - \theta_3)\begin{pmatrix}\dot{u} & u\end{pmatrix}, \tag{8}$$

$$\omega_i(t) = \begin{pmatrix}c(t)\\ k(t)\end{pmatrix}, \tag{9}$$

where the sign in Equation (8) is minus or plus if $i = 2$ or if $i = 4$, respectively.

Referring to Equations (7)–(9), the general expressions to be defined for a generic recursive last squares (RLS) filtering algorithm are [47]:

$$\begin{aligned}\hat{\omega}_i(t) &= \hat{\omega}_i(t-1) + K_i(t)\epsilon_i(t)\,,\\ \epsilon_i(t) &= y_i(t) - \hat{y}_i(t)\,,\\ \hat{y}_i(t) &= \phi_i^T(t)\hat{\omega}_i(t-1)\,,\\ K_i(t) &= Q_i(t)\phi_i(t)\,,\end{aligned} \tag{10}$$

where $\hat{\omega}_i(t)$ and $\hat{y}_i(t)$ are the current estimation values of $\omega_i(t)$ and $y_i(t)$, $\epsilon_i(t)$ is the current prediction error, $K_i(t)$ is the gain determining how much the prediction error affects the update in the parameters estimation, and $\phi_i(t)$ represents the gradient of the predicted model output with respect to $\omega_i(t)$.

Two recursive last squares (RLS) filtering algorithms are applied considering different ranges of values for the parameters to be estimated. The algorithms are then compared to each other to understand how much the different viscous and elastic characteristics of the dynamical system affects the convergence of the estimation algorithms. With respect to the formulation in Equation (10), the two approaches differ for the choice of $Q_i(t)$. Moreover, it is worth noting that the symmetric structure of the dynamics represented by Equations (1) and (2) implies that, under a full state measurement, the results obtained choosing $i = 2$ are equal to the results corresponding to the case $i = 4$. Therefore, firstly the case $i = 2$ and consecutively the case $i = 4$ are addressed. Once equivalence and effectiveness of both cases has been proved, the choice in real applications should be driven by the simplicity of the measurement. For example, the operative technique of torque action on the first jaw only implies that for $i = 4$ no torque measurement is required, strongly simplifying the implementation.

4.1. Forgetting Factor Based RLS

The first method is based on a forgetting factor based RLS algorithm. In Equation (10), the following choice is performed

$$Q_i(t) = \frac{P_i(t-1)}{\lambda + \phi_i{}^T(t)P_i(t-1)\phi_i(t)}\,, \tag{11}$$

where

$$P_i(t) = \frac{1}{\lambda}\left(P_i(t-1) - R_i(t)\right) \tag{12}$$

and

$$R_i(t) = \frac{P_i(t-1)\phi_i(t)\phi_i^T(t)P_i(t-1)}{\lambda + \phi_i{}^T(t)P_i(t-1)\phi_i(t)}\,. \tag{13}$$

It's assumed that the residual $\epsilon_i(t)$ (the difference between the estimated and the measured value of $y_i(t)$) is affected by a white noise with covariance equal to 1. According to previous equations, $\hat{\omega}_i(t)$ is computed in order to minimize the sum of residuals squares

$$\hat{\omega}_i(t) = \arg\min_{\theta} \sum_{k=1}^{t} \lambda^{t-k} \epsilon_2^i(k)\,. \tag{14}$$

In Equations (11)–(14), $\lambda \in R$ is the forgetting factor introduced in order to consider differently the time sequence of the errors $\epsilon_i(t)$, according to an exponentially decreasing weight if $\lambda \in (0,1)$. This choice is effective in case of time varying parameters while, when dealing with constant parameters, the choice $\lambda = 1$ is usually adopted.

The algorithm in Equation (10), with positions (11)–(13), has been applied to the case $i = 2$. In all the simulations, the initial values of the parameters have been chosen far from the real values. The initial covariance, proportional to P_2, has been fixed taking into account that the covariance matrix has to be chosen according to a priori knowledge of the parameters at $t = 0$: very high values of the covariance matrix elements correspond to completely unknown parameters.

Remark 1. *Note that the forgetting factor method is a particular simplified case of the Kalman filter.*

4.2. Normalised Gradient Based RLS

The second approach consists of a normalized gradient algorithm. This technique is based on the choice of $Q_i(t)$ with a simpler form with respect to (11):

$$Q_i(t) = \frac{\gamma}{|\phi_i(t)|^2 + \beta}\,, \tag{15}$$

where γ is the adaptation gain scaled by the gradient $\phi_i(t)$. The bias term β is added to the square norm of the gradient vector in the denominator of the expression, in order to prevent critical situations in the case $\phi_i(t)$ is close to zero. The algorithm (10) with $Q_i(t)$ defined by (15) requires only the initialization for the values of the parameters to be estimated.

Since the presence of noise is not explicitly considered in this formulation, the drawback is a smaller rate of convergence and a larger sensitivity to the presence of noises.

This technique has been simulated making reference to the dynamics represented by Equation (2), that is with $i = 4$.

5. Simulations

Numerical simulations, using Matlab® (MathWorks, Inc., Natick, MA, USA) and Simulink® (MathWorks, Inc., Natick, MA, USA) tools, were performed in order to show effectiveness, benefits and differences of the proposed estimation methods. Three different numerical cases were analysed, considering predominant elastic or dissipative behaviours.

The first case corresponds to a realistic condition with elastic and damping coefficient much greater than the ones of the mechanical structure, with $c = 8.4 \times 10^{-6}$ Nms/rad and $k = 2.5 \times 10^{-3}$ Nm/rad, where the elastic coefficient greater than the damping one.

In the second case, a damping coefficient greater than the elastic one was considered in order to check, by comparison, the dependency of the algorithm convergence from the two different mechanical characteristics. The order of magnitude for the two coefficients have been exchanged, setting $c = 8.4 \times 10^{-3}$ Nms/rad and $k = 2.5 \times 10^{-6}$ Nm/rad.

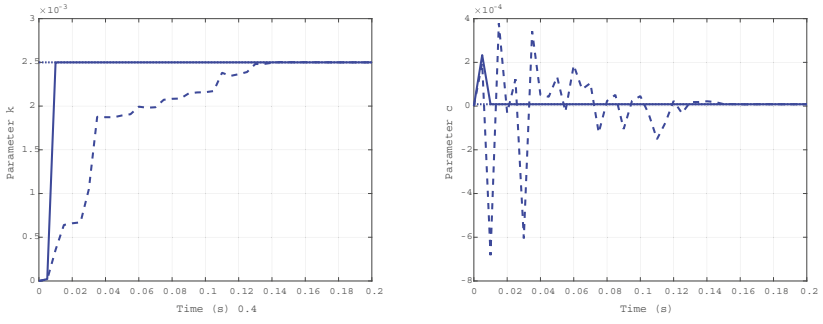
In the last case, a very poorly damped sample has been chosen with $c = 8.4 \times 10^{-11}$ Nms/rad, whereas $k = 2.5 \times 10^{-5}$ Nm/rad.

The initial parameters values, for all the simulations and for both the estimation algorithms have been chosen as $c(0) = 10^{-9}$ Nms/rad and $k(0) = 10^{-7}$ Nm/rad.

For the forgetting factor RLS estimator introduced in Subsection 4.1, the 2×2 square covariance matrix is diagonal, with both the diagonal elements equal to 10^{20}, while the forgetting factor λ is fixed to $\lambda = 0.99$.

For the normalized gradient estimator, presented in Subsection 4.2, the adaptation gain γ has been set as $\gamma = 0.9$ and the normalization bias has been chosen as $\beta = 2.2 \times 10^{-16}$.

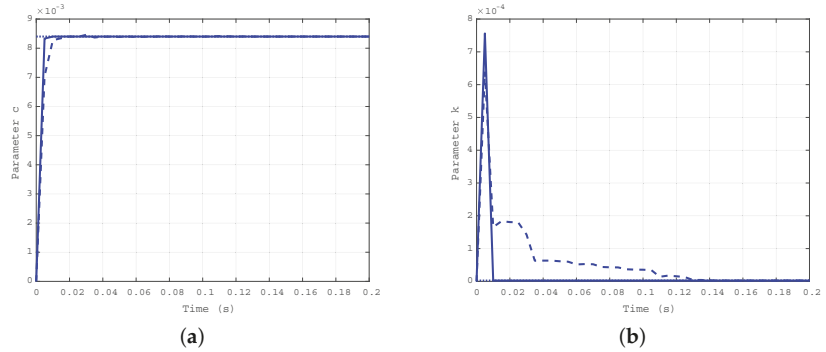
Simulation results obtained for the first case ($c = 8.4 \times 10^{-6}$ Nms/rad and $k = 2.5 \times 10^{-3}$ Nm/rad) are depicted in Figure 4a for the elastic coefficient k and in Figure 4b for the damping coefficient c. The solid line shows the estimation evolution with the first algorithm, whereas the dashed one reports the results of the second algorithm. The dotted line corresponds to the true values of the parameters, plotted as a reference. As expected, both the algorithms converge in a very short time, but the first is faster than the second.



(**a**) (**b**)

Figure 4. Time evolution of the estimated parameters k (**a**) and c (**b**) for the first case: $c = 8.4 \times 10^{-6}$ Nms/rad, $k = 2.5 \times 10^{-3}$ Nm/rad.

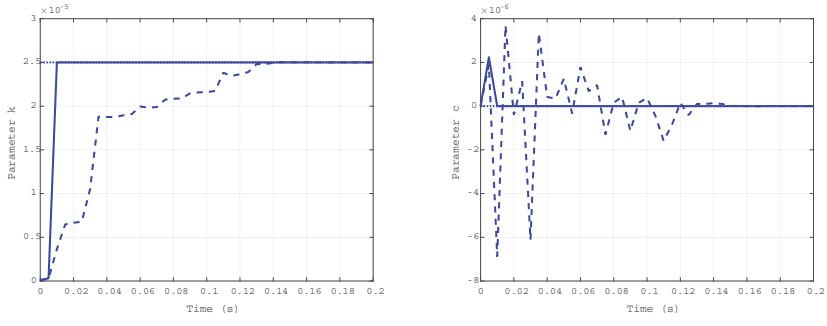
For the second case ($c = 8.4 \times 10^{-3}$ Nms/rad and $k = 2.5 \times 10^{-6}$ Nm/rad), the simulation results are depicted in Figure 5a for the damping coefficient c, and in Figure 5b for the elastic one k. As in the previous case, the solid line refers to the estimation evolution with the first algorithm, the dashed one refers to the second algorithm, and the dotted one is the true reference value. The difference in the convergence rate for the two approaches is confirmed.



(**a**) (**b**)

Figure 5. Time evolution of the estimated parameters c (**a**) and k (**b**) for the second case: $c = 8.4 \times 10^{-3}$ Nms/rad, $k = 2.5 \times 10^{-6}$ Nm/rad.

The results obtained by simulation of the third case ($c = 8.4 \times 10^{-11}$ Nms/rad and $k = 2.5 \times 10^{-5}$ Nm/rad) are reported in Figure 6a for the elastic coefficient k, and in Figure 6b for the damping one c. The difference in the convergence rate for the two approaches is confirmed, and uniformity in the estimation of the damping coefficient c with the two algorithms is the same as in the first case.



(**a**) (**b**)

Figure 6. Time evolution of the estimated parameters k (**a**) and c (**b**) for the third case: $c = 8.4 \times 10^{-11}$ Nms/rad, $k = 2.5 \times 10^{-5}$ Nm/rad.

Numerical results support the effectiveness of the proposed method, tested considering different orders of magnitude for the parameters to be estimated. Clearly, in the present formulation, the state is assumed to be measurable, with possible additive Gaussian noise to model uncertainties and measurement noise.

6. Conclusions

In this paper, the possibility of using a microgripper for the identification of the mechanical properties of biomaterials was investigated. To describe the gripper–sample system dynamics, the pseudo-rigid body model and the Kelvin–Voigt constitutive law of viscoelasticity were considered for the microgripper and for the sample, respectively. The normalised gradient-based RLS and the forgetting factor based RLS algorithms were implemented to solve the parameters estimation problem. Three cases were analysed, assuming different ranges of values for the coefficients to be estimated. The simulation results confirmed the feasibility of the method: both the algorithms converged in a very short time, but the normalised gradient-based RLS algorithm resulted to be faster than the other one. Therefore, the proposed approach could enable the ability to simultaneously perform the manipulation of the biosample and the identification of its mechanical characteristics, i.e., stiffness and damping coefficients. However, the application of the proposed method to a variety of biological samples is limited by the assumption of the Kelvin–Voigt constitutive law of viscoelasticity. From the methodological point of view, further investigations will focus on the implementation of more complex constitutive models and on the design of high-performance algorithms, always attaining robustness with respect to noise and parameter uncertainties.

Author Contributions: P.D.G. and M.L.A. implemented the estimation algorithms and performed the simulations. O.G. and M.V. developed the model adopted for the kinematic and dynamic analysis of the system. All authors contributed in writing the manuscript.

Funding: This research received no external funding.

Acknowledgments: The Authors acknowledge the Micro Nano Facility, Center for Materials and Microsystems of Fondazione Bruno Kesler (FBK-CMM-MNF) for the fabrication of the device and the technical support (Figure 1).

Conflicts of Interest: The authors declare no conflict of interest.

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Preparing and Mounting Polymer Nanofibers onto Microscale Test Platforms

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Received: 7 September 2018; Accepted: 10 October 2018; Published: 15 October 2018

Abstract: Because they can achieve a high degree of molecular chain alignment in comparison with their bulk counterparts, the mechanical and thermal properties of polymer nanofibers are of great interest. However, due to their nanometer-scale size, it is difficult to manipulate, grip, and test these fibers. Here, we demonstrate simple repeatable methods to transfer as-drawn fibers to micrometer-scale test platforms where their properties can be directly measured. Issues encountered and methods to minimize measurement artifacts are also discussed.

Keywords: polymer nanofiber; manipulation; precision mounting; UHMWPE

1. Introduction

Polymer nanofibers have mechanical and thermal properties significantly higher than their bulk counterparts [1–6]. They have a wide range of potential applications in areas such as tissue engineering, sensing, reinforcing composites, ballistic armor, textiles, and in improving thermal management [7–11]. Detailed study of their material properties is necessary to assess their true application possibilities.

However, due to their nanometer-scale size, it is difficult to manipulate and mount individual polymer nanofibers onto a test platform to make measurements. The macroscale approach of manipulating objects with pick-and-place tools has been scaled to smaller sizes using micro- and nanoscale grippers [12–14]. A tweezer-like gripping mechanism in these grippers can be obtained using various actuation mechanisms such as electrostatic, thermal, shape memory, fluidic, magnetic, and piezoelectric [14]. But it is difficult to ensure reliable placement of a long nanofiber onto a horizontal platform using a nanogripper. Optical tweezing is capable of sorting and manipulating an individual DNA molecule. This technique requires a liquid medium, which is not compatible with mechanical measurement platforms operating in laboratory air [15]. Another common approach is to use an atomic force microscope (AFM) cantilever tip or a tungsten tip mounted on a precision XYZ manipulator. This has been successfully used for nanoscale fabrication of polymer nanofibers [16] and precision mounting of nanofibers [1,17]. This approach is low yield and time-consuming. In addition, the process is generally performed in a scanning electron microscope (SEM) [17] because nanofibers are difficult to resolve optically. However, polymer nanofibers are damaged by electron-beam irradiation [18]. Hence, reliable optical methods are strongly needed for manipulation and testing.

In view of the urgent need for such micromanipulation, this article describes and demonstrates a simple specimen collector technique that enables nanofiber placement with ± 1 μm precision onto micromachined test platforms using optical methods. Electrospinning is one common method to make such long fibers [9,19,20]. Here, as an alternative example, we describe a method by which ultra-high molecular weight polyethylene (UHMWPE) nanofibers can be made. Then we show how they can be manipulated onto micromachined test platforms without damage. Next, we demonstrate a nanomechanical test calibration procedure, discuss potential testing artifacts in nanomechanical and

nano thermal measurements, and propose methods to solve them. Detailed characterization, thermal and nanomechanical measurements on highly crystalline nanofibers prepared by this method have been reported in reference [3].

2. Experimental

2.1. Ultra-High Molecular Weight Polyethylene (UHMWPE) Nanofiber Specimen Preparation

Ultra-high molecular weight polyethylene (UHMWPE), a linear chain polymer, is used in this work. However, the processes described here can be generalized to other drawable polymers. To fabricate the nanofibers, we first produced a gel from 0.8 wt % UHMWPE powder (average molecular weight ~3×10^6–6×10^6 g/mol purchased from Sigma Aldrich, St. Louis, MO, USA) and mixed it with decalin solvent [21]. The procedure was carried out inside an argon-filled glove box to avoid oxidation and subsequent molecular degradation. After heating on a hot plate to 145 °C, the mixture became transparent and viscous as the UHMWPE powder dissolved in the solvent while a glass rod constantly stirred the solution. The solution was then quenched in a room temperature water bath, and the gel formed.

Next, we devised a two-stage tip drawing method. First a μm-scale UHMWPE fiber (MF) was drawn, as schematically represented in Figure 1. There, a 5 mm × 5 mm silicon chip with a thin film heater attached on the backside heated the gel to 120–130 °C. The temperature was measured using a thermocouple attached to the thin film heater. A hot plate placed 1 cm below the silicon chip heated the overhead air to 90 °C to prepare for hot-stretching. The translucent gel transformed to a transparent solution as it reached 130 °C. As the solution became clear, a sharp (~10 μm) glass tip was inserted into the solution using an XYZ-axis micromanipulator. It drew a MF of short length (several hundreds of μm). The drawing induces stress-induced crystallization with folded chain lamellae (kebab) on oriented chains (sheesh). Upon drawing, the lamellar structure unfolds into microfibrils that are pulled taut between entanglements [4]. During crystallization, the decalin evaporates, aided by the convective current from the hot plate. Decalin syneresis further facilitates crystallization. The fiber was then further drawn to a length of 1 cm, after which it was quenched to room temperature to minimize relaxation of the extended chain.

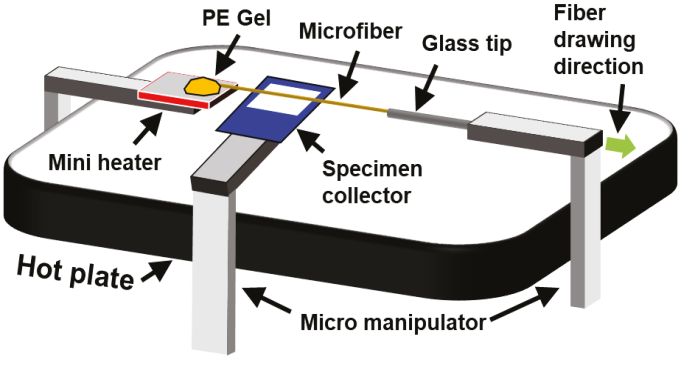

Figure 1. Schematic of tip drawing setup.

To manipulate and further process the microfiber, a bulk-micromachined specimen collector was fabricated by bulk micromachining processes, as detailed in Appendix A. It is a 1 cm × 3 cm silicon chip of 525 μm thickness and a square hole of 7.5 mm × 7.5 mm. Again using a standard XYZ-axis micromanipulator, the specimen collector was raised from under the drawn MF, as depicted in Figure 1, and the MF was gripped onto the specimen collector using epoxy. Because it was stretched when the

specimen collector was raised and glued, the MF remained taut with a moderate tensile stress. The MF was then cut outside the collector edges.

With the specimen collector acting as a carrier, the MF could now be conveniently transferred from the drawing station and positioned under an optical microscope objective. There, a locally etched and bent tungsten wire, used as a microheater, was placed in the same horizontal plane 20 μm from the MF. Details of the microheater construction are given in Appendix B.

A voltage was applied to the microheater, the tungsten wire temperature increased due to Joule heating, and the collected MF was heated during a ~1 s pulse by convection to make a UHMWPE *nanofiber* (NF), as schematically shown in Figure 2. Most advanced thermometers have either temporal or spatial resolution limitations that limit direct temperature measurement during this process. We estimated the maximum temperature attained by the fiber during local heating using the ANSYS/FLUENT computational fluid dynamics finite element analysis software package [22]. First, the microheater temperature profile was calculated. Second, we assumed a MF with a typical 3 μm diameter and placed at a distance of 20 μm from the fiber. A potential bias 0.7 V was applied to the heater, as in the experiment. Both convective and radiative heat losses were considered. At the microscale, the boundary layer is smaller than at the macroscale. Thus, the heat transfer coefficient (h) greatly increases. Here, we used h = 1000 W m^{-2} K^{-1} and emissivity = 0.1 [23]. Figure 3a shows the simulated temperature MF profile. The thermal conductivity of the MF was modeled using gel spun microfibers with k = 20 W m^{-1} K^{-1} [24]. We used an emissivity value of the MF of 0.2 [25]. A constant thermal conductivity of air, k = 0.02 W m^{-1} K^{-1}, was used. The MF temperature reached 450 K (180 °C), and the temperature profile along the fiber is shown in Figure 3b.

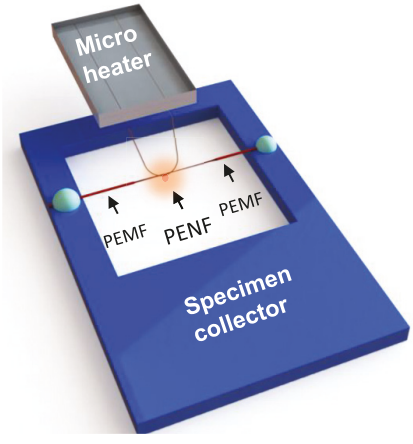

Figure 2. Schematic of the specimen collector and the fabrication of an ultra-high molecular weight polyethylene (UHMWPE) nanofiber (NF). Adapted from reference [3] supplementary information.

During this short temperature cycle, the MF temperature likely approached the melting temperature, and stretched rapidly due to the residual tension mentioned above. As the NF developed, its molecular alignment and crystallization was further increased. This highly aligned structure was then quenched as the NF rapidly cooled after the heater was turned off. The NF, now suspended between MF regions outside the heater area, was itself typically no longer optically observable, but the outer MF regions resided in the same location as before the second drawing stage. This indicated that a NF now had been created and still physically connected the MF regions, as represented in Figure 2.

The geometrical dimensions of NFs created by this fabrication sequence were characterized by scanning electron microscopy (SEM). We found that factors influencing the diameter included the initial MF diameter, the pre-existing tensile stress, and the heater current. The diameters ranged from

20–250 nm and a length of 100 µm. Detailed specimen geometry information is provided in reference [3]. Further characterization data on twinning as determined from transmission electron microscopy and on molecular alignment as inferred from Raman spectroscopy has also been reported [3].

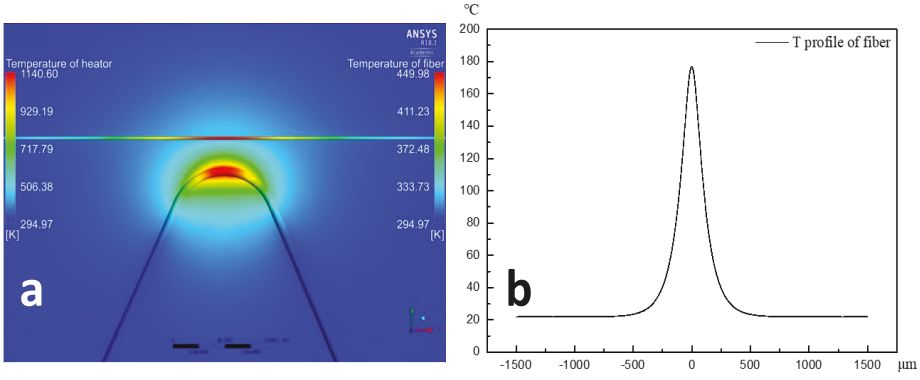

Figure 3. Temperature profile of the W wire heater and the polyethylene microfiber (MF) from an ANSYS/FLUENT simulation. (**a**) Temperature map of the heater and the MF. (**b**) Plot of the MF temperature profile. The maximum temperature reaches 450 K (180 °C). Adapted from reference [3] supplementary information.

2.2. Positioning the Nanofibers on the Test Platforms

2.2.1. Mechanical Test Platform

A micromachined stepper motor, detailed in [26], was selected for the mechanical measurements. The test platform was designed at Carnegie Mellon University and microfabricated by Sandia National Labs, Albuquerque, NM, USA using the SUMMiT V process [27]. The polycrystalline silicon (polysilicon) motor was "released" in the Carnegie Mellon University nanofabrication facility. The release process involves removing silicon oxide layers in hydrofluoric acid. Then a critical point drying process [28] rendered the structure freestanding. The full platform included a trailing load cell and grips, as seen on the left of Figure 4. The motor was actuated electrostatically, and each step was approximately 60 nm. Here the steps were taken at 1 Hz. It can generate an in-plane force of up to 1 mN, substantially more than the ~100 µN needed to stretch and break the NFs. Subpixel interpolation with approximately 5 nm resolution was used to measure displacements, as recorded through a 50× objective in an optical microscope. Because the load spring and the NF specimen are in series, the displacement in each depends on their relative stiffnesses. Here, the load spring stiffness was on the same order as the NFs. The tensile load was estimated by measuring the separation between the load cell combs, while the specimen extension was determined from the comb attached to the moving grip.

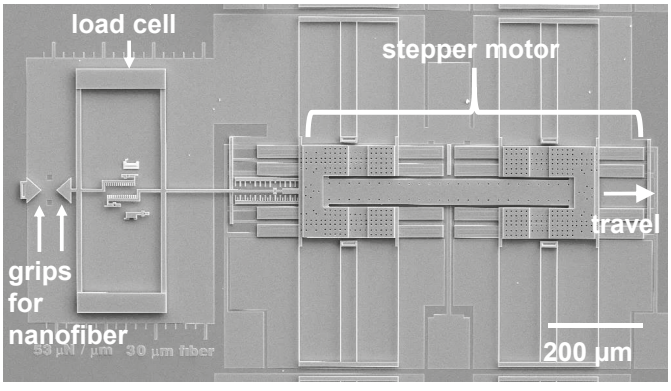

Figure 4. The micromachined test platform used for mechanical measurements.

The minuscule sample size, the limited resolution of the optical microscope and nonlinear forces such as van der Waals, triboelectric, and capillary forces due to moisture are factors that make it difficult to successfully place nanofibers with tip-based methods. Yet, it is critical to locate and position the nanofiber precisely onto the micromachined test platform without damaging it. Electron microscopes cannot be used because of the sensitivity of polymer samples to electron beam radiation [29,30], which scissions bonds and hence degrade the mechanical and thermal property enhancement [31,32]. The manipulation by optical methods is preferred. Regarding alignment, during tensile testing, an unwanted bending moment can cause the load cell to rotate and the fiber force will be underestimated.

The specimen collector with a suspended NF was again used as a specimen carrier, now to align the NF with the mechanical test apparatus. It was maneuvered above the grips, as shown in Figure 5. The suspended MF in the undrawn region was used as a reference to align the nanofiber. Two triangular pads, which must grip the specimen, are behind the load spring, as seen in Figure 4. One pad is fixed to the substrate while the other is connected to the load cell. The first step in mounting the nanofiber involves pre-gluing. A small volume of cyanoacrylate glue (~20 μm × 5 μm × 5 μm) was transferred to the load cell beam that is connected to the specimen displacement-indicating comb using a compliant tungsten tip. This was maneuvered into position with a precise motorized micromanipulator. This glue held the specimen in place while the fiber was aligned and cut. The specimen collector was held by a second micromanipulator and was aligned along triangular pads under a 50× optical objective. This was done quickly (<5 min), as the glue cured in 20 min. The specimen collector was then lowered until the fiber made contact with the glue. It was fixed at that position until the glue cured enough to hold the fiber while cutting. Figure 6 shows a pre-glued PE nanofiber aligned to the grips. In this case, as in other images of fibers presented in this article, the fiber diameter is ≈250 nm so that it is optically detectable. Fibers with diameters <100 nm were also successfully mounted. Though they could not be observed optically, their presence was confirmed during mechanical or thermal testing.

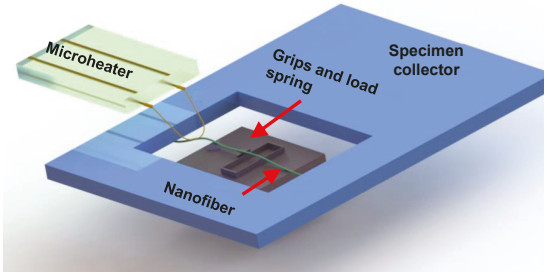

Figure 5. Schematic of alignment of the NF along the platform grips (figure not to scale). Adapted from reference [3] supplementary information.

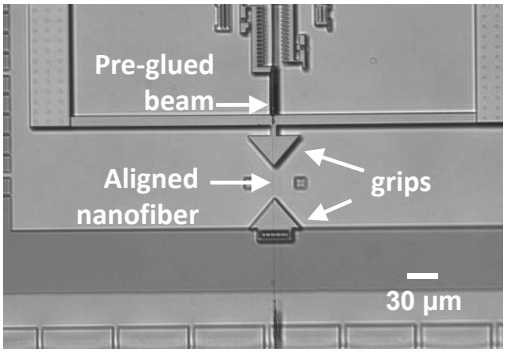

Figure 6. Optical image showing the pre-glued beam and an aligned nanofiber.

Next, the fibers were glued to the grips. Using a sharp compliant tungsten tip, glue was gently placed onto the aligned fiber over the pads. If the fiber came out of alignment, it could be nudged back into position using the same tungsten tip. As the glue became more viscous, manipulating it became more difficult and the success rate diminished. Since the working time of the glue was ~8 min, this process was completed within 6–8 min. Figure 7 shows a sample that has successfully been gripped at both ends.

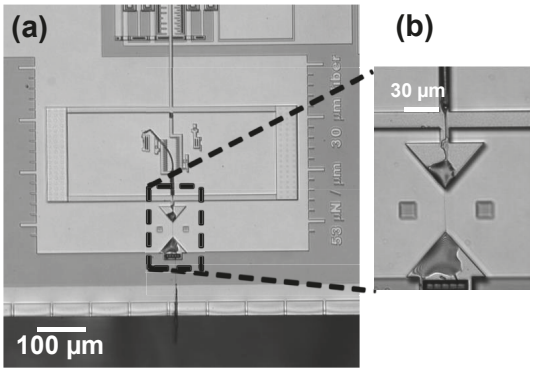

Figure 7. Optical image of UHWMPE nanofiber tor ready for mechanical testing. (**a**) The fiber is glued to both grips pads after the cutting process (**b**) Magnified image of the pads after gluing.

After 15 min, the glue had partially cured. The microheater then cut the fiber to a length of 200–300 μm by Joule heating, much longer than the 30 μm gauge length. A DC power of 8–15 mW cut the fibers. As the heater made contact with the fiber, it melted it locally, thereby cutting it. Due to the tensile stress remaining from the specimen collector, the fiber was pulled towards the pad by a few μm. Similarly, the other end of the fiber was cut, thereby keeping the NF within the grips, with some extra length extending beyond the glue.

After a fiber had been glued, it was cured in laboratory air for 48 h, and was then ready for testing. Figure 8 shows a SEM image of a gripped nanofiber in the mechanical platform.

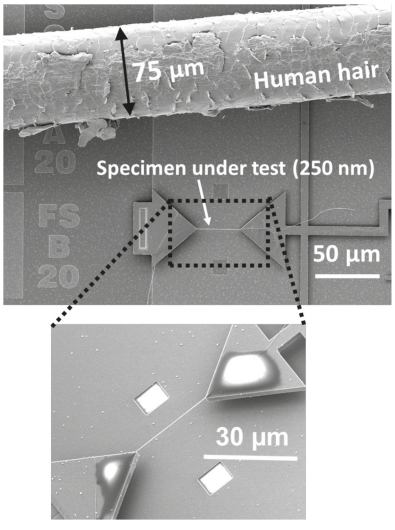

Figure 8. SEM image of a 250 nm diameter NF in comparison with human hair. Inset: detailed view of gripped nanofiber.

2.2.2. Thermal Test Platform

A different microdevice, consisting of suspended platinum resistance thermometers (PRTs) was used to conduct thermal characterization [33]. A total of seven SiN_x beams, with platinum deposited on each, supports each of two suspended SiN_x plates. Serpentine platinum coils are fabricated on top of the SiN_x plates and are connected to electrical pads in a supported region. A lock-in amplifier detects small changes in the resistance of these PRTs. Consequently, the device is sensitive to minute temperature changes. A NF connecting one platform to the adjacent one conducts heat from the heated side to the sensing side. Details of the measurement system are given in reference [33].

The alignment of the nanofiber to the plates was similar to that discussed above for the mechanical device. However, it was found that epoxy glues wetted the Pt wire surface, creating an insulating layer that significantly increased the thermal contact resistance, R_c. Without glue, the cutting process tended to pull the nanofiber off the suspended plates. Focused ion beam (FIB)-deposited Pt would likewise reduce the crystallinity and significantly increase the thermal resistance.

To solve this problem, we took advantage of the surface tension from evaporating isopropanol (IPA), which was able to affix the fiber onto the substrate. As the liquid evaporated, it pulled the fiber into contact with the plates and created sufficient adhesion to keep the nanofiber in place during the cutting process. This adhesion process was termed "capillary-assisted adhesion" and is discussed next.

Steps to achieve capillary-assisted adhesion are schematically shown in Figure 9. In this process, an IPA liquid drop was first placed on top of the thermal device (Figure 9b). Evaporation of IPA from

the device was closely monitored under the optical microscope. After the liquid meniscus connecting the suspended plates broke, the suspending beams (not shown) cause the plates to separate to their nominal 5–10 μm gap. Before the liquid layer evaporated from the surface of the plates, a pre-aligned nanofiber was expeditiously lowered into focus and contact was made with the pads (Figure 9c). The liquid wet the fiber and the evaporating liquid surface pulled the fiber down due to surface tension. The fiber then adhered to the pads (Figure 9d). This surface tension, accompanied by the van der Waals force, increased the contact width of the fiber on the pads. Also, the fiber conformed to the undulating surface, increasing the contact length significantly. Increased contact width and contact length due to capillary-assisted adhesion reduced the thermal contact resistance. Figure 10 shows an optical image of a mounted nanofiber on the device. NFs that are of a significantly smaller diameter and not resolvable under the optical microscope could be manipulated and mounted onto the test platform.

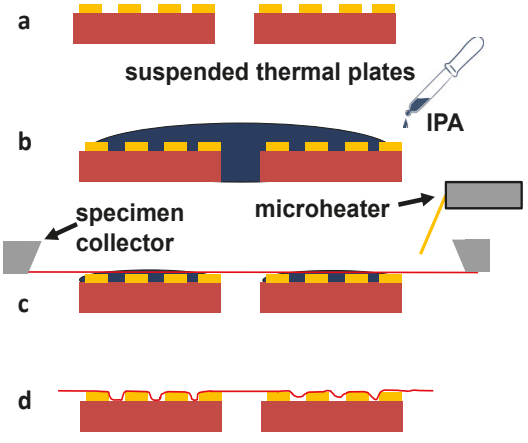

Figure 9. Schematic diagrams representing nanofiber placement onto a platinum resistance thermometer plates. (**a**) suspended plates, (**b**) a small drop of isopropanol (IPA) is placed on the platforms (**c**) after the IPA evaporates, the microheater cuts the fibers, (**d**) The evaporating IPA pulls the fiber into contact with the plate.

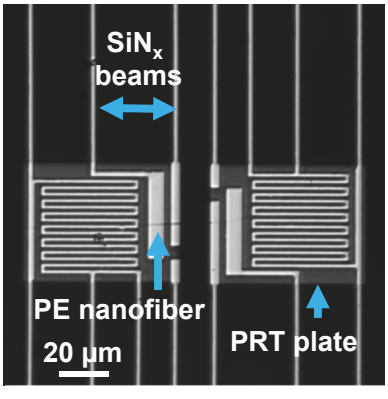

Figure 10. Optical image of a UHWMPE nanofiber sample mounted on the thermal device.

3. Results and Discussion

3.1. Calibration of Mechanical Test Platform Using Silica Fibers

Silica nanofibers were used to validate the nanomechanical measurements reported in [3]. They were prepared using a flame brushing technique [34], where a silica microwire was heated in the center using a butane torch and pulled at other two ends. It was stretched until broken; nanofibers were then obtained. The nanofiber was placed on a PDMS substrate, and in this case, conventional manipulation methods were used to mount the fiber. A tungsten tip with a tip of 1 μm diameter connected to a high precision micro-manipulator was used to manipulate the silica nanofiber sample from the PDMS to the mechanical test platform. Cyanoacrylate glue was used to attach the silica nanofiber to the grips.

Stress and strain were obtained from the force normalized over the cross-section diameter and elongation over the gauge length. Figure 11 shows the stress–strain curve of a silica nanofiber of diameter 108 nm. The Young's modulus was obtained from the slope of the stress–strain curve in the elastic region. For that fiber, it was found to be 67.7 GPa. The strength was 4.5 GPa and its elongation at failure was 9%. A second silica nanofiber was also tested with diameter D = 90 nm (Figure 11). The Young's modulus value was 76.5 GPa, while the ultimate strength was 8.2 GPa, twice that of the previous sample. Elongation at failure was also observed to be higher than the previous sample at 13.8%.

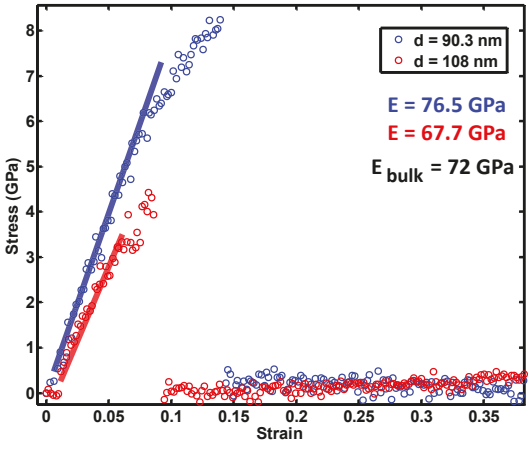

Figure 11. Stress–strain curves for the silica nanofibers.

The Young's modulus values are similar to that of bulk silica (72 GPa) [35]. Silva et al. tested silica nanofibers using an AFM cantilever bending test and found that silica nanofiber Young's modulus for diameters larger than 280 nm was similar to bulk [35]. Wang et al. [36] obtained 27 ± 7 GPa for diameters between 43 nm and 95 nm and Dickin et al. [37] obtained 47 ± 7 GPa for diameters between 80 nm and 98 nm. These measurements were done in the TEM using a resonance method. Recent work [38] on a tensile test of silica nanofibers of dimensions 33.9 nm and 18.7 nm under TEM found that Young's modulus was about 77.6 GPa, close to bulk silica. The values in Figure 11 are in good agreement with the bulk value and with much of the literature.

The strengths are much higher than bulk silica (0.2 GPa) [39]. This is because of nanoscale size effect, where flaw density is significantly reduced and the probability of having a flaw larger than critical flaw is minimal. However, our values of silica nanofiber strength are somewhat lower than that of reference [39], where an average strength of 10 GPa was measured. Our fabrication method of using butane torch and strength test in high relative humidity (~70%) introduces hydroxyl groups in the silica surface that may degrade the silica sample [39,40].

We observed elongation as high as 13.8% in the silica nanofibers. However, SEM imaging of the fractured glass fiber shows a flat fracture surface, consistent with brittle failure. It is likely that the glue slipped to a small degree as failure was approached, as the curve deviated from linearity. However, this only affects the penultimate portion of the stress–strain curve. From this data, it is apparent that the test platform can measure strength and strain if the specimen is firmly gripped.

3.2. Gripping the UHMWPE Nanofibers in Mechanical Testing

For UHMWPE nanofibers, initial test results revealed more severe slip at the grips before their strength was reached. The surface energy of PE is low (31 mJ m^{-2}) and the absence of a polar component (similar to PTFE, 19 mJ m^{-2}) makes it difficult to wet with adhesives [41,42]. There are many commercial including Loctite 3032, Loctite 3035, Scotch weld DP 8005, Scotch weld DP 8010, Loctite plastic bonding system, and TAP poly-weld. They are designed for low surface energy olefins such as polyethylene and polypropylene. However, their viscosity, working life, and/or requirement to surface treat the sample made them unsuitable for this work. All available glues were tried. All slipped to some degree.

Six ultra-drawn NFs with diameters ranging from 85 nm to 255 nm were tested with Loctite® AA 3032/Henkel, which is a specialized glue for PE and was found to be the best glue for this work. From composites theory [43], the maximum force (F_{max}) applied to a specimen of diameter D before it slips can be used to obtain the average interface shear strength, τ_{max}. Accordingly,

$$\tau_{max} = \frac{F_{max}}{\pi D l_{glue}} \tag{1a}$$

which can be rewritten as

$$\frac{F_{max}}{\pi l_{glue}} = \tau_{max} D \tag{1b}$$

Here, l_{glue} is the length of the nanofiber/glue interface. In Figure 12a, $F_{max}/\pi l_{glue}$ is plotted against D. The slope of specimens that slipped before failure, as indicated by the red circles, is 4.7 MPa. This corresponds to τ_{max}. Indeed, video images taken during the testing revealed the slip directly. The τ_{max} value compares reasonably well with the shear strength of Loctite® AA 3032/Henkel, (6 MPa with high density PE). On the other hand, specimens tested until failure (blue circles in Figure 12a) displayed a significantly higher force per unit length. The additional gripping force was contributed by a mechanical locking procedure, as described in [3]. Figure 12b,c shows the schematic of fiber within the glue.

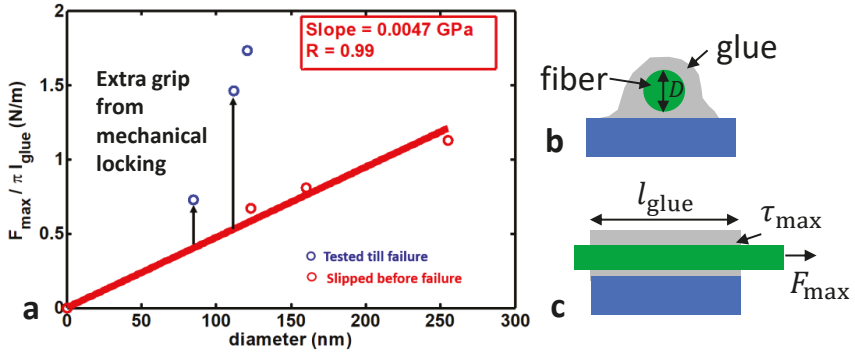

Figure 12. Estimate of glue shear strength. (**a**) Shear force per unit length versus fiber diameter for Loctite® AA 3032/Henkel glue/fiber interface. The slope gives the shear strength of the glue/nanofiber interface. (**b**,**c**) are schematic cross-section and longitudinal views of the fiber in the glue. Adapted from reference [3] supplementary information.

From this data, we can estimate the length of the glued/fiber interface required to grip a typical 100 nm NF at a uniaxial stress of 10 GPa, on par with the tensile strength, σ_{ts}. Then, $l_{glue} = \sigma_{ts} D / 4\tau_{max}$ $\approx$ 53 μm. The gripper length was 30 μm. With greater lengths, it should be possible to test UHMWPE nanofiber specimens to failure without the mechanical locking procedure, and thereby also assess Young's modulus.

3.3. Thermal Contact Resistance

Thermal contact resistance R_c between the sample and the islands will result in an incorrect thermal conductivity value if not correctly assessed. Furthermore, if R_c is much larger than the intrinsic thermal resistance of nanofiber, R_s, the procedure is subject to large errors. The total thermal resistance at steady state is the sum of thermal contact resistances, $2R_c$, and R_s. Then,

$$R_{tot} = R_s + 2R_c \tag{2}$$

For high thermal conductivity samples, such as UHMWPE nanofibers, R_s can be increased by making the nanofiber long and the cross-section area small. However, this will also reduce the heat flux significantly. Consequently, the temperature rise of sensing side will be lessened, making accurate measurement difficult. A platinum or graphite coating, using electron beam or focused ion beam (FIB), has been extensively used in literature to reduce thermal contact resistance [44,45]. However, high-energy electron/ion beam amorphized the NFs and reduced the thermal conductivity down to the bulk value, as shown in Figure 13. There, R_{tot} data is shown over a wide temperature range, but the extracted thermal conductivities of 0.1 to 0.4 W/mK are very low. A better approach is to obtain a condition in which $2R_c \ll R_s$. Values from 60–90 W/mK [3], on par with metals, were measured for NFs attached by the capillary-assisted adhesion method described in Section 2.2.2.

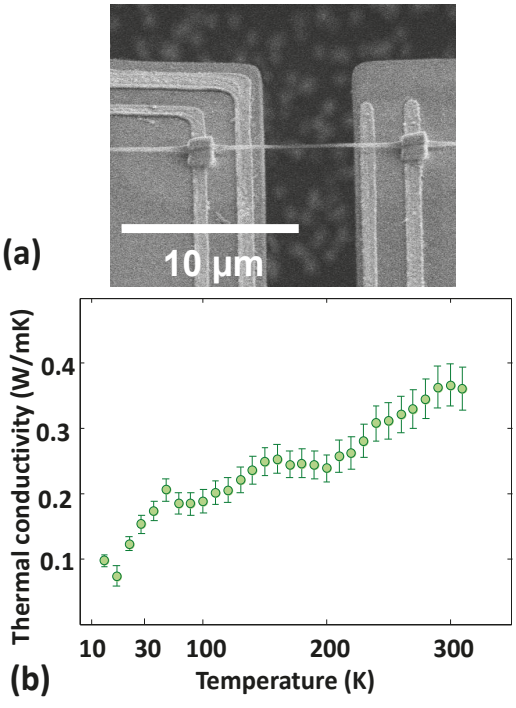

Figure 13. (**a**) NF attached by a platinum coating using FIB. (**b**) The thermal conductivity values of NF prepared by this procedure are similar to bulk PE due to the damage in the NF chains. Adapted from reference [3] supplementary information.

4. Summary and Conclusions

Polymer nanofibers may possess properties much higher than their bulk counterparts, and therefore accurate assessment of these properties is of great interest. However, because of their small size and the omnipresence of van der Waals, electrostatic and capillary forces, it is often difficult to manipulate individual polymer nanofibers on to test platforms that measure their properties. Here, we have detailed the fabrication (Appendix A) and use (Section 2.2.2) of a specimen collector (Figures 1, 2 and 5), which makes it possible to position nanofibers within ± 1 μm of the desired location (Figures 6–8) without exposing them to damaging electron beam irradiation. Even after they are in place, the very high strength and low surface energy of UHWMPE nanofiber render grip strength an obstacle to accurate mechanical testing results (Figure 12). This issue can be surmounted by gripping outside the smallest nanofiber region if extra material has accumulated there. Alternatively, longer grip regions may be effective. With respect to thermal measurements, contact resistance can dominate the values (Figure 13). We have found that a capillary-assisted technique (Figure 9) can significantly reduce contact resistance without damage to the nanofiber. While we find these procedures to be helpful, further manipulation improvements will surely be welcomed by practitioners.

Author Contributions: Conceptualization, all the authors; Investigation, R.S.; Writing—Original Draft Preparation, M.P.d.B.; Writing—Review & Editing, R.S. and M.P.d.B.; Supervision, S.S. and M.P.d.B.

Funding: This work was supported by the US National Science Foundation (NSF) under award CMMI-1334630.

Acknowledgments: We thank Kedar Hippalgaonkar (Institute of Materials Research and Engineering (IMRE), Singapore) for providing us with thermal platform.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A. Specimen Collector Fabrication

The specimen collector was fabricated by bulk micromachining techniques, as shown in Figure A1. The selectivity of a traditional oxide mask for silicon etching is insufficient for a through etch. Four-inch (100)-oriented silicon wafers coated with a 300 nm LPCVD silicon nitride (SiN_x) film on both sides were purchased from WRS materials. The wafers were HMDS vapor-primed at 150 °C for 5 min to promote photoresist adhesion. AZ 4210 photoresist was spin-coated at 4000 rpm, resulting in a 2.1 µm thick layer. This was followed by a soft bake at 95 °C for 1 min.

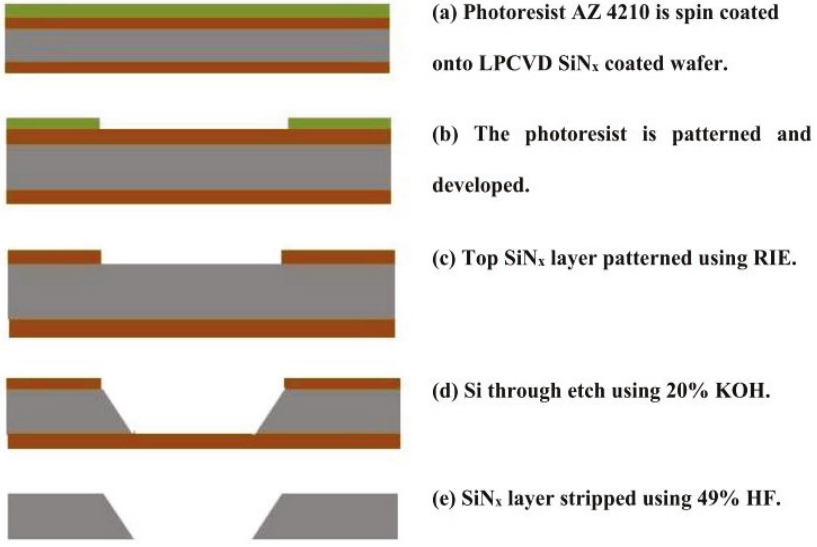

Figure A1. Micro fabrication processes of the sample collector.

The photoresist was patterned using a Karl Suss MA6 aligner with ultraviolet light of 320 nm wavelength at 5 mW/cm^2 for 80 s. The exposed photoresist was then developed by a 1:4 mixture of AZ 400K and deionized (DI) water for 120 s. The wafer was then cleaned in DI water and dried with nitrogen.

The photoresist pattern was transferred into the SiN_x by reactive ion etching (RIE) using a Plasmatherm 790 RIE system. A gas mixture of tetrafluoromethane (CF_4) and oxygen (O_2) at flow rates of 19.5 sccm and 0.5 sccm, respectively and a total pressure of 35 mT. An RF power of 60 watts at a bias of 315 volts was applied to induce plasma. The fluorine in CF_4 reacts with the silicon in silicon nitride to form volatile silicon tetrafluoride (SiF_4). Carbon can form polymer deposit onto the surface, reducing the etch rate. Hence, the fluorine/carbon ratio determines the Si_xN_y etch rate. A higher ratio provides a higher etch rate. Oxygen can react with the carbon forming volatile gas, which increases the fluorine to carbon ratio and thus increase the reaction rate. However, oxygen also etches the photoresist, so an optimum ratio must be found. We obtained an average etch rate of 26 nm/min but the etch rate decreased over the time due to the polymer deposit. After 6 min, the etching was stopped and argon gas was used to purge the chamber for 2 min at 200 mT. This was done to eliminate polymer deposit that would otherwise hinder the SiN etching. The total duration was 12.5 min to ensure all silicon nitride had been etched. Oxygen plasma was then applied for 5 min at 50 mT and 100 W to ash

the photoresist and remove any residual polymer. The chamber was once again purged with argon at 200 mT for 2 min. The wafer samples were then cleaned using acetone, isopropanol alcohol and DI water, respectively.

The wafer was now through-etched using KOH according to SiN_x. pattern. KOH etching depends on the crystal orientation of silicon, the solution concentration and the temperature. Here, KOH pellets were mixed with DI water to obtain 20% solution, an optimum concentration for (100) Si etching [46]. The solution was heated to 80 °C and the wafer was placed in the solution, whereupon hydrogen bubbles, indicative of the etching, were immediately observed. The glassware was covered with aluminum foil to reduce evaporative loss, thereby reducing the KOH concentration increase as the vapor re-condensed into the solution. However, the foil reacted with the vapor and some holes were produced. The total etch duration was 7 h for an average etch rate of 78.5 μm/h, less than the reported etch rate for 20% KOH at 80 °C [46], likely due to the increasing concentration. Etching ceased at the bottom SiN_x layer. KOH etching is an anisotropic etch with the (111) plane etching more slowly than the (110) and (100) planes, resulting in a characteristic pyramid like structure with 54.7° [46]. At the end of KOH etch, we obtained square holes with the expected 54.7° slanted walls. The wafer was rinsed in DI water, and the remaining SiN_x material on the top and bottom was removed after 15 min in 49% hydrofluoric acid (HF). The wafer was then diced into individual specimen collectors.

Appendix B. Microheater Construction

The micro heater is a sharply bent tungsten wire locally etched at the tip to 5–10 μm. It is joule-heated to a temperature exceeding 500 °C (Figure 3) at its tip to locally transform a MF into a NF. It was constructed and assembled as follows: Tungsten wire of 50 μm diameter was purchased from ESPI metals, cut to a 10 cm length, bent near its center, and adhered to a glass slide. The protruding bent edge was submerged in a beaker with 30% H_2O_2 solution that was held at room temperature while it was stirred magnetically. The wire was etched until the final diameter at the tip reached 5–10 μm as shown in Figure A2a. The ends of the wire were attached with copper tape for electrical attachment. Figure A2b shows a fabricated microheater.

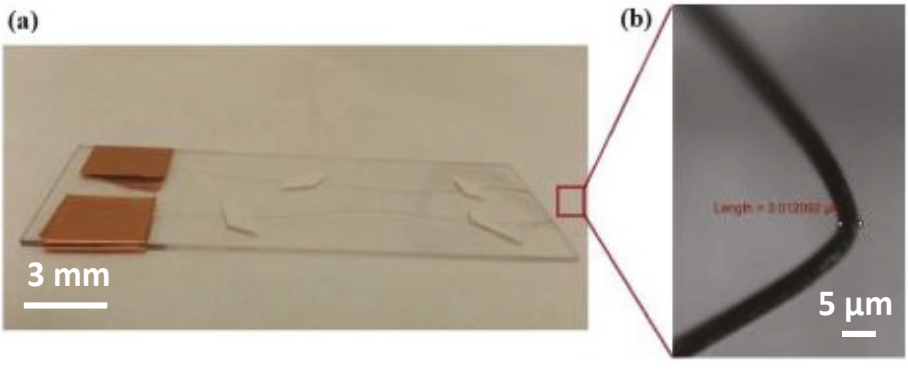

Figure A2. Microheater used for local heat stretching and cutting PE fiber. (**a**) microheater (**b**) tungsten tip of microheater with diameter ~3 μm.

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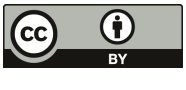

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An Approach to the Extreme Miniaturization of Rotary Comb Drives

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Received: 7 September 2018; Accepted: 6 October 2018; Published: 11 October 2018

Abstract: The evolution of microelectronic technologies is giving constant impulse to advanced micro-scaled systems which perform complex operations. In fact, the actual micro and nano Electro-Mechanical Systems (MEMS/NEMS) easily integrate information-gathering and decision-making electronics together with all sorts of sensors and actuators. Mechanical manipulation can be obtained through microactuators, taking advantage of magnetostrictive, thermal, piezoelectric or electrostatic forces. Electrostatic actuation, more precisely the comb-drive approach, is often employed due to its high versatility and low power consumption. Moreover, the device design and fabrication process flow can be simplified by compliant mechanisms, avoiding complex elements and unorthodox materials. A nano-scaled rotary comb drive is herein introduced and obtained using NEMS technology, with an innovative design which takes advantages of the compliant mechanism characteristics. A theoretical and numerical study is also introduced to inspect the electro-mechanical behavior of the device and to describe a new technological procedure for its fabrication.

Keywords: microactuation; nanoactuation; comb drives; flexure hinge; nanofabrication; electron beam lithography

1. Introduction

The end of the 1960s has witnessed the rise and consolidation of semiconductor fabrication technologies leading, together with the first microchip developments, to the evolution of increasingly complex systems able to connect advanced electronics and engineering systems with the physical world [1]. This process was made possible due to the development of MEMS sensors and actuators, that are currently an essential part of our daily life, being present in information-communication, electromechanical, optical, chemical and biological devices [2–5]. Moreover, the scaling phenomenon from macro- to micro-actuators led to a drastic change in the influence of individual parameters and opened new perspectives with innovative mechanical designs.

MEMS actuators and sensors are many and different. Micromachined structures movimentation can be carried out using various approaches: such as magnetostrictive, thermal, piezoelectric or electrostatic [6]. In this work, attention is given to work-producing actuators, with silicon being the key material, by an electrical and mechanical point of view. State-of-the-art MEMS devices are often actuated taking advantage of electrostatic forces between charged surfaces at a distance. Electrostatic actuation combines high versatility, fast dynamic response and low power consumption with a

simplified design and a fabrication process flow which is made possible by avoiding complex elements like coils or cores and unorthodox materials like shape-memory-alloys or piezoelectric ceramics [7]. It exploits the relation of surface to spacing (not volume to spacing), therefore its force is less affected by scaling [8].

Among all the possible electrostatic actuation techniques, the comb-drive approach is one of the most employed [9], with different possibilities in terms of out-of-plane [10] or in-plane [11] motion.

Moreover, Compliant mechanisms have been successfully adopted to develop MEMS devices in order to replace hinges and improve motion properties: the incorporation of the Conjugate Surfaces Flexure Hinges (CSFH) mechanism into the MEMS technology has opened up new perspectives for a completely new class of microdevices, characterized by a neutral stable configuration with a stationary pose, avoiding lubrication due to the absence of mechanical backlash, and easier to actuate [12]. The outcome is not a combination of different layers and sub-parts, the whole device can be made of a monolithic body of a single material, obtained by a highly simplified fabrication procedure, being at the same time extremely versatile in terms of integration in complex systems and structures for different applications, such as compliant 3 D.O.F. microrobots [13–16], micromechanisms [17,18], micro hinges [19–21] and microgrippers [22–27].

The drastic reduction of complexity made possible by the CSFH principle, together with its minimization of internal stresses and its robustness in operation, can facilitate the scaling process from a Micro-Electro-Mechanical System to a Nano-Electro-Mechanical System (NEMS), with new challenges in terms of mechanical movement, actuation possibilities and applications [28–30]. In this work, the properties of a NEMS rotary comb-drive nano-scaled actuator is introduced, its design is inspected, its electro-mechanical behaviour is theoretically and numerically studied and a new technological procedure for its fabrication is described, making it able to integrate in standard systems as well as in wearable/flexible systems, polymer substrates or in glass devices, with a variety of applications in the field of mechanical manipulation, such as nano-surgery, lab-on-chip bimolecular analysis, or nano-movimentation in space environment.

2. A Nano-Scaled Rotary Comb-Drive Electrostatic Actuator

The presented structure covers a total area of 120×75 μm^2 and is made of 2 main components, as depicted in Figure 1:

- A fixed part (red colored), with interdigitated fingers;
- A suspended part (green colored), composed by a mobile rigid body with interdigitated fingers and a flexure hinge with an anchored part on the left.

The system can be moved by changing the voltage between the static and movable part, which generates an alteration in the electrostatic forces between the two components, which are proportional to the capacitance between the two interdigitated structures: the result is a variation of the gap between the comb fingers and a movement of the rigid suspended body, made possible by the flexible hinge. Figure 2 shows a 3D sketch of the system.

The actual comb-drive mechanism is 50×25 μm^2, the smallest size ever presented in literature (to the best of the authors' knowledge), and has two arrays of 18,600 nm-wide and 1800 nm-spaced fingers, with a length which changes from 4 μm up to 9 μm; the first and last finger of each array is 1200 nm-wide. The whole suspended part is able to move due to a 600 nm wide curved beam flexure hinge, with a circular shape and a radius of 20 μm. The curved beam is considered as a revolute joint, centered in correspondence of the center of the beam elastic weight.

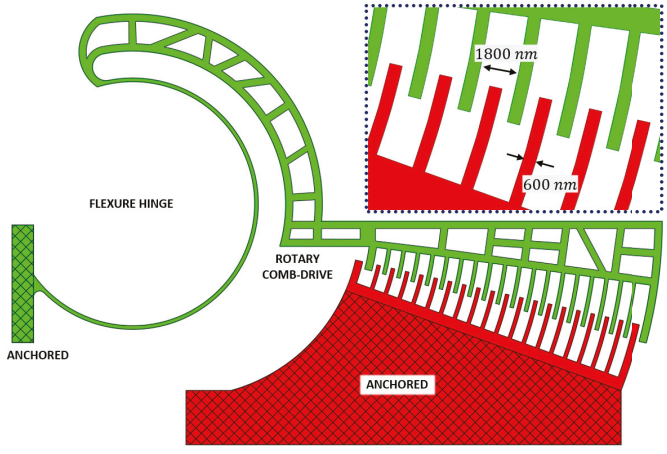

Figure 1. Top view of the proposed device.

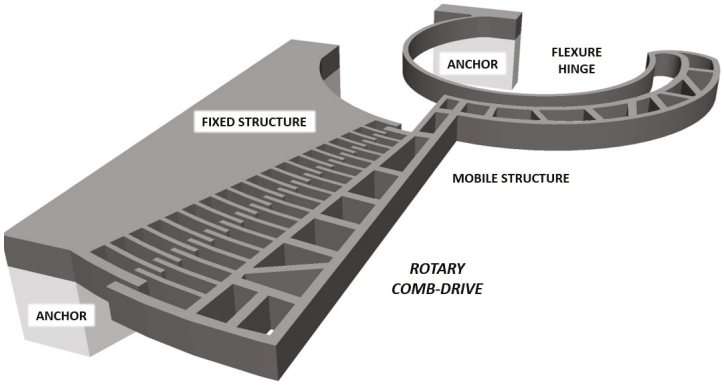

Figure 2. Three-D sketch of the proposed device.

3. Microelectromechanical Simulation

The response of the comb drive to the application of a voltage on the pads connected to the fixed and mobile sets of fingers has been simulated in two steps.

Firstly, considering the geometry of the systems, the number of fingers and the dielectric characteristics of the gap, an elementary force F has been calculated as acting on each one of the n fingers. Force F has the line of action perpendicular to the finger free-end section and can be calculated as [31]

$$F = \frac{\epsilon_0 \epsilon_r h V^2}{g}, \tag{1}$$

where ϵ_0 is the vacuum permittivity (8.8541×10^{-12} Fm^{-1}), ϵ_r is the air relative permittivity (1.00058), h is the thickness of the finger, g is the radial distance between the movable and fixed finger, and V is the applied voltage.

Then, a static force analysis has been performed by means of the commercial Finite Elements Analysis software ANSYS© (Canonsburg, PA, USA). Polycrystalline silicon has been considered as structural material, with Young's modulus equal to 164 GPa and Poisson's ratio equal to 0.22 [32].

Isotropic material formulation and nonlinearity due to large deflections were also considered. A fixed support was introduced to simulate the anchor constraint at the flexure end–section.

Two different layouts for the comb drives have been considered.

3.1. First Layout

For the first layout, $n = 19$ fingers, a thickness $h = 500$ nm and a gap $g = 600$ nm have been considered. The forces were calculated for an applied voltage ranging from 1 to 14 V, with steps of 1 V.

Figure 3 shows the generated mesh, composed of 9484 nodes and 7594 elements, and refined in correspondence to the flexible beam and to the comb-drive fingers.

The rotation of the floating part and the maximum value of the maximum principal stress (MPS), for increasing values of the applied voltage, are reported in Figure 4. The value of 14 V corresponds a rotation of 4.85 °, close to the maximum rotation permitted by the comb-drive geometry (5°). This case is represented in Figure 5, showing the deformed and neutral configurations of the structure and the stress distribution on the flexure. The maximum MPS value is equal to 66.7 MPa. Yield strength of polycrystalline silicon is equal to 1.2 GPa [33,34].

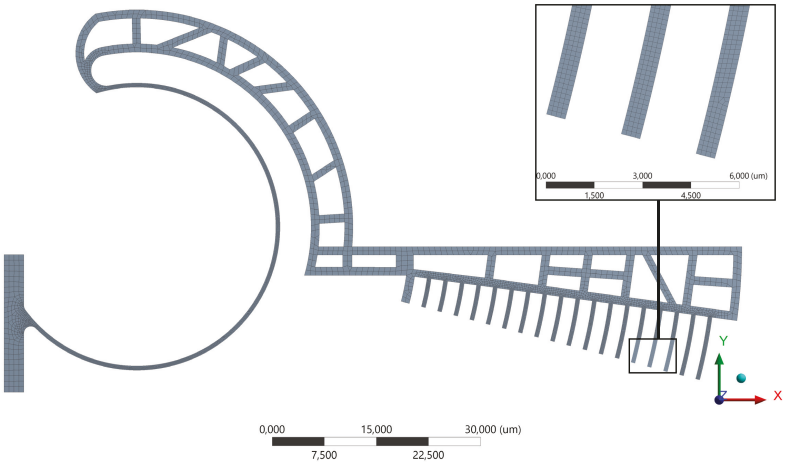

Figure 3. First layout: mesh of the floating structure and detail of the comb-drive fingers.

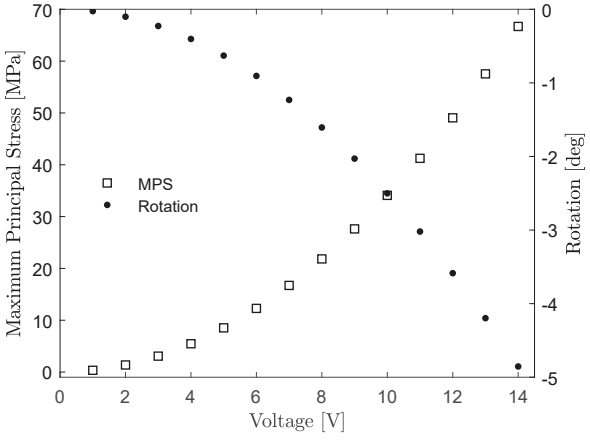

Figure 4. First layout: maximum values of the MPS and rotation angle of the movable fingers for increasing values of the applied voltage V.

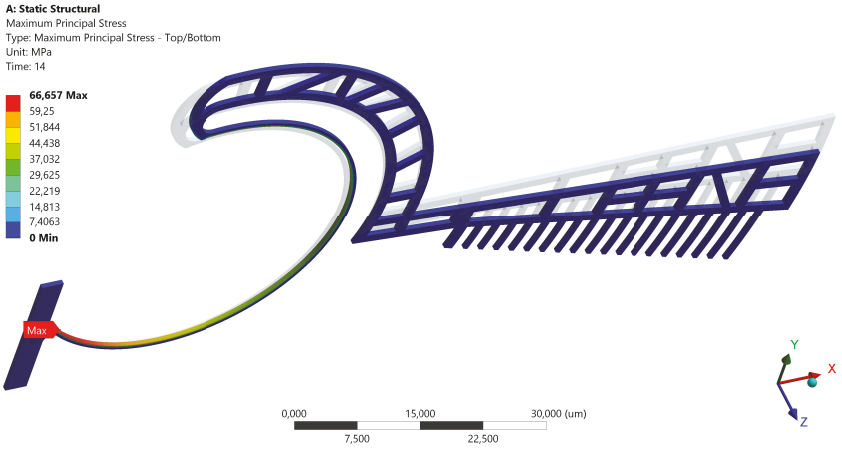

Figure 5. First layout: neutral and deformed configurations, and stress distribution on the flexure.

A modal analysis has been performed with ANSYS to evaluate the dynamic performance of the actuator. The first resonant mode, occurring at 16.5 kHz, corresponds to an out-of-plane motion characterized by an axis of rotation parallel to the *y*-axis (see Figure 5). This occurrence is not really critical for the fingers which are subject locally to a relative translation with no problems for fingers interference.

The second mode occurs at 18.3 kHz, and it is associated to an in-plane oscillation with an axis of rotation parallel to the *z*-axis. This mode corresponds, actually, to the designed working motion for the mobile fingers and therefore there are no interference problems.

The third and sixth modes, corresponding to out-of plane motions, are characterized by natural frequencies equal to 41.6 kHz and 149.3 kHz, respectively.

The fourth and fifth modes, occurring at 56.5 kHz and 126.9 kHz, respectively, are attributed to in-plane motions that could be detrimental because of the possible contact among floating and

anchored fingers. Finally, higher order modes have been neglected because the natural frequencies are very high.

3.2. Second Layout

After the former simulation, a second case has been studied with a gap g between the fingers larger than in the first case. In fact, the gap has been increased by the 30%, in order to avoid surface sticking during the fabrication and the releasing of the device. Furthermore, this could help operational performances during actuation.

To model the comb drive action in this second layout, the force F has been applied to each one of the $n = 13$ fingers. The thickness of the finger h remained 500 nm, as well as the radial distance between the movable and fixed finger (1.2 μm), while the range of applied voltage spans from 1 to 24 V, with 1 V steps.

Figure 6 reports the generated mesh, composed of 8065 nodes and 6423 elements, and refined in correspondence to the flexible beam and to the comb drive fingers.

The rotation of the floating part and the maximum value of the maximum principal stress (MPS), for increasing values of the applied voltage, are reported in Figure 7. The value of 24 V corresponds to a rotation of 4.88 °, close to the maximum rotation permitted by the comb-drive geometry (5°). As for the previous layout, the limit case is reported in Figure 8, that shows the deformed and neutral configurations and the stress distribution. The maximum MPS value is equal to 66.7 MPa.

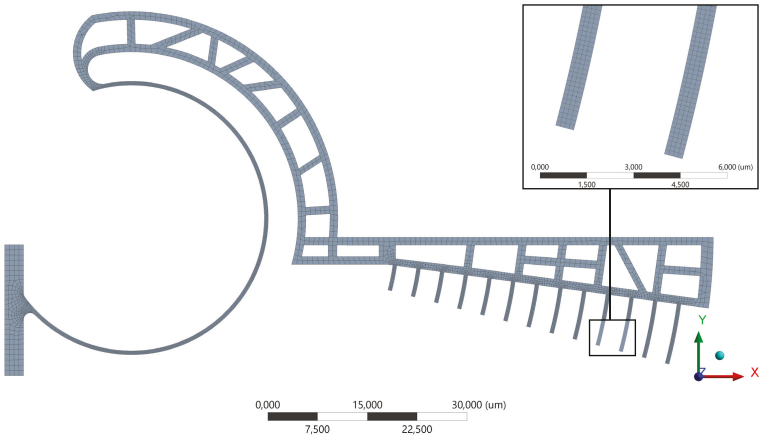

Figure 6. Second layout: mesh of the floating structure and detail of the comb-drive fingers.

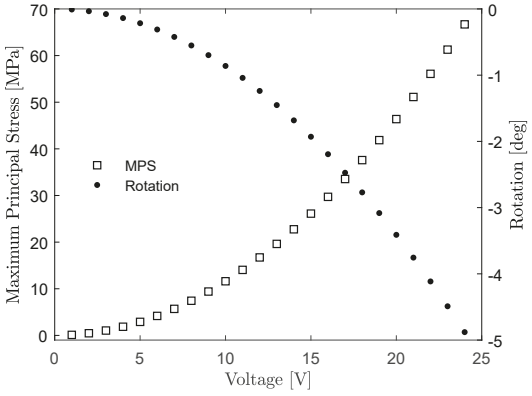

Figure 7. Second layout: maximum values of the MPS and rotation angle of the movable fingers for increasing values of the applied voltage.

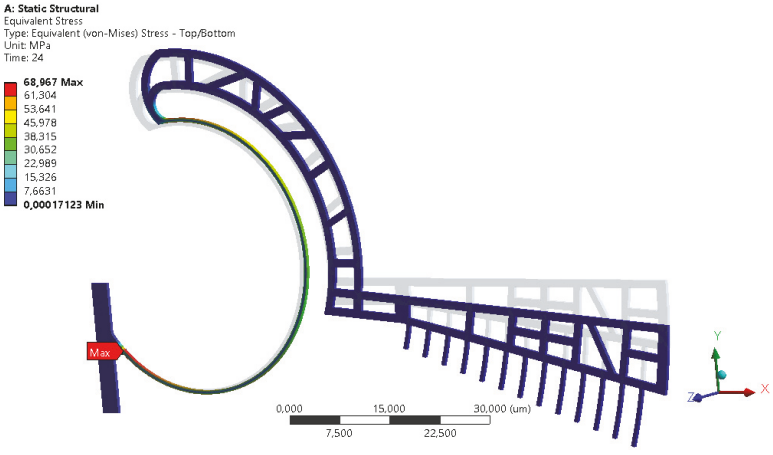

Figure 8. Second layout: neutral and deformed configurations, and stress distribution on the flexure.

As for the first layout, modal analysis has been conducted to calculate the most affecting natural modes and so the first six natural frequencies have been considered, having, respectively, values equal to 16.7, 18.6, 40.4, 57.7, 127.1 and 147.9 kHz. The corresponding modes are similar to the ones described in the previous case.

4. Fabrication

Microelectronic manufacturing technologies, and its constant evolution in time, opened new possibilities for the fabrication of micro- and nano-scaled electro-mechanical systems, allowing to obtain a highly complex mechanism through a sequence of iterations of simple steps:

- Deposition/growth of a chosen material on a substrate in form of thin/thick film, using various kinds of methods, such as Physical or Chemical Vapor Deposition (PVD/CVD), Atomic Layer Deposition (ALD) and Epitaxy;
- Geometry definition of the latter, usually via lithographic techniques or similar;
- Material patterning, via wet/dry etching procedures.

The result is a device assembled by a composition of different layers overlapped, each shaped with the corresponding geometry [35,36].

MEMS manufacturing and production, for this kind of devices, usually employs a standard Silicon-on-Insulator (SOI) stack. For this work, an unconventional material stack is designed, as sketched in Figure 9.

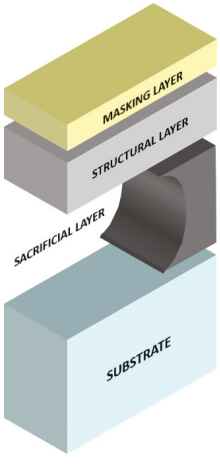

Figure 9. Three-D sketch of the proposed device.

In particular, the stack is composed by:

- A glass substrate;
- A Ti/W metallic alloy as intermediate sacrificial layer;
- Doped hydrogenated amorphous silicon (a-Si:H) as structural layer: a highly versatile material, suitable for large-area and low-cost electronic devices, biologically compatible, while monolithically integrable on rigid or flexible, metallic, dielectric or polymeric substrates [37–42]. It also shows interesting mechanical features and was chosen to build bridge-like structures and cantilevers based devices [43].
- Chromium as masking layer for the structure patterning.

The fabrication procedure was carried out as follows:

- PVD of the sacrificial layer;
- CVD of the amorphous silicon layer;
- Geometry definition on a PVD obtained chromium film by lithographic technique: in particular, Electron Beam Lithography (EBL) was required, in order to obtain submicrometric resolution, on an electron-sensible polymeric film [44];
- Structure patterning by Reactive Ion Etching (RIE) technique, with the previously shaped chromium film as masking layer;
- Structure releasing via isotropic and selective removal of the intermediate layer, in a wet etching solution. Etch holes in the rigid suspended bodies were considered in order to promote the under-etching phenomenon and optimize the releasing process.

Figure 10 shows first examples of the nanoscaled rotary comb-drive during its fabrication process. Since the structure geometries reach the sub-micrometric scale, it was impossible to appropriately identify and discern all the features of our device with an optical microscope: therefore Scanning Electron Microscope (SEM) pictures were taken.

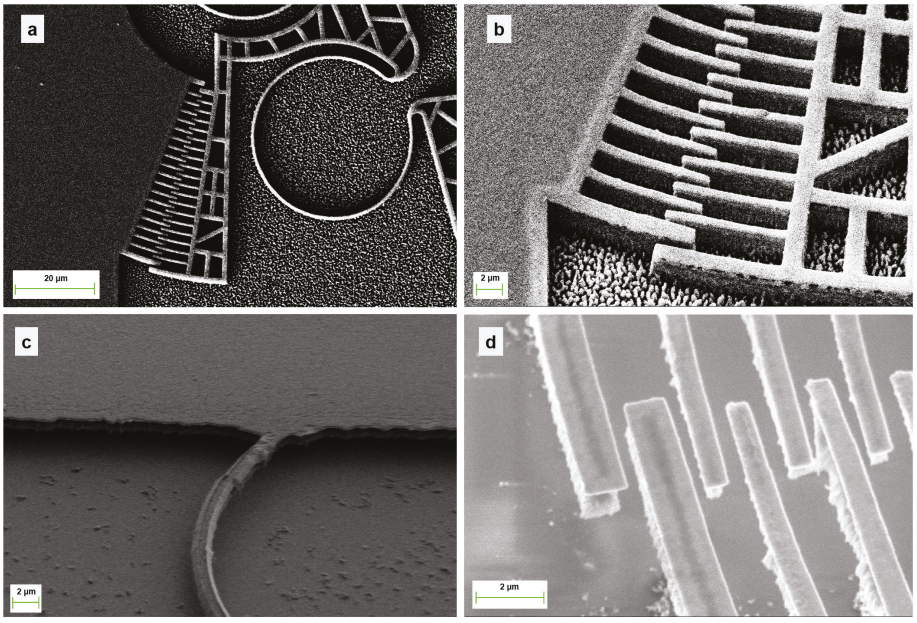

Figure 10. SEM pictures of the nano comb-drive during its fabrication process: top view of the whole patterned device (**a**) with a 30° tilted enlargement on the comb-drive fingers (**b**); 45° tilted view of an anchored flexure hinge (**c**) and fingers (**d**) released from the glass substrate and suspended. These images were taken with an accelerating voltage (EHT) of 10 kV, 10 kV, 2 kV and 5 kV respectively; a magnification of 2630×, 8630×, 8000× and 2281×, respectively; a working distance of 5.5 mm, 10 mm, 5.9 mm and 10.0 mm, respectively, and a beam current of 10 pA.

Figure 10a,b show a top view (top left) of the whole device during the structure patterning phase, with an enlargement on the comb fingers (top right). Figure 10c,d show the crucial parts of the device suspended and detached from the substrate after the releasing process: the attachment between the hinge of the actuator and the anchor (bottom left), and the comb-drive fingers (bottom right). The displayed devices have n-doped a-Si:H, deposited by Plasma Enhanced Chemical Vapour Deposition (PECVD), as structural layer, with the following parameters:

- A substrate temperature of 200 °C;
- A SiH_4 flow of 45 sccm and a PH_3 flow of 10 sccm into the process chamber, with a process pressure of 0.3 Torr;
- A glow discharge with a power density of 25 mW/cm^2, for 1 h.

5. Conclusions

This paper has shown the feasibility of a new nano fabrication process to build nano actuators, based on electrostatic actuation. The construction process has been applied to a rotary comb-drive which appears promising as a tool for microactuation and micromanipulation. Furthermore, the numerical simulations gave also encouraging results in terms of operational capability. Two models have been studied with different characteristics and the results showed that operational capabilities can be increased by reducing the gap between the fingers and their size, which makes the systems less robust. Therefore, an optimal design should find a balance of mediation between the two different factors. The developed actuator seems to be one of the smallest actuators available as state-of-the-art, and this could lead to new applications, especially in the biomedical field.

Author Contributions: A.V., A.B., M.H., and E.G. were involved in the fabrication and design of the rotary comb drives; M.V. and N.P. B. have performed the numerical analysis and design; F.F., G.d.C. and N.P.B coordinated the work and supervised the project.

Conflicts of Interest: The authors declare no conflict of interest.

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An Image Analysis Approach to Microgrippers Displacement Measurement and Testing

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Received: 5 September 2018; Accepted: 20 September 2018; Published: 24 September 2018

Abstract: The number of studies on microgrippers has increased consistently in the past decade, among them the numeric simulations and material characterization are quite common, while the metrological issues related to their performance testing are not well investigated yet. To add some contribution in this field, an image analysis-based method for microgrippers displacement measurement and testing is proposed here: images of a microgripper prototype supplied with different voltages are acquired by an optical system (i.e., a 3D optical profilometer) and processed through in-house software. With the aim to assess the quality of the results a systematic approach is proposed for determining and quantifying the main error sources and applied to the uncertainty estimation in angular displacement measurements of the microgripper comb-drives. A preliminary uncertainty evaluation of the in-house software is provided by a Monte Carlo Simulation and its contribution added to that of the other error sources, giving an estimation of the relative uncertainty up to 3.6% at 95% confidence level for voltages from 10 V to 28 V. Moreover, the measurements on the prototype device highlighted a stable behavior in the voltage range from 0 V to 28 V with a maximum rotation of 1.3° at 28 V, which is lower than in previous studies, likely due to differences in system configuration, model, and material. Anyway, the proposed approach is suitable also for different optical systems (i.e., trinocular microscopes).

Keywords: microgripper; displacement; measurement; uncertainty; image analysis

1. Introduction

Microgrippers are MEMS technology-based devices, able to manipulate objects with dimensions of the order of 10^{-6} m. The development and diffusion of such devices is increased considerably thanks to the advent of D-RIE (Deep Reaction Ion Etching) manufacture technology, which led the construction even on a micrometer scale. The microgrippers under study in this article are made up of flexible beams, and the mechanism that allows the movement of these devices is based on a particular and innovative hinge, called a conjugate surface flexure hinge (CSFH), which can be built as a monolithic body and easily integrated into any mechanical structure of a MEMS. These devices are actuated by innovative capacitive rotary comb-drives that can generate a torque when driven by an applied voltage (Figure 1). Based on the theoretical model in [1], to perform an electro-mechanical characterization of these devices, in [2] the behavior of Comb-Drive as studied when a potential difference is applied. In this study, however, some important characteristics such as mass and material effects were neglected and the rotary capacitors were assumed as parallel plates. To validate the theoretical results and obtain a more realistic and accurate model, a numerical simulation based

on Finite Element Method (FEM) was also proposed. Moreover, through the Direct Method, it was possible to validate the theoretical behavior, solving the problem of offering a good angle of rotation while minimizing the maximum internal stress. Through a Finite Element Analysis (FEA) simulator, the displacement of the clamps according to the applied potential difference [3] were studied. In addition, in [4], simulations have been conducted on V-Rotation Comb-Drive and V-Force relationships. Thanks to these studies a quadratic behavior of the relationship between the applied voltage and the angular rotation of the Comb-Drive was determined. Despite the great variety of applications involving microgrippers, such as biology [5,6], surgery [7,8] or aerospace [9,10], very few works reported measurement data for the experimental characterization of such devices [2] and the quality of the results is not established through an in-depth uncertainty analysis. To give a contribution to this issue, in this study an experimental approach to measure the microgripper displacements is proposed and applied to verify the theoretical relationship between the applied voltage and the Comb-Drive rotation. To this aim an uncertainty analysis of the quantities measured is also performed: an example of measurements and testing on a microgripper prototype is described here to confirm the theoretical displacement of its comb-drives by means of an optical system (profilometer) and an in-house image analysis software.

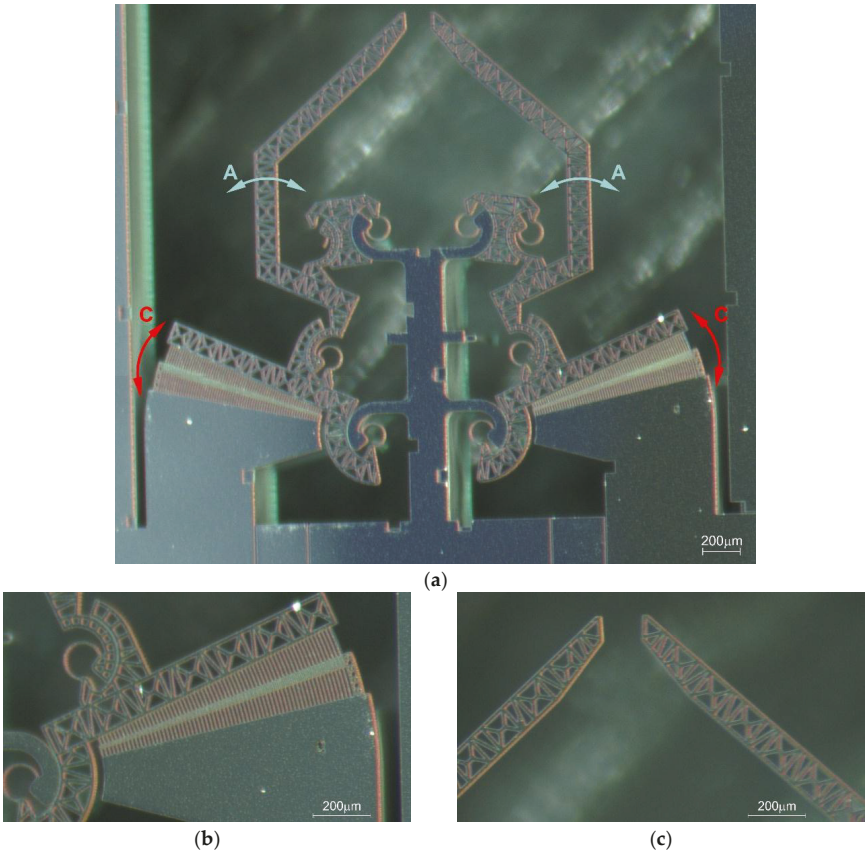

Figure 1. Images of a microgripper captured by a trinocular microscope. (**a**) microgripper, the displacement of the arm A is provided by the rotation of the Comb-Drive C (**b**) Hinges and rotary Comb-Drive (**c**) pliers.

2. Materials and Methods

With the aim to properly characterize and test the microgripper behavior, many kinematic and dynamic output quantities should be measured (e.g., from position and displacement to forces and torques) and related to the input quantities supplied to the device (e.g., voltage, current), therefore a first systematic approach may consider the displacement of one or more of its components taking into account the characteristics of the experimental set up (Table 1).

Table 1. Experimental setup.

Device	Characteristics
Power supply	Keithley 236, Range settable to 1.1/11/110/1100 V with respectively 0.1/1/10/100 mV resolution, accuracy 0.06 V at F.S.
Micropositioners	n.1 MP25L, n.1 MP25R, range X/Y/Z 10/10/10 mm with 5 μm resolution
Probes (supply)	PA-C-1M with tungsten needles
Device Under Test (DUT)	Silicon microgripper (Silicon type P. dopant Boron, orientation <100>, electrical resistivity 0.005–0.03 Ohm×cm), module dimensions 2000 μm × 1500 μm, device thickness 40 μm, insulated layer thickness 3 μm, handle thickness 400 μm, capacitive comb-drives
DUT stage	The wafer containing the DUT is placed on the profilometer working surface fixed by an adhesive tape
Optics: 3D Optical Profilometer	field of view from 7.2 mm × 5.4 mm to 80 nm × 60 nm, maximal lateral resolution 0.6 μm
Digital Image	768 × 580 pixels, 8 bit, 0.6 px/μm
Image Processing Software	In-house software developed in MATLAB (2017a, MathWorks)
Notebook pc	Intel core i7-2670, 6 Gb RAM, Nvidia GeForce GT 520 MX

In particular, with reference to Table 1 the following components have been considered:

a. The microgripper under study (Device Under Test or DUT), made up of flexible beams and hinges, is actuated by capacitive rotary comb-drives that generate a torque when a voltage is applied (Figure 1).
b. A constant voltage source should be provided to evaluate the dependencies of the mechanical output on the voltage. In our study, the power supply device is a Keithley 236 SMU with 0.055% F.S. output voltage accuracy.
c. Two micropositioners embedded with probe arms and tungsten needles have been used (Figure 2) to apply the voltage to the DUT.
d. The microgripper displacements have been measured from images acquired and collected by a Fogale Zoomsurf 3D optical profiling system (Figure 3). The maximum resolution is limited by the diffraction as in conventional optical microscopes to 0.6 μm and the vertical resolution reaches 0.1 nm. The digital image resolution is 0.6 pixel/μm therefore each pixel corresponds to about 1.6 μm. Taking into account that to clearly discriminate 2 point objects in a digital image, 3 pixel are needed in the best case [11,12], the total lateral accuracy into the digital image has been estimated to be ±2 pixel (about ±3.3 μm).
e. The acquired images have been processed by an in-house software developed in MATLAB® and a notebook PC. From the analysis of the image sequences, the movement of the components has been measured by means of a template-matching algorithm. A set of images has been acquired, each one corresponding to a specific voltage setting: to calculate the angular displacement,

the first acquired image (0 V) has been compared with the others (i.e., 2 V, 4 V, ..., 28 V) through the following steps:

- Step 1:
  - The initial coordinates of the Comb-drive ICR (Instant Center of Rotation) are estimated. As shown in Figure 4, four points are manually selected on the edges of the Comb-Drive, and the intersection of the two corresponding lines coincided with the ICR of the Comb-Drive. The second phase involves the manual selection of a particular Region of Interest (ROI) on the image. The Comb-Drive under examination, has been designed in such a way that it has a static part anchored to the structure of the MEMS device and a mobile part. In Figure 4 the lower part is the static one. As shown in Figure 5, a ROI can be selected on the mobile part of device's Comb-Drive to determine the angular displacement (see step 2). Figure 6 shows the superposition of two images in the sequence, one corresponding to no actuation (the neutral configuration) and one corresponding to a 28 V applied voltage.
- Step 2:
  - The coordinates of the following three points are used: the most distant point from the ICR on the fixed Comb-Drive part, the ICR of the Comb-Drive and the center of the ROI. They determined a triangle, where the vertex ICR corresponds to the angular opening of the Comb-Drive.
  - A match is found between the coordinates of the selected ROI on the first image (see step 1) and on all the subsequent images. Through this operation the in-house software can detect the new coordinates of the ROI's center of gravity for each subsequent image and therefore for each applied voltage. On the hypothesis of no deformation due to the movement of the Comb-Drive, the rigid movement of the device keeps unchanged the points on the bottom left and that corresponding to the ICR. Therefore, for each subsequent image in the sequence, the coordinates of the center of the ROI are detected and used to identify a new triangle and consequently a new angular opening of the Comb-Drive for each applied voltage.
  - From the angular aperture of the Comb-Drive at each voltage, the corresponding angular displacement is obtained.

The above steps can be easily adapted to measure the movement of other elements of the microgripper and are suitable on digital images produced also by systems different from that used in this work. On the other hand, it is important to consider the errors due to the variability of the parameters related to the manual selection of the operator, together with the uncertainty of the code itself. For this reason, an uncertainty analysis has been made considering the above issues by means of Monte Carlo Simulation (MCS).

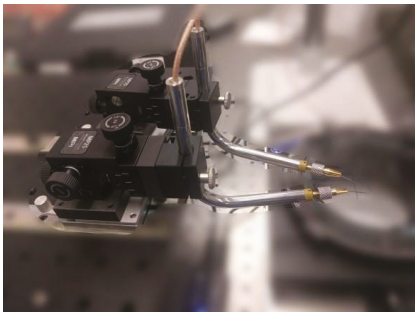

Figure 2. Micro-positioner.

Figure 3. Optical profiling system.

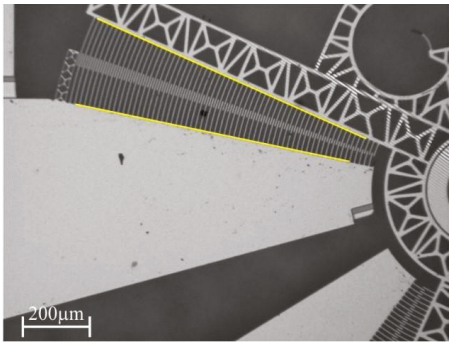

Figure 4. Microgripper under study (DUT): image of the Comb-Drive provided by the 3D optical profiling system. From the intersection of the yellow lines the ICR is determined.

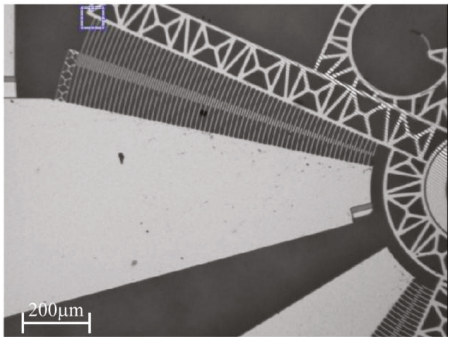

Figure 5. Image of the DUT provided by the 3D optical profiling system. ROI identification on the mobile arm of the Comb-Drive.

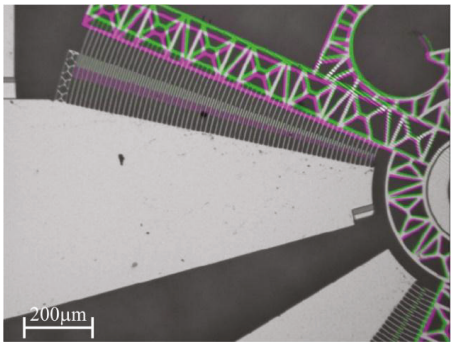

Figure 6. Superposition of two images of the DUT Comb-Drive, corresponding to 0 V and 28 V applied voltage, respectively.

First of all, an uncertainty analysis of the errors due to the experimental setup has been done, therefore the main uncertainty sources have been identified and evaluated, as shown in Table 2 where for each source a probability density function PDF together to the uncertainty type, mean m and standard deviation σ is determined.

- Power supply uncertainty has been evaluated trough the data sheet and a Gaussian distribution has been assumed, therefore the corresponding standard deviation σ is evaluated from the declared uncertainty u as $\sigma = u/2$.
- DUT stage: orientation and vibrations of the DUT stage are related to a change in image focusing during measurements. The corresponding uncertainty has been assumed negligible as the setup is mounted on an optical table with a pneumatic suspension system and at each step a correct image focusing is maintained.
- Optical system and digital image: as described before the uncertainty due to the use of the optical profilometer has been evaluated considering its maximal lateral resolution and the digital resolution of the acquired image, i.e., 0.6 µm and 1.6 µm/px, respectively. As shown in Figures 7–10, two point objects in a digital image are resolved if they are separated by 1 or more pixel depending if they are centered or not on a picture element. In the best case (i.e., for high contrast and Signal to Noise Ratio) the minimum distance needed between them should be about 3 pixels [11,12], therefore in this study a uniform uncertainty distribution with a 2 pixels semi-amplitude is assumed.

- Software Processing: two uncertainty contributions can be determined: the former is related to the step 1 (manual selection), while the other is due to the template-matching algorithm. As in previously works [14–17] the total uncertainty has been evaluated by an MCS.

Table 2. Uncertainty sources.

Uncertainty Source	Probability Density Function (PDF) [1]	Type [2]	m	σ	Unit
Power Supply: Voltage accuracy	N (m, σ)	B		0.03	V
DUT Stage: Plate planarity and vibration, focusing plane variation	-	-	-	-	-
Optical System: Profilometer Maximal lateral resolution due to diffraction	U (m, σ)	B		0.4	μm
Digital Image: digital conversion	U (m, σ)	B		1.9	μm
Calibration: Included in the nominal uncertainty of the Optical System	-	-	-	-	-
Image Processing Software: uncertainty in point identification uncertainty in ROI position (template-matching) uncertainty in ROI size (template-matching)	U (m, σ)	A		±2 ±2 ±2	pixel pixel pixel
uncertainty in the template-matching algorithm	A Monte Carlo Simulation is used.				

[1] N(m, σ) is a Gaussian PDF with mean m and standard deviation σ; U(m, σ) is a uniform PDF with mean m and standard deviation σ; [2] Type A and type B uncertainty as in [13]

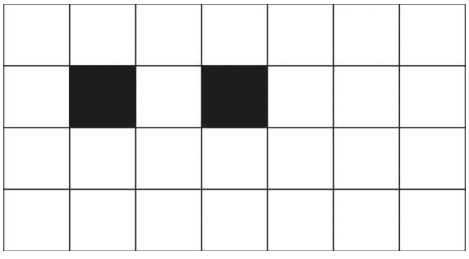

Figure 7. Two pixel-centered point elements separated by 2 pixels.

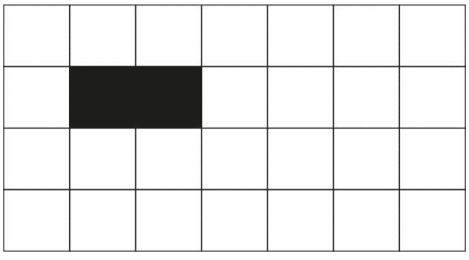

Figure 8. Two point elements in 2 near pixels.

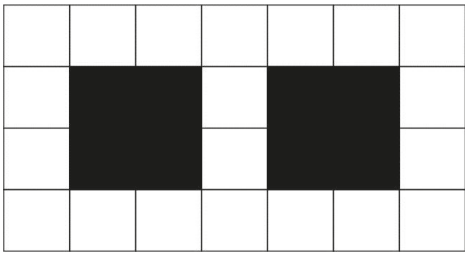

Figure 9. Two point elements not pixel-centered separated by 3 pixels.

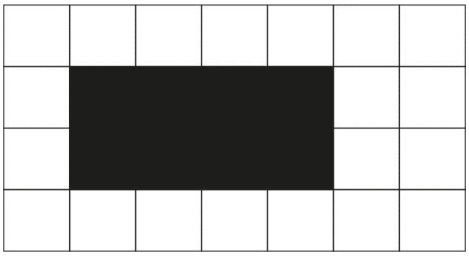

Figure 10. Two point elements not pixel-centered separated by 2 pixels.

To evaluate the accuracy of image processing software, a simulation was implemented in MATLAB® in accordance with the MCS Method. In this procedure the uncertainty sources were introduced by the variability of the parameters related to the manual selection of the operator. In particular:

- Uncertainty due to point selection: with the aim to determine the ICR, the operator manually selects some points on the image. This operation is performed only once at the beginning of the measurement process, and even if the coordinates of these points are confined to a limited region of space (the mobile edge of the Comb-Drive), there is a variability introduced by the nature of the manual process. Under the same conditions (same image selected and same operator), a set of 10 consecutive tests were carried out and a dispersion of ± 2 pixels was estimated. A uniform distribution U (m, σ) was also assumed with semi-amplitude of 2 pixels and mean m from the average of the 10 tests.
- Uncertainty related to ROI position in the template-matching algorithm: the operator arbitrarily chooses the ROI position in the image. Although the ROI lies in a limited region of the image (i.e., the mobile part of the Comb-Drive), an uncertainty is introduced by the variability of the manual selection. Also, in this case, a set of 10 consecutive tests were carried out and a dispersion of ± 2 pixels was determined: a uniform distribution U (m, σ) was assumed with a 2-pixel semi-amplitude and mean m from the average of the 10 tests.
- Uncertainty related to ROI size in the template-matching algorithm: in the same way, the ROI size selected by the operator may vary. Under the same conditions (same image selected and same operator), also in this case a set of 10 consecutive tests were carried out and a ± 2 pixels dispersion was estimated. A uniform distribution U (m, σ) was also assumed with mean value equal to the average of the 10 trials and a 2-pixel semi-amplitude.
- Uncertainty of template-matching algorithm: also, the template-matching gives a contribution to uncertainty as described in [13], nevertheless it is included in the MCS results, as the final data dispersion is affected by this source of error too.

Once the distributions related to the variability of manual selection processes have been established, the process of calculating the angular displacement of the Comb-Drive is performed 10,000 times (i.e., 10,000 iterations in the MCS), for an applied potential difference of 28 V. This procedure is carried out to verify the distribution of the angular displacements when the above parameters change.

The total uncertainty u_c of the displacement measurement is evaluated as suggested in [13] with the following expression:

$$u_c = \sqrt{u_A^2 + u_B^2} \tag{1}$$

where u_A and u_B are the type A and type B overall uncertainty respectively. In particular, u_A is the uncertainty contribution evaluable with a statistic analysis of the measurements dispersion obtained with the system and u_B is the overall uncertainty from the main error sources related to the experimental setup. To also evaluate the uncertainty contribution of the power supply the relation f between the voltage V and the angular displacement α has been considered. The u_B can be written as:

$$u_B = \delta\alpha = \sqrt{(\delta\alpha_t)^2 + \left(\frac{\partial f}{\partial V}\delta V\right)^2} \qquad \frac{\delta\alpha_t}{\alpha_t} = \sqrt{\left(\frac{\delta\alpha_p}{\alpha}\right)^2 + \left(\frac{\delta\alpha_s}{\alpha}\right)^2} \tag{2}$$

where α_p are the angle measurement uncertainty due to the profilometer system (optical and A/D conversion), α_s is the angle measurement uncertainty due to the implemented software system, δV the uncertainty on the supply voltage and $\partial f/\partial V$ is the derived function that described the correlation between the supply voltages and the Comb-Drive rotations [13]. This last uncertainty contribution has been evaluated through an MCS. To evaluate the α_p contribution, the angle has been measured by means of the triangular properties as shown in Figure 11 and (3).

$$\alpha_p \simeq tg\alpha_p = \frac{a}{b} \tag{3}$$

From Figure 11 the values a and b are obtained from the Δx and Δy lengths, following the expressions in (4):

$$a = \sqrt{\Delta x_a{}^2 + \Delta y_a{}^2},\ b = \sqrt{\Delta x_b{}^2 + \Delta y_b{}^2} \tag{4}$$

Therefore, the uncertainty due to the profilometer error in Table 2 can be obtained as:

$$\frac{\delta\alpha_p}{\alpha_p} = \sqrt{\left(\frac{\delta a}{a}\right)^2 + \left(\frac{\delta b}{b}\right)^2} \tag{5}$$

where:

$$\delta a = \sqrt{\left(\frac{\partial a}{\partial x}\delta x\right)^2 + \left(\frac{\partial a}{\partial y}\delta y\right)^2} = \sqrt{\left(\frac{\Delta x_a}{\sqrt{\Delta x_a{}^2 + \Delta y_a{}^2}}\delta x\right)^2 + \left(\frac{\Delta y_a}{\sqrt{\Delta x_a{}^2 + \Delta y_a{}^2}}\delta y\right)^2} \tag{6a}$$

$$\delta b = \sqrt{\left(\frac{\partial b}{\partial x}\delta x\right)^2 + \left(\frac{\partial b}{\partial y}\delta y\right)^2} = \sqrt{\left(\frac{\Delta x_b}{\sqrt{\Delta x_b{}^2 + \Delta y_b{}^2}}\delta x\right)^2 + \left(\frac{\Delta y_b}{\sqrt{\Delta x_b{}^2 + \Delta y_b{}^2}}\delta y\right)^2} \tag{6b}$$

The quantities a and b in (6) depend on the image resolution and size. For this study, the lengths of the Comb-Drive triangle in the image are considered for the maximum rotation, i.e., Δx_a = 25 pixel, Δy_a = 150 pixel, Δx_b = 680 pixel and Δy_b = 150 pixel, therefore a = 152 pixel, and b = 700 pixel.

From Table 2 and the maximum measured angle is 1.3° and the relative uncertainty is up to 3.6% for voltages higher than 10 V.

In a second step of our study some tests have been conducted to evaluate how good the microgripper behavior fits the theoretical model [2]. With this aim it has been supplied with different voltages, i.e., from 0 V to 28 V at 2 V steps, forward and backward 6 times.

In Figure 6 two superimposed images of the same Comb-Drive at different voltages are shown: the in-house software can evaluate the angle difference between them.

For each voltage value the standard deviation and the average of the angular displacement of the Comb-Drive have been measured as above. Finally, all the measurement data have been processed and interpolated to provide a curve fitting of the actual motion of the Comb-Drive depending on the voltage supply.

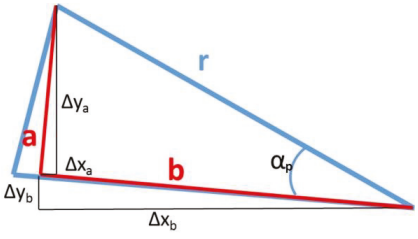

Figure 11. Angle measurement for the uncertainty evaluation related to the profilometer image.

3. Results

From the MCS the image processing software accuracy has been estimated: as shown in Figure 12 the distribution of results followed a Gaussian curve with $\pm 0.02°$ interval at a 95% confidence level (percentile values for 2.5% and 97.5% are 1.29° and 1.32° respectively). This value has been used to evaluate the total measuring system uncertainty following (1).

The uncertainty analysis above has been applied to angular measurements on the microgripper at different voltages (Figure 13, Table 3). Measurement data showed that the interpolating curve (voltage vs angular displacement) was similar to the numerical results and to measurements reported in other works on similar devices [2,16], but the angles measured here were usually lower for the same voltages. This different behavior is likely due to the different microgripper used (a different model, configuration, and material). Indeed, in the studies above mentioned [2,16], different microgripper configurations and models provided different angular behaviors, as shown in Table 3.

In Figure 13 the measurement results proposed here seem to be more similar to [16] than [2]. The different materials and configuration may be related to different elastic resistances in the hinges translating in different angular displacements. Anyway, a best fitting curve with a $R^2 = 0.999$ demonstrated a coherent behavior along all the voltages: this is likely due also to the systematic approach and to the in-house software applied in this study to perform the measurements with repeatability and low-operator-dependence.

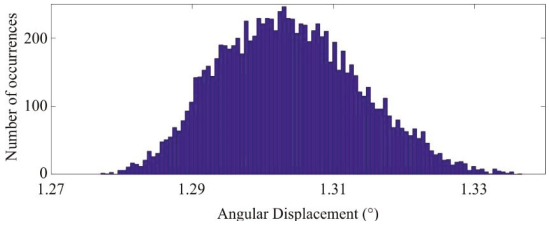

Figure 12. Example of results of the Monte Carlo Simulation for software uncertainty analysis (10,000 cycles).

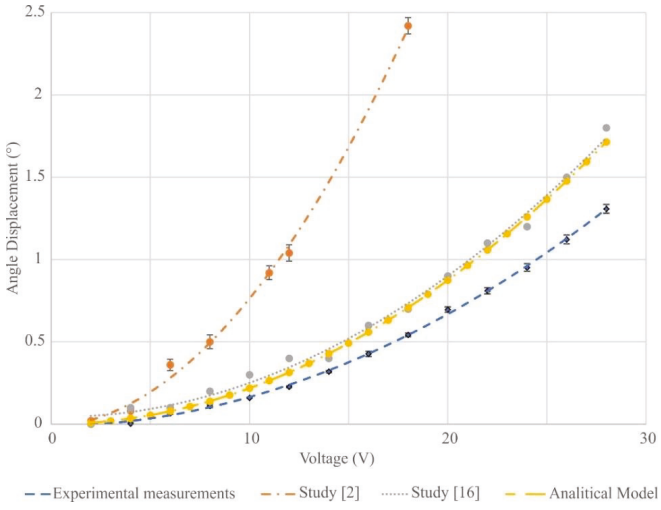

Figure 13. Curve and points obtained for measurements made with the proposed system (blue dashed line) compared with that from the analytical model (yellow dashed line) and the experimental data (orange dot-dashed line) in [2,16] (dotted line).

Table 3. Results on angular displacement measurements (Figure 13).

Supply Voltage (V)	Angle Rotation Mean Val. (°)	Standard Deviation (°)	Angle [2] (°)	Angle [16] (°)
2	0.005	0.005	0.02	0
4	0.011	0.008	0.08	0.1
6	0.065	0.004	0.24	0.1
8	0.11	0.011	0.5	0.2
10	0.1603	0.004	0.8	0.3
12	0.227	0.006	1.08	0.4
14	0.3190	0.007	1.65	0.4
16	0.427	0.015	2.08	0.6
18	0.5425	0.011	-	0.7
20	0.696	0.016	-	0.9
22	0.810	0.018	-	1.1
24	0.952	0.024	-	1.2
26	1.122	0.027	-	1.5
28	1.308	0.027	-	1.8

4. Conclusions

With the aim to characterize and test the behavior of a microgripper a systematic approach to measurements is proposed and applied. An experimental setup has been developed and the main sources of error have been evaluated to estimate the measurement uncertainty in angular displacements of the microgripper Comb-Drive depending on the applied voltage.

The results showed a stable behavior of the Comb-Drive with angular displacement lower than in other works, that was likely due to the different MEMS configuration and model, as the different

materials and configuration could imply a different elastic resistance in the hinges evolving in different angular displacement.

The software here implemented can work on different components of the gripper (clamp instead Comb-Drive). Moreover, the method proposed can be easily applied to images provided by systems that provide bidimensional images of the microgripper in operational conditions, e.g., it can work also with images from a trinocular microscope equipped in a probe station. In this case, an in-depth analysis of the probe station characteristics and its calibration should be done. Although the overall measurement uncertainty may be higher than in the case described here, it must be pointed out that the optical microscope is a measurement system less complex and expensive than the 3D optical profilometer, therefore it may be a good trade-off between costs and results accuracy. Future works will be focused on this kind of testing system and a study on the clamp motion behavior in comparison with the Comb-Drive rotation is going to be conducted to further contribute to a more complete approach to microgripper experimental characterization and testing.

Author Contributions: Conceptualization, A.S.; Methodology, F.O., F.V. and A.S.; Software, F.O. and F.V.; Validation, F.O., F.V. and A.S.; Formal Analysis, F.O.; Investigation, F.V. and F.O.; Resources, S.A.S. and R.C.; Data Curation, F.V.; Writing-Original Draft Preparation, F.O., F.V. and A.S.; Writing-Review & Editing, A.S., F.O., F.V. and S.A.S.; Visualization, F.O. and F.V.; Supervision, A.S. and S.A.S.; Project Administration, A.S. and S.A.S.; Funding Acquisition, S.A.S.

Conflicts of Interest: The authors declare no conflict of interest.

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Article

Scalloping and Stress Concentration in DRIE-Manufactured Comb-Drives

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Received: 23 July 2018; Accepted: 4 September 2018; Published: 5 September 2018

Abstract: In the last decades, microelectromechanical systems have been increasing their number of degrees of freedom and their structural complexity. Hence, most recently designed MEMSs have required higher mobility than in the past and higher structural strength and stability. In some applications, device thickness increased up to the order of tens (or hundred) of microns, which nowadays can be easily obtained by means of DRIE Bosch process. Unfortunately, scalloping introduces stress concentration regions in some parts of the structure. Stress concentration is a dangerous source of strength loss for the whole structure and for comb-drives actuators which may suffer from side pull-in. This paper presents an analytical approach to characterize stress concentrations in DRIE micro-machined MEMS. The method is based on the linear elasticity equations, the de Saint-Venant Principle, and the boundary value problem for the case of a torsional state of the beam. The results obtained by means of this theoretical method are then compared with those obtained by using two other methods: one based on finite difference discretization of the equations, and one based on finite element analysis (FEA). Finally, the new theoretical approach yields results which are in accordance with the known value of the stress concentration factor for asymptotically null radius notches.

Keywords: scalloping; stress concentration; compliant mechanisms; DRIE

1. Introduction

Interest in microelectromechanical systems devices has steadily increased in the last years. Micro-scale technologies have proven to be very effective, playing a prominent role in a wide variety of applications from different fields, such as drug delivery [1–5], aerospace [6,7], medical diagnosis [8,9], surgical applications [10–12], and cell manipulation [13–15].

Since 2013 [16,17], a new class of MEMS, equipped with Conjugate Surface Flexure Hinges (CSFH) has been developed [18,19] and fabricated [20]. A sample of CSFH is depicted in Figure 1. These systems consist essentially of micro *compliant mechanisms* where the flexure hinges are manufactured as CSFHs. In such kind of flexure, the thickest *rigid* elements are linked by a certain number of CSFHs, where each CSFH consists of a curved beam together with a portion of a conjugate-profile pair, as illustrated in Figure 1. According to the CSFH design, the center of each arc is coincident with the center of the elastic weights of the curved beam [21].

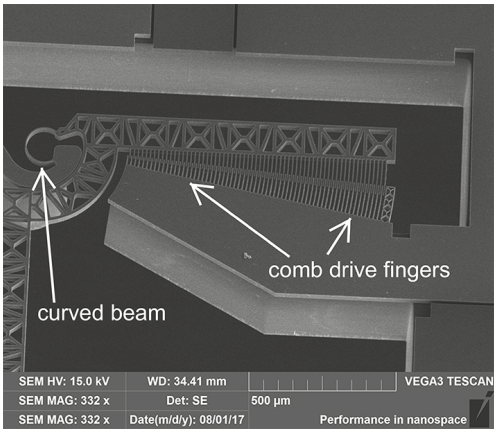

Figure 1. Curved beam adopted in CSHF hinges and comb-drive fingers.

The availability of CSFH hinges gave rise to the design and fabrication of new devices, such as microrobots [16,17,22–24] micromechanisms [17,25], microgrippers [18,26–28], and even a microtribometer [29].

In-plane MEMSs perform properly only if their flexible elements offer high compliance selectivity around the perpendicular axis. Hence, high aspect ratio structures are desirable. While surface micro-machining offers, typically, thickness from 2 to 5 µm, the Deep-Reactive Ion Etching (DRIE) process is used to obtain thickness of 50 µm and much more.

Since 1993 [30], Bosch DRIE process has introduced an effective method to provide, simultaneously, high mask selectively and very high anisotropy of the etched structure. In 1999, the scalloping observed on vertical walls during time multiplexed deep etching (TMDE), the roughness of horizontal surfaces exposed to the glow discharge, and the radius at the bottom of etched features were highlighted [31]. Few years later, it was demonstrated that important features of the samples, such as achievable anisotropy, etching uniformity, fillet radii, and, mainly, surface roughness, are strongly dependent on chamber pressure, applied coil and electrode power, and reactant gases flow rate [32]. This experimental observation lays the groundwork for the present investigation. In fact, scalloped sidewall roughness affects the performance of DRIE-manufactured comb-drives from both an *electrical* and a *mechanical* point of view. An example of DRIE-manufactured rotary comb-drive is depicted in Figure 1.

Considering the electrical aspect, scalloping determines, along the thickness of the device layer, gap variations between the facing surfaces of the fixed and movable fingers. Therefore, it affects the force or the torque exerted by the linear or rotary comb-drive, respectively. Furthermore, charge concentrations occur in sharp edges [33], determining variations of the electric field along the thickness direction.

From a mechanical perspective, it was observed that [31,32] post etch behavior for specimens with high surface roughness always indicated low fracture strength, whereas for specimens with better surface quality there was a wider distribution in sample strength. In addition, recent experimental tests [34] showed that some teeth are subject to lateral pull-in, with possible large deflection and instability.

The present investigation focuses on the mechanical aspect.

As discussed in the next paragraph, one source of strength decay is the stress concentration factor, which holds in those zones where surface discontinuities appear. In high aspect ratio DRIE Bosch process, problems and challenges regarding scalloping attenuation have been reported in various investigations [35–40], although the problem presented several difficulties.

More recently, it has been shown that the sidewall roughness varies with the depth and depends on the trench width. In addition, the surface upper region exhibits a scalloping morphology, while the rougher lower region shows a curtaining morphology [41]. Although cryogenic-DRIE (see for example Refs. [42–45]) has recently been employed to fabricate trenches with highly vertical sidewalls and obtained relatively smooth surfaces, Bosch DRIE remains of great interest for the construction of many devices.

In 2012, a study [46] dedicated to Through-Silicon Via (TSV) technology showed that an etch method not based on Bosch process can get quite smooth sidewalls with no scalloping, together with a good control of the etched profile.

Since scalloping is not acceptable in specific applications, a dedicated effort has been provided to control and reduce it, mainly based on the optimization of DRIE process parameters [47–50], or, alternatively, removing it after fabrication [38].

Unfortunately, indentations and notches on the surface of a DRIE manufactured device give rise to high stress concentration that leads the samples to break at an unexpectedly low load.

The present work intends to contribute in filling this gap and to investigate about the shear stress distribution in the neighborhood of each single scalloped discontinuity on the walls device.

Hence, to develop a theoretical model, a uniform cantilever beam with circular cross section has been considered. To model the scalloping effect, a circular region, or *notch*, has been subtracted from the beam cross section. The center of the notch lies on the cross section profile.

By applying the linear elasticity theory to the beam with notched cross section, the shear stress distribution can be obtained *analytically* for the case of pure torsion. The idea of studying the beam torsional state can be justified by considering that stress concentration is *actually* due to tangential tensions. In addition, the shearing force is quite likely eccentric with respect to the center of shear of the cross-section. This implies a twisting moment acting on the beam, which is dangerous because of the very low torsion rigidity of thin-walled open profiles. Furthermore, the equations are integrated also by using a numerical approach based on finite discretization of the section and finite differences. Finally, the results are compared to those obtained with finite element analysis (FEA).

2. Motivation of This Work

The Bosh DRIE process is performed by means of pulsed regimes in two steps: in the first one, plasma etches the wafer along mainly the vertical direction, while in the second phase a passivation layer (Polytetrafluoroethylene) is chemically deposited on the silicon surface thanks to the action of a source gas (usually Octafluorocyclobutane). The two steps are iterated in such a way that, after each surface passivation phase, an etching phase follows, during which ions attack selectively the passivation layer at the bottom of the trench but not along the side walls. This iterative process permits very deep vertical etching [30]. However, the etching phase affects also the lateral walls and so the side is not perfectly planar, but it is rather undulating, as pictorially represented in Figure 2. The first image, reported in Figure 2a, represents a simplified scheme of a portion of a SOI wafer, after four cycles (note that the stop layer, represented in gray color, is reached after one more cycle only). The central cantilever beam appears after the wet etching phase, as illustrated in Figure 2b,c, during which the oxide is partially removed.

As known, any geometric irregularity on the cross section of a beam is responsible for stress concentration, which decreases its actual strength. The capability of micro-electromechanical systems (MEMS) to move is a direct consequence of their embedded flexures. According to the adopted process, a SOI (Silicon on Insulator) is used to obtain a suspended so-called *device layer* that can move along *in-plane* directions. The geometry of the device is designed and then transferred to the device layer by means of a mask. In the device layer, there are some *pseudo-rigid* parts that achieve motion capability because of the presence of flexible micro-beams. It is therefore clear that the strength of the micro-beams affects performance and reliability of the whole microsystem.

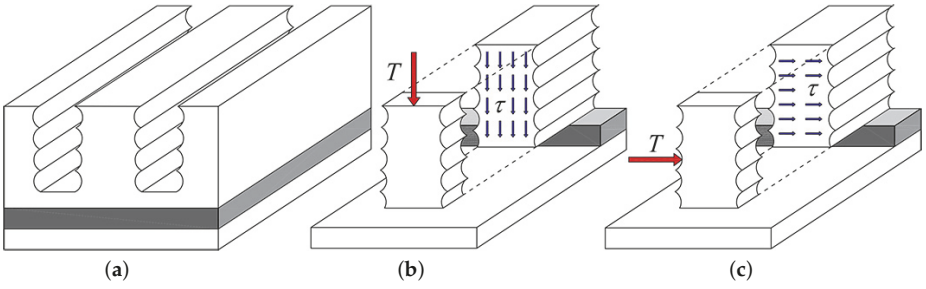

Figure 2. A simplified representation of the SOI wafer after four cycles of the D-RIE process (**a**), and of a cantilever beam after oxide wet etching, subject to out-of-plane (**b**) and in-plane (**c**) shear tension.

Given the large variety of applications, micro-beams in MEMS are used for many different purposes. For example, they can be used either to create the fingers (linear or curved) of a comb-drive actuators or as a structural element (once again linear of curved) of a flexure. Therefore, these micro-beams are generally subject to local or distributed external forces that induce shear tensions on the beam cross-section. These forces may act along either an *out-of-plane* or an *in-plane* direction. Out-of-plane tensions are represented in Figure 2b, while in-plane tensions are depicted in Figure 2c. In the case of comb-drive actuators, fingers are subject to the *lateral* pull-in effect, and therefore they are inflected around an axis that is orthogonal to the pane of motion (in-plane inflection). On the other hand, surface forces, such as electrostatic forces, may act perpendicularly to the plane of motion, giving rise to out-of-plane inflection.

Furthermore, since the gaps among fingers can be reduced to as small as one micron, the possible inflection is responsible for stitching, which makes the MEMS practically impossible to use any further. The mechanics of adhesion this context has been extensively discussed in a recent review [51], where some fundamental parameters for microscale adhesion in MEMS are discussed in detail.

Figure 3 shows scalloping over nine fingers of a circular comb-drive that the authors use for MEMS actuation. Although comb-drives fingers do not present direct contacts, mechanical impacts, vibrations or electrostatic field forces can still play a certain action on the fingers.

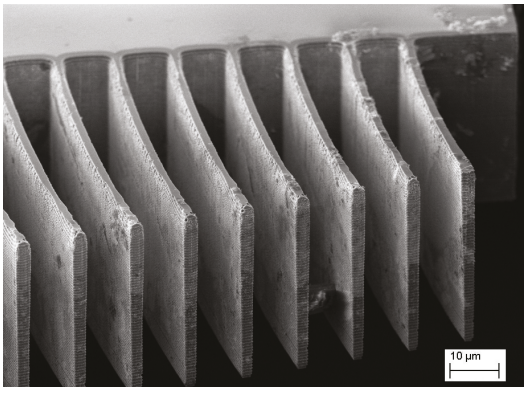

Figure 3. Comb-drive finger region.

However, a much greater stress state is generated on the curved beam which sustains the moving parts. For example, in the case of CSFH hinges [34], the curved beam is bent around an axis which is orthogonal to the plane of motion.

Figure 4 shows how scalloping produces roughness on one finger surface.

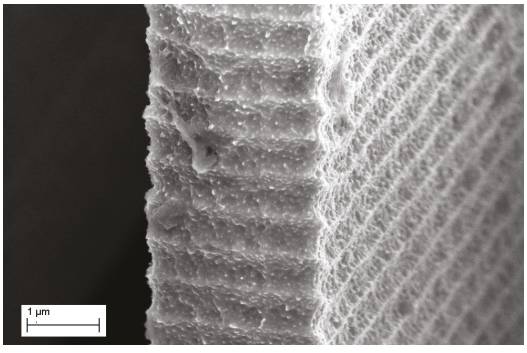

Figure 4. Scalloping details in one finger.

3. Theoretical Background

We refer to a prismatic beam as a region bounded by an exterior cylindrical surface and two terminal plane sections perpendicular to the generators [52]. The beam axis has a mild curvature compared to its thickness, so the theory for straight beams can be extended to this case of study. According to Ref. [53], this assumption is true when the ratio of the curvature radius to thickness is greater than 5.

The generic cross-section is depicted in Figure 5. Axes x and y, not necessarily principal of inertia, intersect each other in the centroid S and form an orthogonal Cartesian frame with z lying along the direction of the beam axis.

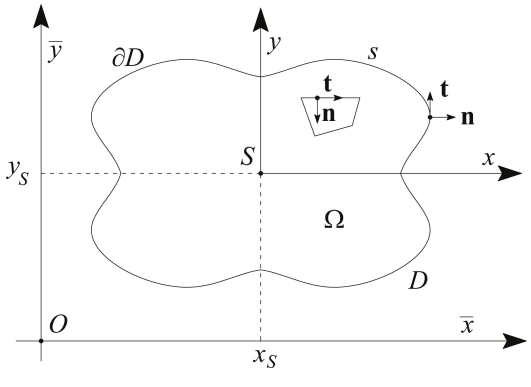

Figure 5. Cross-section with nomenclature.

The present formulation applies with no restriction to cross-sections of arbitrary shape and also to hollow beams. However, the case of simply-connected domain is analyzed here.

In the case of two axes of symmetry, the section centroid coincides with the center of shear, and extension, bending , and torsion are uncoupled if forces are applied to S [54–58].

The cross-section is a domain D of area Ω, bounded by the exterior directrix s whose direction is uniquely defined asking the associated outward normal $\mathbf{n}$ and tangent $\mathbf{t}$ unit vectors to form a right-handed system with the unit vector $\hat{e}_z$.

In the following sections, shear stresses is determined from the theory of linear elasticity. Once the stress and strain tensors are defined, the semi-inverse method of Saint-Venant [59–61] is applied by setting [52,54,55,57,58]

$$\sigma_x = \sigma_y = \tau_{xy} \equiv 0\,,$$

the equilibrium equations, neglecting body forces, read

$$\frac{\partial \tau_{xz}}{\partial z} = 0\,, \tag{1}$$

$$\frac{\partial \tau_{yz}}{\partial z} = 0\,, \tag{2}$$

$$\frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0\,. \tag{3}$$

Assuming the exterior surface to be free of stress corresponds to setting

$$\tau_{xz} dy - \tau_{yz} dx = 0 \tag{4}$$

on ∂D, with $dy = n_x ds$ and $dx = -n_y ds$.

The hypotheses of isotropy and linear elastic behavior of the material lead to the stress-strain relations:

$$\sigma_z = E\epsilon_z = E\frac{\partial w}{\partial z}\,, \tag{5}$$

$$\tau_{xz} = G\gamma_{xz} = G\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right)\,, \tag{6}$$

$$\tau_{yz} = G\gamma_{yz} = G\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right)\,, \tag{7}$$

where $\mathbf{u} = [u, v, w]^T$ is the small displacement field, and E and G are Young's and shear modulus, respectively.

The stress resultants on the cross-section are the normal and shear forces T_z, T_x, T_y and the twisting and bending couples M_z, M_x and M_y:

$$\begin{aligned} T_z &= \int_D \sigma_z \, dxdy\,, & M_x &= \int_D y\sigma_z \, dxdy\,, \\ T_x &= \int_D \tau_{xz} \, dxdy\,, & M_y &= -\int_D x\sigma_z \, dxdy\,, \\ T_y &= \int_D \tau_{yz} \, dxdy\,, & M_z &= \int_D (x\tau_{yz} - y\tau_{xz}) \, dxdy\,. \end{aligned} \tag{8}$$

4. Shear Stresses in Torsion

As mentioned in Section 1, we focus on the shear stress concentration due to a scallop-like geometry in a beam subject to pure torsion. To compute the shear stresses, the solution of Poisson's problems, with Neumann or Dirichlet boundary conditions, is required. Such problems imply the use of Laplace operator in two dimensions [52,54–58,62].

Pure torsion is defined as the partial solution of the problem of shear stresses where $\sigma_z \equiv 0$ (implying the absence of shear forces and bending moments) [52]. The shear stresses $\tau = [\tau_{xz}, \tau_{yz}]^T$ satisfy the simplified equilibrium equation:

$$\operatorname{div} \tau = \frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} = 0. \tag{9}$$

When a twisting couple M_z is applied to the free end of a clamped beam, the flatness of the section is preserved only if the beam has circular cross-sections. In all other cases, the longitudinal displacement w is not linear in the variables x and y, causing a warping of the section.

Assuming the torsion angle of the cross-section $\omega_o = \omega_o(z)$ to be small, the displacement field $\mathbf{u}$ in pure torsion is devoid of rigid contributions,

$$u = -\omega_o(z)y\,, \quad v = \omega_o(z)x\,, \quad w = \theta g(x,y)\,, \tag{10}$$

where the constant parameter $\theta = \frac{d\omega_o}{dz}$ is the twist of the centroidal fiber and $g(x,y)$ is the so-called *warping function*.

Generally, the function $g(x,y)$ is not linear in x and y and causes the loss of flatness of the section. It is defined only up to an additive constant and can be made unique by the constraint

$$\int_D g\,dxdy = 0.$$

Deriving shear strains as shown in Equations (6) and (7) allows us to write shear stresses as:

$$\tau_{xz} = G\theta\left(\frac{\partial g}{\partial x} - y\right), \quad \tau_{yz} = G\theta\left(\frac{\partial g}{\partial y} + x\right). \tag{11}$$

By substituting Equation (11) into the equilibrium condition, Equation (9), leads to

$$G\theta\left(\frac{\partial^2 g}{\partial^2 x} + \frac{\partial^2 g}{\partial^2 y}\right) = 0\,.$$

Together with the boundary condition in Equation (4) it determines the following Neumann's problem:

$$\begin{cases} \nabla^2 g = 0 & \text{in } D, \\ \frac{\partial g}{\partial n} = \frac{1}{2}\frac{\partial}{\partial s}\left(x^2 + y^2\right) & \text{on } \partial D, \end{cases} \tag{12}$$

with ∇^2 Laplace operator in the plane of the domain D. The solution $g(x,y)$ of this boundary value problem is unique up to an additive constant, and can be found in a closed form only for some simple cases (e.g., circular cross-sections, where $g = 0$ is an exact solution and fulfills the boundary condition [52]). Once the problem is solved, shear stresses are determined by means of Equation (11).

Another formulation of the problem can be derived defining *Prandtl's stress function* $\Phi(x,y)$ such that

$$\tau_{xz} = G\theta\frac{\partial \Phi}{\partial y}, \quad \tau_{yz} = -G\theta\frac{\partial \Phi}{\partial x}. \tag{13}$$

The equilibrium condition, Equation (9), is satisfied when the stress function Φ is a solution of the problem:

$$\begin{cases} \nabla^2 \Phi = -2 & \text{in } D, \\ d\Phi = 0 & \text{on } \partial D. \end{cases} \tag{14}$$

Since the addition of a constant to Φ does not affect stresses, the constant value Φ_0 on the outer boundary ∂D_0 can be set equal to 0. In hollow beams (multiple connected domains), the values of Φ_{0i} on the inner boundaries ∂D_i cannot be chosen arbitrarily, but must satisfy the circuital conditions

$$\int_{\partial D_i} \frac{\partial \Phi}{\partial n}\,ds = 2\Omega_i, \qquad i = 1, 2, \ldots, n$$

where Ω_i denotes the area of the ith cavity. This constraint, which is necessary to ensure the displacement w has one value, cannot in general be solved explicitly within numerical analysis [56,63].

The torsion cross sectional rigidity J, entering the linear constitutive relationship for the twisting couple,

$$M_z = GJ\theta\,,$$

can be determined by integrating Prantdl's stress function over the section [52,54–58,62]:

$$J = 2\int_D \Phi\, dxdy.$$

5. Circular Notched Cross-Section

As recalled in previous sections, the shear stress field is the solution of elliptic boundary value problems defined on the cross-section of the beam. Such problems are generally solved by means of finite differences or FEA.

However, the scheme presented here takes advantage of the simple form that the stress functions present for some particular geometries.

The warping function $g(x,y)$ and its harmonic conjugate $Z(x,y) = \Phi(x,y) + \frac{1}{2}\left(x^2+y^2\right)$ are two functions which are coupled by the Cauchy-Riemann relations, i.e.,

$$\begin{aligned}\frac{\partial g}{\partial x} &= \frac{\partial Z}{\partial y},\\ \frac{\partial g}{\partial y} &= -\frac{\partial Z}{\partial x}.\end{aligned}$$

Simple analytical solutions of Laplace equation are given by polynomials in the complex variable $x+iy$, such that the generic m-th term is

$$g + iZ = (p_m + iq_m)(x+iy)^m, \tag{15}$$

with [52]

$$\begin{aligned} g &= p_m\left(x^m - \frac{m(m-1)}{2}x^{m-2}y^2 + \ldots\right) + \\ &- q_m\left(mx^{m-1}y + \ldots\right), \\ Z &= p_m\left(mx^{m-1}y + \ldots\right) + \\ &+ q_m\left(x^m - \frac{m(m-1)}{2}x^{m-2}y^2 + \ldots\right). \end{aligned}$$

In the case of negative values of m in Equation (15), the function $g+iZ$ will present *poles*, i.e., singularity points where the function approaches an infinite value. Thus, negative exponents m are possible in Equation (15), provided that the generated poles fall out of the elastic region D. For example, for $m=-1$, there is one simple pole in the origin:

$$g + iZ = \frac{(p_{-1} + iq_{-1})(x-iy)}{x^2+y^2}\,.$$

In this case, the stress function

$$\Phi(x,y) = \frac{-hr^2x}{x^2+y^2} - \frac{1}{2}\left(x^2+y^2\right) + hx + \frac{1}{2}r^2 \tag{16}$$

is regular inside the domain D, vanishes on the boundaries $x^2+y^2 = r^2$ and $(x-h)^2+y^2 = h^2$ and satisfies the problem in Equation (14) (it is Prandtl's stress function for the torsion of a circular

cross-section beam, of radius h, with a circular notch of radius r centred on its surface, see Figure 6). Considering the details shown in Figure 4, the circular shape of notches is assumed to be a first approximation fitting the real profile.

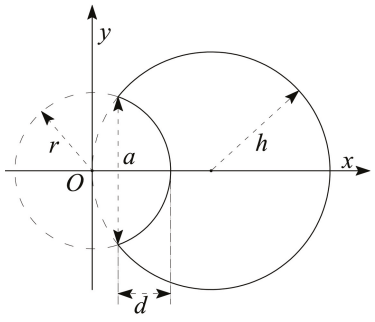

Figure 6. Reference frame, cross-section with radius h, and notch with radius r.

The associated warping function is, up to a constant,

$$g(x,y) = \left(1 - \frac{r^2}{x^2+y^2}\right) h\, y. \tag{17}$$

Thus, for a beam with such a cross-section undergoing pure torsion, shear stresses can be easily evaluated with Equation (11) or (13) by partial differentiation of either g or Φ.

6. Theoretical Aspects in the Finite Element Formulation

The second order partial differential equation problem (with Dirichlet or Neumann boundary conditions) can be written, in its strong formulation, as:

$$\begin{cases} \nabla^2 \varphi = f_0(x,y) & \text{in } D \\ \dfrac{\partial \varphi}{\partial n} = f_1(x,y) \text{ or } \varphi(s) = f_2(s) & \text{in } \partial D \end{cases} \tag{18}$$

in the previous equations,

$$\begin{aligned} f_0(x,y) &= 0, \\ f_1(x,y) &= \frac{1}{2}\frac{\partial}{\partial s}\left(x^2+y^2\right), \end{aligned}$$

in the case $\varphi(x,y) = g$, while

$$\begin{aligned} f_0(x,y) &= -2, \\ f_2(x,y) &= \Phi_0, \end{aligned}$$

in the case $\varphi(x,y) = \Phi$.

The weak form of the problem, which can be implemented by most finite element solvers, can be obtained by a weighted residuals method, introducing a set of weighting functions, η. The integral sum over the domain D in the product of the weights η by the residual $R = -\nabla^2\varphi + f_0(x,y)$ is required to vanish

$$G(\varphi,\eta) = \int_D R\eta \, d\Omega = \int_D \left(-\nabla^2\varphi + f_0\right) \eta \, d\Omega = 0\,. \tag{19}$$

Integration by parts, and Neumann's boundary condition, provide the weak form of the boundary value problem:

$$G(\varphi, \eta) = \int_D \nabla\varphi \cdot \nabla\eta + \eta f_0 \, d\Omega - \int_{\partial D} \eta f_1 \, ds = 0\,, \tag{20}$$

with ∇ the gradient in the plane of the domain D.

The condition for η is $\eta = 0$ on ∂D. The integral over the boundary will then vanish in the case of Dirichlet boundary conditions.

Equation (20) can be solved approximately using the finite element method; the domain D is first decomposed into a number ℓ of sub-domains, each one having n nodes.

The nodal approximations for the coordinates vector $\mathbf{x} = [x, y]^T$, the unknown function φ and the weighting function η are built using the same shape function N_I. Once their values in the nodes are known, the interpolations which derive are:

$$\begin{aligned} \tilde{\mathbf{x}} &= \sum_{I=1}^{n} N_I \mathbf{x}_I\,, \\ \tilde{\varphi} &= \sum_{I=1}^{n} N_I \varphi_I\,, \\ \tilde{\eta} &= \sum_{I=1}^{n} N_I \eta_I\,, \end{aligned} \tag{21}$$

where the ~ (tilde) sign denotes the value of the indicated function.

Inserting the nodal approximations in Equation (21) into the weak form of Equation (20), and summing over the ℓ subdomains leads to the finite element problem:

$$G(\varphi, \eta) = \sum_{e=1}^{\ell} G^e = \sum_{e=1}^{\ell} \sum_{I=1}^{n} \sum_{K=1}^{n} \eta_I \left(K_{IK}^e \varphi_K - P_I^e\right) = 0, \tag{22}$$

with the global matrix K_{IK} and the vector P_I formed by the elementary matrices and vectors:

$$K_{IK}^e = \int_{D^e} \left(\nabla N_I \cdot \nabla N_K\right) \, d\Omega^e \tag{23}$$

$$P_I^e = \int_{D^e} f_0(x, y) N_I \, d\Omega^e + \int_{\partial D} f_1(x, y) N_I \, ds. \tag{24}$$

To compute shear stresses, we must solve the linear system in Equation (22). To this end, the cross-section D properties, the second order moments of inertia, I_{xx}, I_{xy}, I_{yy}, and the centroid coordinates x_S and y_S, as well as the derivatives of the shape functions N with respect to x and y, must be known.

7. Stress Analysis

The presented methods can be specialized for the study of the above-mentioned class of CSFH-based microdevices. A member of this class is characterized by the presence of an embedded CSFH hinge, which is composed of a flexible curved element connecting a couple of conjugate profiles. One possible configuration is depicted in Figure 7. Relative motion among the pseudo-rigid parts i and j is possible because the thin cross-section of the curved element k behaves as a flexible rod.

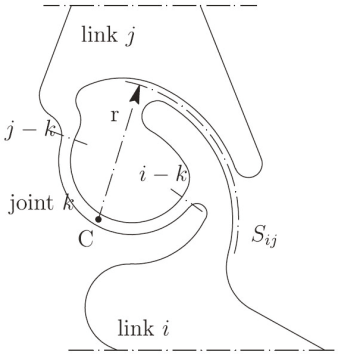

Figure 7. Flexural hinge geometry (see also Refs. [15,64]).

The technological limitations or the application complexity do not guarantee that the flexible curved beam k is subject only to *pure bending*. On the contrary, the tension field is rather complex and also includes shear component on the cross section. In fact, if a curved beam is considered, in-plane shear stress (as in Figure 2b) is always present, while out-of-plane, or *transverse*, shear stress (Figure 2c) is likely to be present for several reasons. One of them, already pointed out in Section 1, is that the shearing force on the beam is quite likely eccentric with respect to the shear center, thus implying a twisting couple. Considering that beams are thin-walled, we know this implies remarkable shear stress.

Since the aim of this paper is restricted to the analysis of stress concentration, it is reasonable to simplify the general case of *curved beam* by a *straight axis beam with in-plane shear* at one end. For this reason, the stress distribution over a cross section of the beam is investigated, and increase in the overall tension maximum value is monitored to avoid unpleasant damages or fractures at unexpectedly low loads.

In the following paragraphs, the elementary beam theory is applied to the case of a circular cross-section shaft with a circular shaped notch undergoing pure torsion. Stresses are computed by using the analytical solution of the equations shown in Section 5. Then, two more methods, both having numerical nature, are used to compare their results with the analytical solutions: finite differences and finite elements.

7.1. Analytical Solution

For the geometry in Figure 6, it has been shown that Prandtl's stress function

$$\Phi(x,y) = \frac{-hr^2x}{x^2+y^2} - \frac{1}{2}\left(x^2+y^2\right) + hx + \frac{1}{2}r^2\,,$$

and the warping function

$$g(x,y) = \left(1 - \frac{r^2}{x^2+y^2}\right) h\,y\,,$$

have to satisfy Equations (11) and (13):

$$\tau_{xz} = G\theta\left(\frac{\partial g}{\partial x} - y\right) = G\theta\frac{\partial \Phi}{\partial y}\,,$$
$$\tau_{yz} = G\theta\left(\frac{\partial g}{\partial y} + x\right) = -G\theta\frac{\partial \Phi}{\partial x}\,.$$